

**Honeywell**

---

**TPN Bridge**  
**Overview and Implementation**  
**Guide**

TN08-100c

R100

04/04

**TotalPlant**

## Notices and Trademarks

**Copyright 2001-2004 by Honeywell Inc.  
Release 100 – April 2004**

While this information is presented in good faith and believed to be accurate, Honeywell disclaims the implied warranties of merchantability and fitness for a particular purpose and makes no express warranties except as may be stated in its written agreement with and for its customers.

In no event is Honeywell liable to anyone for any indirect, special or consequential damages. The information and specifications in this document are subject to change without notice.

Honeywell and **TotalPlant** are U.S. registered trademarks of Honeywell International Inc.

Other brand or product names are trademarks of their respective owners.

Honeywell International  
Industry Solutions  
12500 West Union Hills  
Phoenix, AZ 85027  
**1-800 343-0228**

# About This Document

## References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

| Document Title                                       | Doc. ID |
|------------------------------------------------------|---------|
| <i>TPN Bridge Planning, Installation and Service</i> | TN30    |

## Contacts

### World Wide Web

The following Honeywell web sites may be of interest to Industry Solution customers.

| Honeywell Organization | WWW Address (URL)                                                                       |
|------------------------|-----------------------------------------------------------------------------------------|
| Corporate              | <a href="http://www.honeywell.com">http://www.honeywell.com</a>                         |
| Industry Solutions     | <a href="http://www.acs.honeywell.com">http://www.acs.honeywell.com</a>                 |
| International          | <a href="http://content.honeywell.com/global/">http://content.honeywell.com/global/</a> |

### Telephone

Contact us by telephone at the numbers listed below.

| Organization             |                                                         | Phone Number     |                   |
|--------------------------|---------------------------------------------------------|------------------|-------------------|
| United States and Canada | Honeywell International Inc.                            | 1-800-343-0228   | Sales             |
|                          | Industry Solutions                                      | 1-800-525-7439   | Service           |
|                          |                                                         | 1-800-822-7673   | Technical Support |
| Asia Pacific             | Honeywell Asia Pacific Inc.<br>Hong Kong                | (852) 23 31 9133 |                   |
| Europe                   | Honeywell PACE<br>Brussels, Belgium                     | [32-2] 728-2711  |                   |
| Latin America            | Honeywell International Inc.<br>Sunrise, Florida U.S.A. | (954) 845-2600   |                   |

## Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

| Symbol                                                                              | Definition                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | This CAUTION symbol on the equipment refers the user to the Product Manual for additional information. This symbol appears next to required information in the manual.                                                                        |
|    | This WARNING symbol on the equipment refers the user to the Product Manual for additional information. This symbol appears next to required information in the manual.                                                                        |
|    | WARNING: risk of electrical shock. This symbol warns the user of a potential shock hazard where HAZARDOUS LIVE voltages greater than 30 Vrms, 42.4 Vpeak, or 60 VDC may be accessible.                                                        |
|  | ATTENTION, Electrostatic Discharge (ESD) hazards. Observe precautions for handling electrostatic sensitive devices                                                                                                                            |
|  | Protective Earth (PE) terminal. Provided for connection of the protective earth (green or green/yellow) supply system conductor.                                                                                                              |
|  | Functional earth terminal. Used for non-safety purposes such as noise immunity improvement. NOTE: This connection shall be bonded to protective earth at the source of supply in accordance with national local electrical code requirements. |
|  | Earth Ground. Functional earth connection. NOTE: This connection shall be bonded to Protective earth at the source of supply in accordance with national and local electrical code requirements.                                              |
|  | Chassis Ground. Identifies a connection to the chassis or frame of the equipment shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.                              |

# Contents

|                                                                       |           |
|-----------------------------------------------------------------------|-----------|
| <b>1. INTRODUCTION .....</b>                                          | <b>11</b> |
| <b>1.1 Overview .....</b>                                             | <b>11</b> |
| Summary.....                                                          | 11        |
| Terminology and acronyms .....                                        | 11        |
| <b>1.2 Bridge Technology.....</b>                                     | <b>13</b> |
| Network segmentation .....                                            | 14        |
| <b>1.3 Network Architecture of TPN Bridge.....</b>                    | <b>15</b> |
| <b>1.4 TPN Performance Issues.....</b>                                | <b>16</b> |
| Qualifying a site.....                                                | 16        |
| Recording and analyzing TPN data.....                                 | 17        |
| Review current network configuration (customer) .....                 | 18        |
| Overview of Network Analysis Program .....                            | 19        |
| Presenting analysis report to customer .....                          | 19        |
| Discuss options and recommendations.....                              | 20        |
| <b>1.5 Implementation of the TPN Bridge .....</b>                     | <b>21</b> |
| <b>2. PLANNING THE TPN BRIDGE SWITCHOVER.....</b>                     | <b>23</b> |
| <b>2.1 Preliminary.....</b>                                           | <b>23</b> |
| Planning goals.....                                                   | 23        |
| Overview of TPN Bridge switchover .....                               | 24        |
| Scope of network analysis.....                                        | 25        |
| Obtaining the report .....                                            | 26        |
| The report.....                                                       | 27        |
| Report description .....                                              | 27        |
| Breaking cables A and B .....                                         | 29        |
| Inserting bridges, switches, and terminators .....                    | 30        |
| <b>3. CONFIGURING THE TPN BRIDGE SYSTEM TIME.....</b>                 | <b>31</b> |
| <b>3.1 TPN Bridge System Time Synchronization .....</b>               | <b>31</b> |
| Overview .....                                                        | 31        |
| Three steps to configuring the TPN Bridge system clock.....           | 32        |
| <b>3.2 Time Link and GPS Options.....</b>                             | <b>41</b> |
| Options.....                                                          | 41        |
| For further information.....                                          | 41        |
| Implementation of three segments with Time Link and GPS Options ..... | 42        |

|           |                                                                   |           |
|-----------|-------------------------------------------------------------------|-----------|
| <b>4.</b> | <b>IMPLEMENTING THE TPN BRIDGE SYSTEM.....</b>                    | <b>45</b> |
| 4.1       | Overview .....                                                    | 45        |
|           | Summary .....                                                     | 45        |
|           | Overview of procedures .....                                      | 45        |
|           | Software description.....                                         | 46        |
| 4.2       | Using the Displays.....                                           | 47        |
| 4.3       | LCNP Status .....                                                 | 49        |
| 4.4       | Bridge Status .....                                               | 49        |
|           | TPN Bridge Personality State (Bridge Status Window) .....         | 49        |
|           | Message buffering on TPN Bridge nodes.....                        | 50        |
|           | Internal table addresses .....                                    | 50        |
|           | Status registers .....                                            | 52        |
|           | Individual Status Values .....                                    | 53        |
|           | Raw / Rate Counters .....                                         | 53        |
|           | Network communication subsystem state .....                       | 54        |
|           | Bridge Operation Checklists.....                                  | 57        |
| 4.5       | Bridge Operation Checklists .....                                 | 58        |
|           | Bridge Status display.....                                        | 58        |
|           | LCNP Status applet display.....                                   | 61        |
| 4.6       | Capturing Failure Status for TAC Analysis.....                    | 62        |
|           | Overview .....                                                    | 62        |
| 4.7       | Configuring the Nortel (BayStack) Switch .....                    | 64        |
|           | Purpose.....                                                      | 64        |
|           | Prerequisites .....                                               | 64        |
|           | Disable "snooping" on Nortel (BayStack) switch .....              | 64        |
| 4.8       | Installing and Configuring the TPN System Software .....          | 66        |
|           | Installing "Greater than 64 Node" keyfile option .....            | 66        |
|           | Adding the TPN Bridge to the NCF .....                            | 67        |
| 4.9       | TPN Bridge and NT Configuration .....                             | 68        |
|           | Overview .....                                                    | 68        |
|           | Initializing the node.....                                        | 69        |
| 4.10      | Configuring the Bridge Segments Using MMC .....                   | 80        |
|           | Overview .....                                                    | 80        |
|           | Configuration using TPN Bridge configuration/status snap-In ..... | 80        |
| 4.11      | Setting TPN Bridge Time Sync Options.....                         | 94        |
|           | Overview .....                                                    | 94        |
|           | Setting the time synchronization option.....                      | 94        |
|           | Turning off the time synchronization option.....                  | 96        |
|           | Recording TPN clock sources .....                                 | 96        |

|           |                                                                |            |
|-----------|----------------------------------------------------------------|------------|
| 4.12      | <b>Adding a New Segment to an Existing Bridge System .....</b> | <b>97</b>  |
|           | Overview .....                                                 | 97         |
| <b>5.</b> | <b>TPN BRIDGE HOT INSERTION.....</b>                           | <b>105</b> |
| 5.1       | <b>Overview .....</b>                                          | <b>105</b> |
|           | Summary.....                                                   | 105        |
|           | Hot insertion concepts.....                                    | 106        |
| 5.2       | <b>Special Considerations .....</b>                            | <b>107</b> |
|           | TPN Bridge shut-down .....                                     | 107        |
|           | TPN Bridge node failover .....                                 | 107        |
| 5.3       | <b>About TPN Bridge Cables .....</b>                           | <b>108</b> |
|           | Cable breaks .....                                             | 108        |
|           | Cable error interpretation .....                               | 108        |
|           | Cable error symptoms and causes.....                           | 109        |
| 5.4       | <b>First Segmentation of a TPN (LCN) .....</b>                 | <b>111</b> |
|           | Configuring the NCF for the TPN Bridge.....                    | 111        |
|           | Configuring NT on the TPN Bridge nodes .....                   | 111        |
|           | Configuring the TPN Bridge node software .....                 | 112        |
|           | Preparing the LCN for TPN Bridge insertion .....               | 112        |
|           | Inserting hot insertion switchbox tool .....                   | 113        |
|           | Offline testing of Bridge operation .....                      | 115        |
|           | Checking the Bridge node communications on the LCN .....       | 119        |
|           | Breaking the LCN.....                                          | 128        |
|           | Removing the TPN Bridge hot insertion switchbox .....          | 131        |
|           | Setting Time Sync options.....                                 | 133        |
| 5.5       | <b>Creating Additional Segments.....</b>                       | <b>134</b> |
|           | Configuring the NCF for the TPN Bridge.....                    | 134        |
|           | Configuring the NT on TPN Bridge nodes.....                    | 134        |
|           | Configuring the TPN Bridge Node software .....                 | 134        |
|           | Preparing the LCN for TPN Bridge insertion .....               | 135        |
|           | Inserting Hot Insertion switchbox tool.....                    | 135        |
|           | Offline testing of Bridge operation .....                      | 138        |
|           | Checking the Bridge Node communications on the LCN .....       | 142        |
|           | Breaking the LCN.....                                          | 142        |
|           | Removing the TPN Bridge Hot Insertion switchbox.....           | 146        |
| 5.6       | <b>Replacing an LCNE with a TPN Bridge .....</b>               | <b>149</b> |
|           | Overview .....                                                 | 149        |
|           | Offline testing of Bridge operation .....                      | 149        |
|           | Checking the Bridge node communications on the LCN .....       | 155        |
|           | Breaking the LCN.....                                          | 164        |
|           | Setting Time Sync options.....                                 | 167        |

## Contents

---

|            |                                                                     |            |
|------------|---------------------------------------------------------------------|------------|
| <b>5.7</b> | <b>Replacing the Network Card.....</b>                              | <b>168</b> |
|            | Overview .....                                                      | 168        |
|            | Replacing the dual NIC network card in a TPN Bridge pair .....      | 168        |
|            | IMPORTANT – Perform this procedure before you begin .....           | 169        |
|            | Remove the computer from the cabinet .....                          | 170        |
|            | Install the dual NIC network card.....                              | 171        |
|            | Update Dual NIC network adapter driver.....                         | 172        |
|            | Configure MAC Addresses for dual NIC network adapter ports .....    | 173        |
|            | Verify LCNP and TPN Bridge status.....                              | 177        |
| <b>5.8</b> | <b>Replacing a Bridge and Restoring Current Configuration .....</b> | <b>178</b> |
|            | Overview .....                                                      | 178        |
|            | Recovering the hard drive .....                                     | 178        |
|            | Restore the Bridge configuration .....                              | 181        |
| <b>5.9</b> | <b>Removing and Replacing the Ethernet Switch .....</b>             | <b>191</b> |
|            | Replacing the TPN Bridge System ethernet switch .....               | 191        |
| <b>6.</b>  | <b>HEALTH CHECK OF TPS LCN SYSTEM .....</b>                         | <b>193</b> |
| <b>6.1</b> | <b>TPS LCN System Checkout .....</b>                                | <b>193</b> |
|            | Importance of .....                                                 | 193        |
|            | Checking TPS LCN system .....                                       | 193        |
| <b>6.2</b> | <b>Normal Operation TPN Bridge Health Check.....</b>                | <b>195</b> |
|            | Expected values for TPN Bridge status .....                         | 195        |
|            | Expected values for TPN Bridge .....                                | 196        |
| <b>6.3</b> | <b>TPN Bridge Hints and Troubleshooting .....</b>                   | <b>198</b> |
|            | Hints .....                                                         | 198        |
|            | Troubleshooting.....                                                | 199        |
| <b>7.</b>  | <b>RECOVERY GUIDELINES .....</b>                                    | <b>201</b> |
| <b>7.1</b> | <b>Overview .....</b>                                               | <b>201</b> |
|            | Purpose .....                                                       | 201        |
|            | Responsibilities .....                                              | 201        |
|            | TPS LCN recovery requirements.....                                  | 201        |
|            | Required files for FAST LOAD cartridges.....                        | 202        |
| <b>7.2</b> | <b>Scenario Recovery Procedures .....</b>                           | <b>204</b> |
|            | Overview .....                                                      | 204        |
|            | Scenario # 1: No problems encountered .....                         | 204        |
|            | Scenario # 2: Some TPS nodes go ISOLATED.....                       | 204        |
|            | Scenario # 3: Segment(s) ISOLATED from each other .....             | 207        |
|            | Scenario # 4: All nodes are ISOLATED or in the FAIL state .....     | 210        |

---

## Tables

|            |                                                                              |     |
|------------|------------------------------------------------------------------------------|-----|
| Table 1-1  | Network Terms Defined for TPN Bridge .....                                   | 11  |
| Table 2-1  | Segment and Cluster Equivalency Table.....                                   | 28  |
| Table 3-1  | System Timing Requirements and Sync Source .....                             | 35  |
| Table 3-2  | Clock Sources and Configuration Bits .....                                   | 37  |
| Table 4-1  | Queue Status .....                                                           | 49  |
| Table 4-2  | Internal Table Addresses .....                                               | 51  |
| Table 4-3  | Status Registers.....                                                        | 52  |
| Table 4-4  | Individual Status Values.....                                                | 53  |
| Table 4-5  | Raw / Rate Counters.....                                                     | 53  |
| Table 4-6  | Network Communication Subsystem State .....                                  | 54  |
| Table 4-7  | Bridge Operation Checklist – LCN Display .....                               | 57  |
| Table 4-8  | Bridge Operation Checklist – Status Display .....                            | 58  |
| Table 4-9  | TPN Bridge Node Identification.....                                          | 72  |
| Table 4-10 | Ethernet Switch IP Addressing (Maintenance Use Only <i>Not Required</i> ) .. | 72  |
| Table 4-11 | TPN Clock Sources.....                                                       | 96  |
| Table 5-1  | Cable Error Symptoms and Causes .....                                        | 109 |
| Table 5-2  | TDC Status Display.....                                                      | 127 |
| Table 6-1  | TPN Bridge Status: Expected Values .....                                     | 195 |
| Table 6-2  | Expected Values .....                                                        | 196 |
| Table 7-1  | Recovery Guidelines for ISOLATED TPS Nodes (some).....                       | 204 |
| Table 7-2  | Recovery Guidelines for ISOLATED Segments .....                              | 207 |
| Table 7-3  | Recovery Guidelines for All Nodes in ISOLATED or FAIL State .....            | 210 |

### Figures

|             |                                                                  |     |
|-------------|------------------------------------------------------------------|-----|
| Figure 1-1  | TPN Bridge Architecture.....                                     | 15  |
| Figure 1-2  | TPN Bridge Hardware Interfaces .....                             | 21  |
| Figure 2-1  | LCN before Segmentation .....                                    | 23  |
| Figure 2-2  | LCN after Segmentation .....                                     | 24  |
| Figure 2-3  | Identifying the Cable Breaks for Insertion .....                 | 29  |
| Figure 2-4  | Overview of Installation of TPN Bridges.....                     | 30  |
| Figure 3-1  | Time Links and GPS Option .....                                  | 41  |
| Figure 3-2  | Three Segments with Time Link and GPS Options.....               | 43  |
| Figure 5-1  | Location of Switch Box, Cables and Tees.....                     | 112 |
| Figure 5-2  | Hot Insertion Switchbox Tool.....                                | 113 |
| Figure 5-3  | Private LCN Setup.....                                           | 115 |
| Figure 5-4  | Connection of the MAUs to Primary & Secondary Bridge Nodes ..... | 119 |
| Figure 5-5  | Hot Insertion Switchbox Removed and Cables Terminated.....       | 133 |
| Figure 5-6  | Location of Switch Box, Cables and Tees.....                     | 135 |
| Figure 5-7  | Hot Insertion Switchbox Tool.....                                | 136 |
| Figure 5-8  | Private LCN Setup.....                                           | 138 |
| Figure 5-9  | Hot Insertion Switchbox Removed and Cables Terminated.....       | 148 |
| Figure 5-10 | Private LCN Setup .....                                          | 149 |
| Figure 5-11 | LCNE Replacement — Bridge Equipment Collocated.....              | 154 |
| Figure 5-12 | LCNE Replacement — Bridge Segments Remote .....                  | 154 |
| Figure 5-13 | XLCNE2 Board Assembly.....                                       | 163 |
| Figure 5-14 | Dual NIC Adapter Ports for TPN Bridge .....                      | 171 |

# 1. Introduction

## 1.1 Overview

### Summary

The following information provides an overview of the TPN Bridge product and highlights the tasks that will be performed prior to the actual physical and software installation. Most of the front-end analysis tasks will be performed by Honeywell's Network Services. It is important to understand their role and the decisions to be made during the network qualification process before implementing the final hardware and software installation of the TPN Bridge.

### Terminology and acronyms

The use of the acronyms LCN (Local Control Network) and TPN (Total Plant Solutions Network) are used interchangeably at times. In referencing the legacy equipment and specifically the physical cable and cable connections, the use of LCN may be used. In referencing the TPS system, the use of TPN is considered the proper acronym.

With the introduction of the TPN Bridge into the network, there is the possibility of confusion with the established definition of terms. The following table was developed to clarify the definition of terms used and the context to which it applies.

**Table 1-1 Network Terms Defined for TPN Bridge**

| Term    | TPN (LCN)                                                                                                                                                                                                                           | TPN Bridge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLUSTER | A cluster is a group of nodes that are common to the control of a particular area or operating unit. This cluster normally involves one or more AMs, one or more gateways, one or more human interface nodes, and a history module. | <p>A cluster is a group of nodes that must not be separated on different TPN Bridge segments. This is for reasons of reliability and real-time performance. Examples are redundant node pairs and HMs, which collect history for particular nodes.</p> <p>Typically, LCN clusters should not be broken into multiple TPN Bridge clusters, as separating their members would place the control of the relevant area or operating unit at risk in case of failures.</p> <p>Furthermore, if a single operator typically controls two or more LCN clusters, then it would make sense to combine them into a single TPB Bridge cluster for reliability reasons.</p> |
| SEGMENT | An LCN segment is a section of contiguous coax LCN cable. Multiple segments may be combined in a single LCN using LCN Extenders (LCNEs).                                                                                            | A TPN Bridge segment is all LCN segments, which are directly attached to the LCN Interfaces of a redundant bridge pair. Different TPN Bridge segments have no direct connectivity other than that provided by the bridges themselves.                                                                                                                                                                                                                                                                                                                                                                                                                          |

## 1. Introduction

### 1.1. Overview

---

**Table 1-1      Network Terms Defined for TPN Bridge**

| Term   | TPN (LCN)                                                                                                                                                                                                                              | TPN Bridge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LINEAR | An ILLEGAL configuration of LCN cables connected together by LCN Extenders (LCNEs), in which LCNEs are "daisy-chained" together rather than all connected to a common backbone LCN segment.                                            | <p>(As opposed to Star) An arrangement of TPN Bridge clusters into TPN Bridge segments, which only allows physically adjacent sections of LCN cable to be put into a single TPN Bridge segment. Other arrangements, which would include multiple non-adjacent clusters, are not permitted.</p> <p>Linear arrangements of clusters into segments are somewhat less flexible than star arrangements, and may provide somewhat less traffic reduction than star arrangements. However, their advantage is that they do not require the addition of any LCN cable to the existing system (which would otherwise be used to connect non-adjacent pieces of LCN). For running systems into which bridges are "hot inserted", this is the only feasible approach.</p> |
| STAR   | An arrangement of LCN extensions, each made by situating LCN Extenders (LCNEs) on a single backbone LCN segment, and then connecting LCN segments at the remote end of each LCNE. Note that a single central LCN backbone is required. | <p>(As opposed to Linear) An arrangement of TPN Bridge clusters into TPN Bridge segments, which allows physically non-adjacent sections of LCN cable to be put into a single TPN Bridge segment. This is accomplished through adding additional LCN cable that is used to join the previously non-adjacent sections of LCN into a single TPN Bridge segment.</p> <p>Star arrangements are more flexible and often result in improved traffic reduction over linear arrangements, but they require the addition of LCN cable. For this reason, a site may sometimes prefer to restrict their consideration of segmenting options to linear arrangements.</p>                                                                                                    |

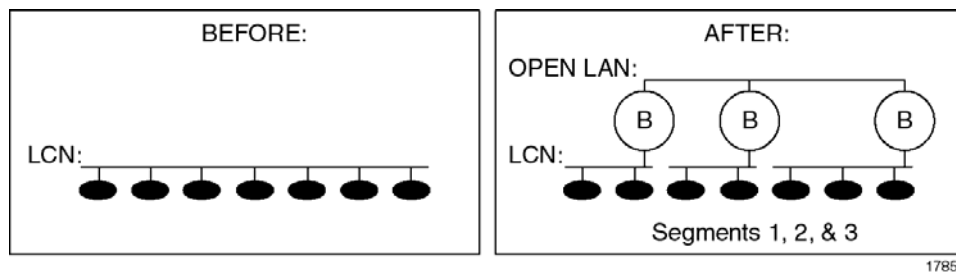
## 1.2 Bridge Technology

Honeywell process control networks (or TPS Process Networks-TPNs) are faced with an ever-increasing demand on usage and bandwidth. These demands come from the introduction of newer nodes on the existing legacy control networks. The addition of nodes to the control networks increases the network density and thereby decreases the throughput. This can create bottlenecks (traffic congestion) and will affect the overall performance of the networks.

A solution to these performance issues is now available. The TPN Bridge will increase the performance of the control networks. The TPN Bridge uses switching technology which allows for increased throughput by filtering of packets, knowing which packets to keep in the local segment and which packets to pass on to other segments, and providing full duplex capability.

The introduction of TPN Bridges provides a number of enhancements over the control network. The foremost enhancement is the ability to divide the control networks into smaller and faster segments. The TPN Bridge will examine each packet, determine where that packet is destined and then forward that packet to only those ports to which the packet needs to go. The TPN Bridge filtering capability knows which messages must be delivered and to which segments. This filtering activity is designed to keep certain problems isolated to a single segment and thus prevents them from disrupting other network activity. The TPN Bridge will be able to do all these tasks without adding noticeable delay.

The driving force behind the TPN Bridge is the goal of significantly reducing the traffic loading on the LCN. This is achieved by breaking up the existing monolithic LCN network into a number of segments, or sub-networks, and then connecting all of the segments back together with TPN Bridges. The following diagram illustrates this idea.



17857

## 1. Introduction

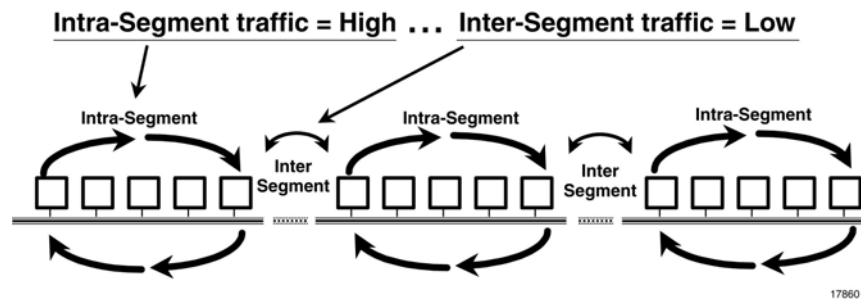
### 1.2. Bridge Technology

---

#### Network segmentation

A basic concept toward accommodating LCN traffic load is "segmentation," which uses bridges (and bridge network) to transparently break up a large single LCN into segments. This concept is shown in the above illustration.

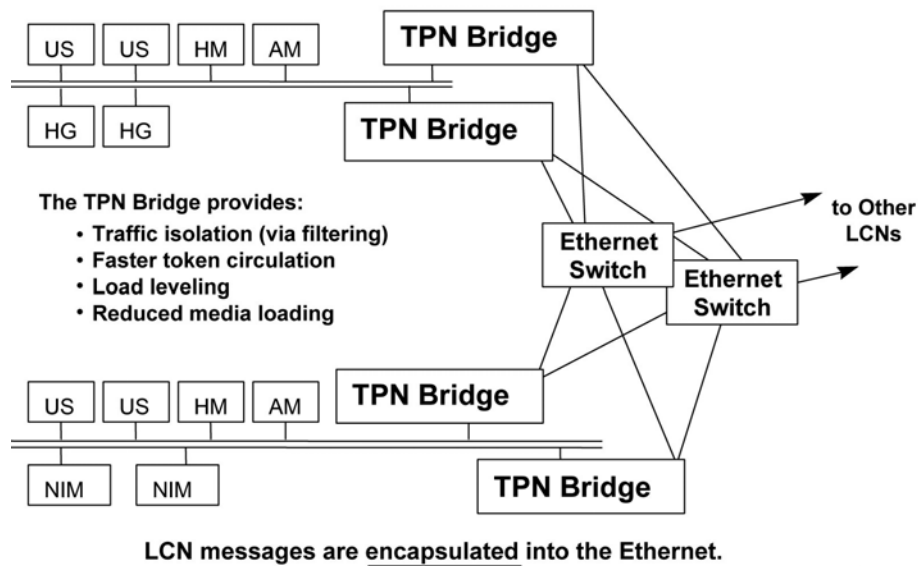
Traffic that currently shares a single common LCN media is not evenly divided among all possible source and destination pairs. As shown in the illustration below, "segments" of nodes tends to have high traffic among them. There is much lower traffic between segments.



Because the bridge acts as a message traffic filter, which separates intra-segment from inter-segment traffic, it improves LCN cluster load capacity and lowers the overall amount of traffic on a given LCN segment. Further, because the bridging reduces the total number of nodes on a LCN segment, it speeds up the LCN token circulation, hence improving LCN response time.

## 1.3 Network Architecture of TPN Bridge

The TPN Bridge facilitates a smooth migration for existing large-system customers to TPS and into the future of Honeywell's networking technology. The TPN Bridge is not a LAN, but is a dedicated computer that reduces network traffic. The TPN Bridge is a device that transparently breaks up large LCN systems into segments compatible with this goal. As previously described, the bridges provide improved LCN cluster load capacity because the bridge acts as a traffic filter, which eliminates unnecessary inter-segment traffic. This lowers the overall amount of traffic on a given LCN segment while speeding up the token circulation on each segment.



17861

Figure 1-1 TPN Bridge Architecture

## 1.4 TPN Performance Issues

The legacy network (formerly referred to as LCN (Local Control Network)) was designed to meet and handle normal systems traffic on the control network. The LCN has a very simple but elegant design; it is deterministic (token ring - messages sent unidirectional), media access is always available (redundancy), and it promotes good data integrity. However, on many larger systems this traffic can reach nearly 100% of the theoretical loading for several seconds. This loading can occur without the installation of additional nodes of the current TPS products, such as GUS, APP, or PHD. It is evident there is the possibility of performance degradation when these nodes are added to the network.

### Qualifying a site

A method to promote a graceful migration into Honeywell's control network future is to optimize the network as new nodes are added. The TPN Bridge can assist this optimization by providing the following:

- Reduced Network Traffic by Traffic Filtering
- Faster Token Circulation
- Improved Signaling via Reduced Media Loading
- Reduced Traffic Variability / Improved System Stability
- System Enlargement
- Topological Flexibility
- Graceful Migration

Initially, the TPN Bridge will be implemented on existing large and heavily loaded systems that are qualified. The qualification process determines the suitability of the TPN Bridge product on a given customer LCN system. The purpose of the determination is to provide an accurate estimate of the traffic load reduction that could be achieved with the TPN Bridge. It also provides a series of recommended TPN Bridge configurations that result in network traffic reduction.

The technique employed in qualifying the site can be a simple and technically satisfying one. There is a three phase process used to determine the suitability of the TPN Bridge product on a given customer system. This three-phase process is summarized below:

| Process Phase                           | Activity                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phase I</b><br>(At customer site)    | <ul style="list-style-type: none"> <li>• Connect a Honeywell Traffic Recorder Tool to the customer LCN system (done by Honeywell representative)</li> <li>• Record and archive LCN traffic on customer's existing unbridged, monolithic LCN system using Traffic Recorder Tool</li> <li>• Gather and document site topology of existing LCN configuration by customer</li> <li>• Verify accuracy of site topology through discussions with site personnel</li> </ul> |
| <b>Phase II</b><br>(Onsite or offsite)  | <ul style="list-style-type: none"> <li>• Perform 3-step decision process using topology map</li> <li>• Run the Traffic Analysis Program for all possible configurations accounting for any outside constraints</li> </ul>                                                                                                                                                                                                                                            |
| <b>Phase III</b><br>(Onsite or offsite) | <ul style="list-style-type: none"> <li>• Determine suitability of TPN (LCN) under study for TPN Bridge</li> <li>• Analyze traffic patterns and traffic reduction using segmentation</li> <li>• Determine loading of LCN</li> <li>• Produce recommendations and segmentation options</li> </ul>                                                                                                                                                                       |

### Recording and analyzing TPN data

Before the analysis starts, the customer produces a topology map that identifies to the Honeywell service engineer the LCN nodes, node definition and desired cluster configurations. The Honeywell service engineer then inputs this data into the Traffic Analysis Tool prior to going to the customer site. The customer identifies the desired nodes due to the unique information that only the customer possesses (e.g., where cable insertions are acceptable to the customer for either logistical, networking, or other practical reasons).

The analysis process begins when a uniquely loaded computer (the "recorder") is brought to the customer site for capture of the existing LCN traffic. This recording of LCN traffic data does not impact LCN operations. LCN T-connectors must be added to the running LCN to allow access to its message traffic. Optionally, an existing engineering LCN node may be taken offline and its T-connectors used to install the Recorder.

## 1. Introduction

### 1.4. TPN Performance Issues

---



#### ATTENTION

Sites should avoid having unused LCN taps. For installing the recorder, Honeywell recommends adding a 3-meter cable at the end of the LCN coax cable, which contains a T-connector and terminator for cables A and B.

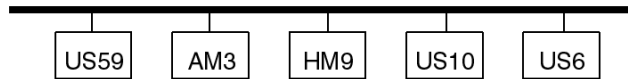
---

### Review current network configuration (customer)

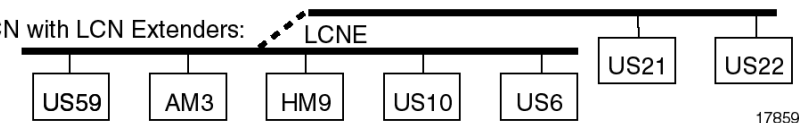
One of the most important factors of network optimization is gathering customer network configuration information. This part of the overall process is by far the most complex and should be given full attention. Any errors that are introduced in this activity may result in a meaningless analysis by the program!

The first thing that must be obtained from the customer is a topology map of the LCN. This must take the form of a diagram (or a list from which a diagram can be made), which lists every node on the LCN, and the order in which nodes are physically connected. In some cases, with the use of LCN Extenders, the diagram may be shaped like a "star" rather than a "linear" arrangement. Shown below are two examples of customer layout diagrams. Note that these diagrams are smaller than what is typical of large LCN systems.

Simple Linear LCN:



"Star" LCN with LCN Extenders:



Once the customer site diagram is available, a three-step decision process begins. The following enumerates those decisions:

Decision 1: Shall we use the existing cabling arrangement and location of the LCN nodes, or rearrange the nodes, which belong in common clusters into nearer physical (directly-connectable) proximity to one another?

Decision 2: How should the system be clustered?

Decision 3: Should the arrangements of clusters into segments by the Traffic Analysis Program be restricted to "linear" arrangements, or shall it be allowed to consider "star" arrangements as well? Review Table 1-1 Network Terms Defined for TPN Bridge for "star" versus "linear" relationship within TPN Bridge.

The final interpretation of these decisions and output of the Traffic Analysis Program is reserved for the network specialists.

### **Overview of Network Analysis Program**

The purpose of the Traffic Analyzer Program is to assess the possible benefits of TPN Bridge technology for LCN traffic reduction. These benefits are based upon the following:

- The network traffic recorded during the data-recording phase.
- The customer site configuration information.

Once the program is started and it completes its initialization, a blank configuration screen displays. At this point, the network specialist may either create a new configuration from scratch or read back a previously saved configuration. Four selectable tabs define the analysis operations. These operations consist of defining the clusters, optimizing those clusters, generating the results, and selecting various options. The analysis may take a long time, depending on the number of clusters defined and the amount of traffic recorded. Notices and progress bars show the analysis proceeding through its various stages.

### **Presenting analysis report to customer**

The customer is presented with two types of reports from the "Detailed Segmenting Reports." The first is the "Results Summary." It indicates the best segmentation result for each of the possible numbers of segments. It gives both the best linear, and the best star configuration, for each segment.

The second is the "Results Details." It contains a detailed breakdown of every combination of clusters that was tried. It provides an indication of which combination gave the best result for each level of segmentation.

Additional reports are generated to provide crosstalk tables for All Messages, Broadcast, Logical, and Physical reports. These tables can then be represented in Excel for a variety of customer-specified graphs and charts.

## **1. Introduction**

### **1.4. TPN Performance Issues**

---

#### **Discuss options and recommendations**

The network specialist will discuss with the customer all of the data files containing the results of the recording and analysis. These recommendations will look at the best linear and star segmentation options. It will then provide a breakdown of every combination of clusters that can be tried applying those constraints within the customer's network environment. Also discussed will be node to node common communication with additional insight into the customer's traffic patterns.

The core of the analysis is totally dependent on accurate information gathered from the customer's network environment. The data files generated from the recording and subsequent analysis is only a small part of the total options and recommendations discussed with the customer. The data collection does require the personnel to be on site but final analysis and recommendations can be discussed and presented offsite.

## 1.5 Implementation of the TPN Bridge

The hardware side of the TPN Bridge (TPNB) is comprised of a pair of (redundant) PCs with a TPNB board installed in each. Each PC is a minimal NT node (which constitutes the least expensive hardware and lowers the cost of developing special communication stacks). An abbreviated software personality is hosted in the TPNB, which acts as a primary traffic filter. Segment messages, which can be addressed physically or logically, are encapsulated into Ethernet packets in the NT node.



### ATTENTION

Although the LCNP4 and the TPNB share a common board layout and parts list, the TPNB board cannot be used as an LCNP4 because of internal firmware changes.

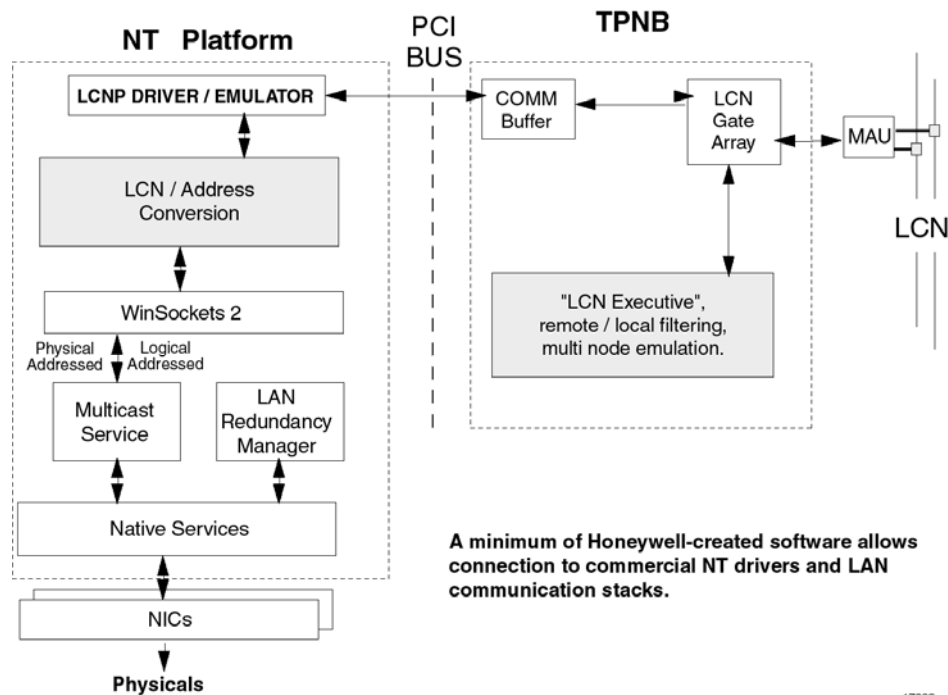


Figure 1-2 TPN Bridge Hardware Interfaces

## **1. Introduction**

### **1.5. Implementation of the TPN Bridge**

---

## 2. Planning the TPN Bridge Switchover

### 2.1 Preliminary

#### Planning goals

There are two goals in planning how an LCN installation should be segmented. The primary goal is to minimize the inter-segment traffic, which improves the LCN performance in terms of throughput and response time. The secondary goal is to limit the segmentation cost (bridging hardware) to the minimum possible. In fact, the two goals may result in opposite effects, i.e., a better segmentation in terms of minimizing the inter-cluster traffic may require more bridges or vice versa. To achieve a balance between the two goals, LCN traffic must be carefully monitored and analyzed. The tradeoff between performance optimization and bridging cost must be presented to the customer. This process is part of qualifying the site and it was discussed in an earlier section.

#### Current TPN Topology

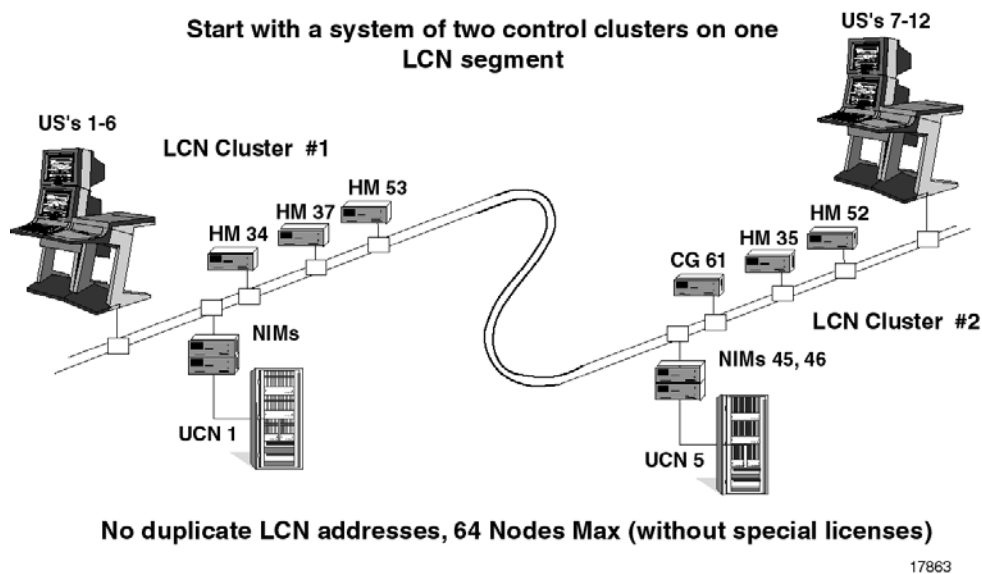


Figure 2-1 LCN before Segmentation

## 2. Planning the TPN Bridge Switchover

### 2.1. Preliminary

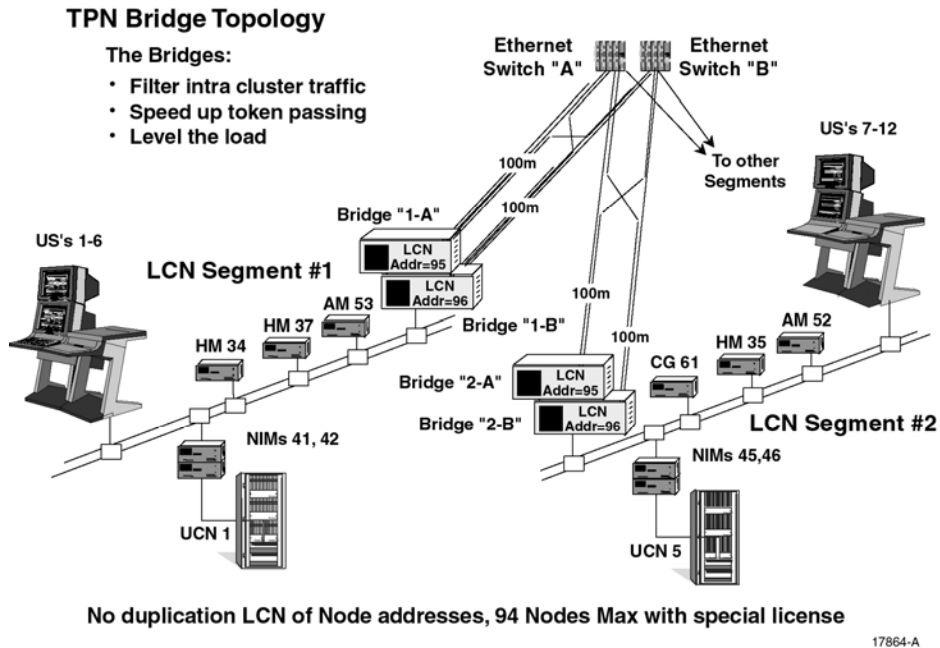


Figure 2-2 LCN after Segmentation

### Overview of TPN Bridge switchover

The blueprint for segmenting the LCN comes from the report generated by the network analysis. This report will be used to determine where the physical breaks will occur in the segment(s). The scope of the network analysis and the content of the network report will be discussed in the paragraphs that follow this overview.

When dividing an LCN into multiple segments, bridges are inserted one redundant pair at a time rather than all at once. After each redundant pair is inserted, proper operation of the bridge pair and the LCN network is confirmed. Only after the proper operation of each bridge pair and the LCN network is confirmed, is the next pair inserted. In addition, between insertions of bridge pairs, enough time should elapse to enable any transient LCN errors to clear (as seen on the System Status display). The stepwise insertion means that an LCN that is intended to be divided into 4 segments, for example, goes up through the steps of 2 segments, 3 segments, and finally 4 segments. It does NOT go directly from the unsegmented condition to 4 segments.

The configuration and hot insertion of redundant TPN Bridge nodes into an operational system consists of three phases. The following describes those phases:

- The TPN system must be at R610.4 or greater, and required NCF changes implemented. The TPN Bridge nodes must be software configured to set the system specific functions to the proper values. This will be done using the **TPN Bridge Configuration/Status Snap-In** tool.
- The TPN Bridge node status must be monitored continuously to prepare for insertion in the LCN segment.
- Performing the actual break in the LCN segment and re-establishing the segment-to-segment communications.

#### Scope of network analysis

The network qualification for TPN Bridge implementation begins with measuring and documenting the way the customer is using their system. This baseline report indicates how much of the system's available capacity is being used. This knowledge provides a reliable basis for planning control system upgrades to TPS and future network enhancements. Additionally, it can help to prevent future performance problems.

The following outlines the steps involved and the deliverables. The report, which includes the recommended site topology, will be used to launch the procedures containing those tasks to install, configure and implement the TPN Bridge(s) on the network.

## 2. Planning the TPN Bridge Switchover

### 2.1. Preliminary

---

#### Obtaining the report

| Qualification Phases           | Description                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------------------|
| Data Collection                | Traffic data is collected from the system and stored on a cartridge using the TPN Bridge Recorder tool. |
| Analysis and Report Generation | The TPN Bridge Recorder data is analyzed, and a report is generated.                                    |
| Presentation of Findings       | Results and recommendations are presented.                                                              |

A written report will consist of text, drawings, graphs, and tables, organized similar to the following:

|                                               |                                                                                                                                                                      |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Executive Summary</b>                      | <ul style="list-style-type: none"><li>• Overview</li></ul>                                                                                                           |
|                                               | <ul style="list-style-type: none"><li>• Objectives and Goals</li></ul>                                                                                               |
|                                               | <ul style="list-style-type: none"><li>• IT Plan</li></ul>                                                                                                            |
|                                               | <ul style="list-style-type: none"><li>• Constraints/Priorities</li></ul>                                                                                             |
|                                               | <ul style="list-style-type: none"><li>• Summary of Path Forward Recommendations</li></ul>                                                                            |
| <b>Summary of Performance and Utilization</b> | <ul style="list-style-type: none"><li>• Overview</li></ul>                                                                                                           |
|                                               | <ul style="list-style-type: none"><li>• Explanation of Network Metrics</li></ul>                                                                                     |
|                                               | <ul style="list-style-type: none"><li>• Traffic Analysis Reports</li></ul>                                                                                           |
|                                               | <ul style="list-style-type: none"><li>• Cross-talk Tables</li></ul>                                                                                                  |
|                                               | <ul style="list-style-type: none"><li>• Recommendations (using new Topology Map)<br/>Physical changes to cable-ordering of nodes<br/>TPN Bridge placement.</li></ul> |

## The report

Traffic: C:\Program Files\Honeywell\TPS\TPN Analysis\sample.trf (and .lat)  
Project: C:\Program Files\Honeywell\TPS\TPN Analysis\sample.prj

Nodes in each cluster:

Cluster 0 is nodes 1 2 3 4 5 6  
Cluster 1 is nodes 52 61 45 46 10 11 12 7 8 9  
Cluster 2 is nodes 38 49 50 13 14 15 16 19 20 59  
Cluster 3 is nodes 21 22  
Cluster 4 is nodes 54 31 17 18 23 24 29 30 25 26 27 28  
Cluster 5 is nodes 62 33 35 60  
Cluster 6 is nodes 34 37  
Cluster 7 is nodes 51 53 32 36 41 42 43 44

The unsegmented load is 45.7%

The best LINEAR result with 2 segments is 36.6%: Seg-1: 0 1 2 3 Seg-2: 4 5 6 7

The best STAR result with 2 segments is 29.0%: Seg-1: 0 2 3 7 Seg-2: 1 4 5 6

The best LINEAR result with 3 segments is 26.1%: Seg-1: 0 1 Seg-2: 6 7 Seg-3: 2 3 4 5

The best STAR result with 3 segments is 25.4%: Seg-1: 3 7 Seg-2: 0 1 6 Seg-3: 2 4 5

The best LINEAR result with 4 segments is 23.8%: Seg-1: 2 Seg-2: 7 Seg-3: 0 1 Seg-4: 3 4 5 6

The best STAR result with 4 segments is 23.6%: Seg-1: 0 Seg-2: 7 Seg-3: 1 5 Seg-4: 2 3 4 6

The best LINEAR result with 5 segments is 23.6%: Seg-1: 0 Seg-2: 1 Seg-3: 2 Seg-4: 7 Seg-5: 3 4 5 6

The best STAR result with 5 segments is 23.6%: Seg-1: 0 Seg-2: 1 Seg-3: 2 Seg-4: 7 Seg-5: 3 4 5 6

The best LINEAR result with 6 segments is 23.6%: Seg-1: 0 Seg-2: 1 Seg-3: 2 Seg-4: 3 Seg-5: 7 Seg-6: 4 5 6

The best STAR result with 6 segments is 23.6%: Seg-1: 0 Seg-2: 1 Seg-3: 2 Seg-4: 3 Seg-5: 7 Seg-6: 4 5 6

The best LINEAR result with 7 segments is 23.6%: Seg-1: 0 Seg-2: 1 Seg-3: 2 Seg-4: 3 Seg-5: 4 Seg-6: 7 Seg-7: 5 6

The best STAR result with 7 segments is 23.6%: Seg-1: 0 Seg-2: 1 Seg-3: 2 Seg-4: 3 Seg-5: 4 Seg-6: 7 Seg-7: 5 6

The best LINEAR result with 8 segments is 23.6%: Seg-1: 0 Seg-2: 1 Seg-3: 2 Seg-4: 3 Seg-5: 4 Seg-6: 5 Seg-7: 6 Seg-8: 7

The best STAR result with 8 segments is 23.6%: Seg-1: 0 Seg-2: 1 Seg-3: 2 Seg-4: 3 Seg-5: 4 Seg-6: 5 Seg-7: 6 Seg-8: 7

## Report description



### ATTENTION

TPN Bridge LCN segments must not cross Process Console boundaries. All nodes defined within a Console must be completely contained within a single LCN segment.

Selecting the Detailed Segmenting Reports button on the Results tab generates the above reports in two new windows. The Notepad application is used to send the reports to the printer for analysis and interpretation. The network analysis team performs this activity.

As part of the planning phase, a copy of this report will be received from the network analysis team. A decision should already have been made as to the site qualification and the specific segmentation map to be used.

## 2. Planning the TPN Bridge Switchover

### 2.1. Preliminary

---

The clusters are referenced by number rather than by name. They will start with 0. You will need to know the cluster configuration, cluster name, and the cluster number assignment. The following is an example of cluster number assignments.

**Table 2-1      Segment and Cluster Equivalency Table**

| # | Cluster Name  |
|---|---------------|
| 0 | = BLEACH 1    |
| 1 | = BLEACH 2    |
| 2 | = PULPER 1    |
| 3 | = PULPER 2    |
| 4 | = ELECTRIC    |
| 5 | = ENGINEERING |
| 6 | = BLARKER     |
| 7 | = OTHER       |

In addition to the numbers representing the cluster utilization to be expected, the specific segmentation map is shown. The following is an example:

S1: 0 1 2   S2: 3 4

This is a map showing a solution with segment S1 containing cluster 0, 1 and 2 while S2 is another segment, which contains cluster 3 and 4. A cluster is a group of nodes that have been configured as being together and inseparable.

The printout of these two text files shown above contains the results of the traffic analysis of all the possible combinations of segments that were considered.

The Results\_Summary.TXT file contains the best results obtained for each of the possible segmentations, from a single segment (the existing, unbridged LCN case) to the case with as many segments as the user has configured for analysis. Both the star and linear analysis results are presented. The decimal number represents the average bandwidth utilization value for segments one through four.

The Results\_Detail.TXT file shows the result of every combination evaluated. It also provides more information about the specifics of each case, such as how much inter-segment and intra-segment data was to result from the particular combination. At the end of each segmentation solution there is an indication of which of the segmentation options was the best solution for that particular number of segments.

As you examine the report you will note that as the number of segments increase the bandwidth goes down. At some point, typically between 3 and 5 segments, the point of no increased benefit is reached. Up to that point you can see the relationship between adding bridges and the cost they incur and getting a reduction in bandwidth consumed.

### Breaking cables A and B

Once the area for TPN Bridge insertion has been located, there are some physical activities that will occur to complete the TPN Bridge install. Part of the activity is the cabling and terminating of the TPN cables. A Hot Insertion procedure must be followed exactly to make the switchover successful. A Hot Insertion Switchbox tool will facilitate the breaking of these cables. (See Section 5, "TPN Bridge Hot Insertion" for details.)

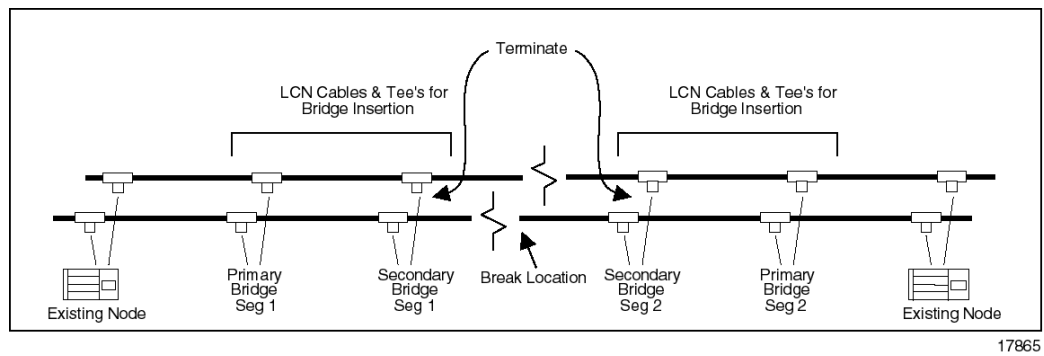


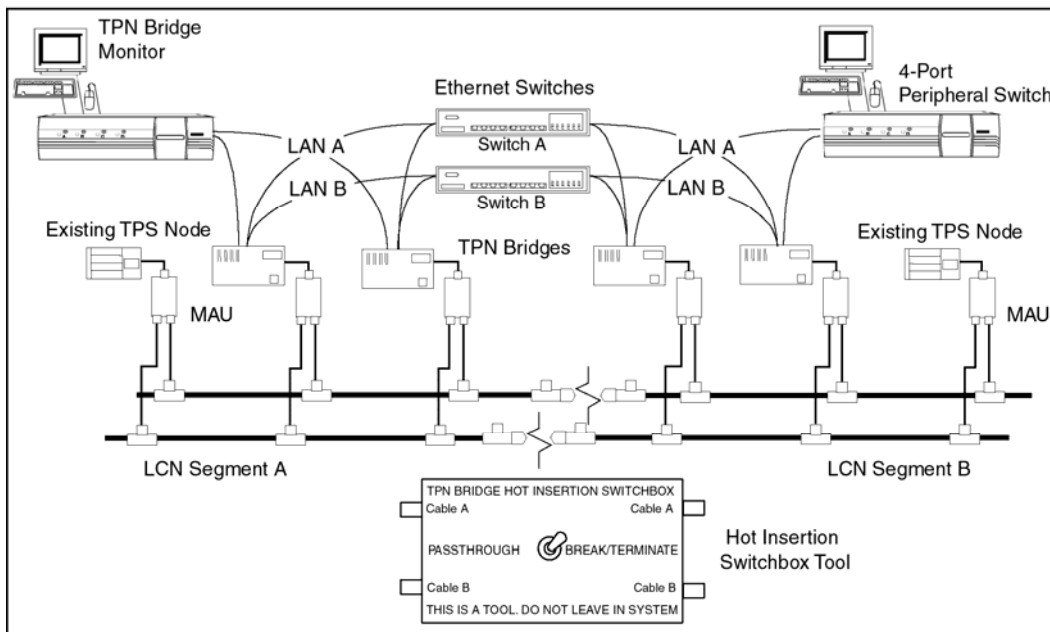
Figure 2-3 Identifying the Cable Breaks for Insertion

## 2. Planning the TPN Bridge Switchover

### 2.1. Preliminary

#### Inserting bridges, switches, and terminators

The following illustration shows the typical installation of the bridges, switches and terminators.



17866

Figure 2-4 Overview of Installation of TPN Bridges



#### ATTENTION

All segments must contain a minimum of 3 LCN nodes in addition to the Primary and Secondary TPN Bridge nodes (5 nodes total). This is an LCN network requirement for reliable token passing.

## 3. Configuring the TPN Bridge System Time

### 3.1 TPN Bridge System Time Synchronization

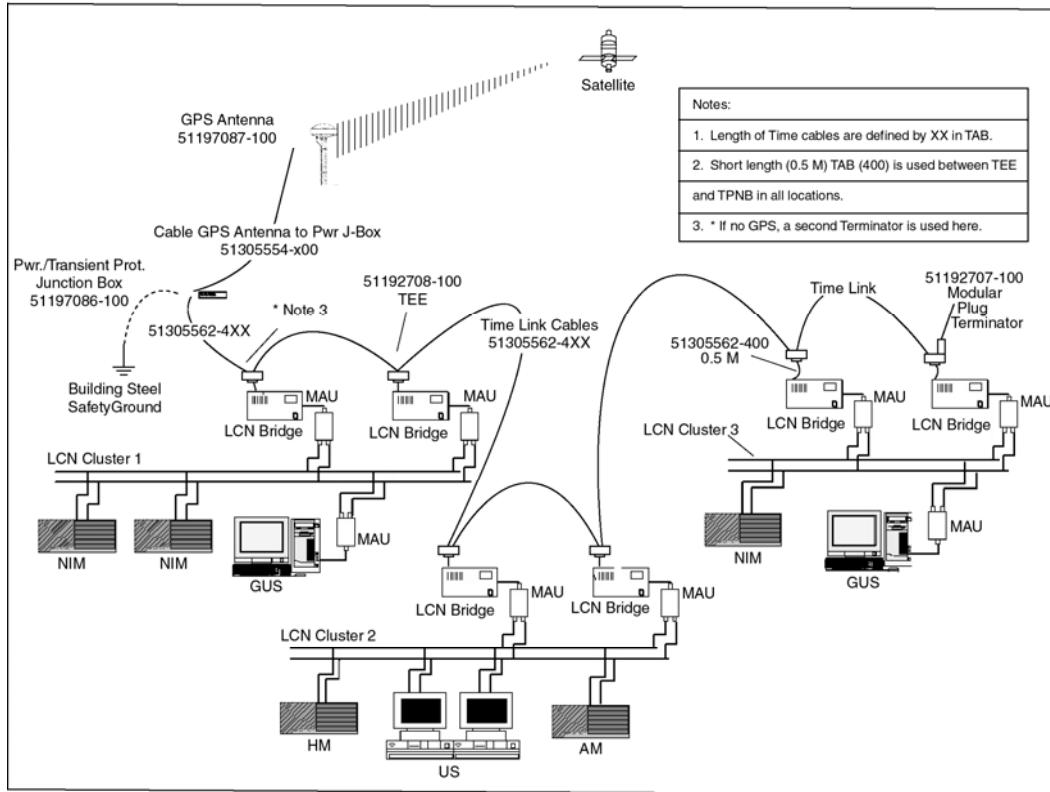
#### Overview

The TPN Bridge provides a method of system time synchronization across LCN segments while maintaining compatibility with current clock subsystems. The existing clock hardware has specific expectations in regards to the repeatability of the time synchronization reference data within specific LCN segments. The TPN Bridge for each LCN segment manages these requirements.

LCN time synchronization messages will be handled in a special manner providing rapid and more accurate time synchronization than can be achieved by treating the time messages as normal bridge traffic that simply passes through the bridges. The variable nature of the store and forward latency in the bridges will, however, not allow the current accuracy of LCN, single segment, Sequence of Events (SOE) timing.

### 3. Configuring the TPN Bridge System Time

#### 3.1. TPN Bridge System Time Synchronization



17867

#### Three steps to configuring the TPN Bridge system clock

The following steps are associated with properly configuring the system for the desired timing attributes. These steps are:

1. Determine the system timing requirements (allowed skew segment to segment, long term system time drift, correlation to "off-site" events in real time, etc.).
2. Understand how the Sync reference options relate to the system timing requirements determined in step 1.
3. Prepare the bridges for physical installation (including the optional Time Link cabling and GPS if required). Make the Time Link Master / Segment Master configuration entries based upon the results of step 2. Make the clock source node address changes in the NCF (if the GPS option is utilized or the node addresses of the clock sources are changing as a result of this).



#### REFERENCE - EXTERNAL

For further information on the Clock System for the TPN, please consult the following publications:

- [Local Control Network Planning](#)
  - [LCN Guidelines Implementation, Troubleshooting, and Service](#)
  - [TPN Bridge Planning, Installation and Service](#)
  - [Time Link & GPS Antenna User's Guide](#)
- 

#### Step 1—Determine System Timing Requirements

The amount of time skew allowed between LCN segments is a function of the application. For TPN system applications that can tolerate up to 16ms of segment-to-segment skew, no Time Link cabling is required.

If low (+/- 300  $\mu$ s) skew between LCN segments is required, but long term system time accuracy is not required, then the addition of the shielded twisted pair (STP) Time Link cable fills this need.

If the +/- 16 ms segment to segment skew is acceptable, but long term accuracy of the system time is required, this requirement may be met by the connection of the GPS antenna to 2 Bridge partners (TPNBs) or to 2 GUS nodes with the LCNP4 assemblies that are GPS capable. This is particularly useful in systems without TPS ac reference capability (all PC based systems.)

If low (+/- 300  $\mu$ s) skew between LCN segments is required and long term accuracy of the system time is required, then the GPS antenna and Time Link STP connection is extended to additional pairs of TPN Bridge nodes.

The system software (R NOS) that is running on the LCN nodes has specific expectations in regards to calendar time ownership, clock mode and synchronization state management, and fail-over recovery. The TPN Bridge operates in a manner that allows the software on existing LCN nodes to operate without modification. When it is desirable to replace the existing LCN clock source node operation with a GPS reference, an NCF configuration change is required to set the clock source node addresses to the Bridge Node address assignments.

### 3. Configuring the TPN Bridge System Time

#### 3.1. TPN Bridge System Time Synchronization

---

#### ***Step 2—Understand How the System Timing Requirements Relate to Sync Reference (Clock Source) Options***

The following synchronization reference (source) options are supported:

1. Synchronization to Local Crystal. (This is generally only used as a backup mode when system clock source synchronization is not active).
2. Synchronization to existing (LCN) clock sources allows the segmentation of existing LCN systems to be performed without modification to the NCF (Network Configuration File) and the associated re-load of nodes to pick up the new NCF configuration.
3. Synchronization to a GPS timing reference is an option that allows the TPN system time to track UTC. This may be accomplished in two ways:
  - GPS antenna connection to a redundant pair of TPN Bridge nodes – OR
  - GPS antenna connection to two GUS or APP nodes with LCNP4/GPSs.

The nodes (GUS, APP or TPN Bridge) that are to act as the interface between the GPS antenna and the system MUST be configured in the NCF as clock sources. If the node addresses of the selected GUS or TPN Bridge nodes are different from the current NCF assignments, then the NCF must be updated to show this new configuration. Only 1 set of clock sources is allowed per system and if GPS is utilized, the nodes that act as the interface between the GPS antenna and the system must be the clock sources.

As you can see, it is not a low segment-to-segment skew requirement that drives the re-configuration of the system clock sources in the NCF, but a system requirement for system time that has a deterministic relationship to UTC. The following table is a summary of how the system timing requirements drive the sync source (clock source) selection.

**Table 3-1 System Timing Requirements and Sync Source**

| System Timing Requirement |                 | Existing LCN Based Clk Sources       | GPS Connection to TPN Bridges                   | GPS Connection to GUS/APP w/GPS Capable LCNP4          | Seg to Seg Time Link Cabling |
|---------------------------|-----------------|--------------------------------------|-------------------------------------------------|--------------------------------------------------------|------------------------------|
| UTC Reference             | Seg to Seg Skew |                                      |                                                 |                                                        |                              |
| No                        | +/- 16 ms       | Okay                                 | No                                              | No                                                     | No                           |
| No                        | +/- 300 $\mu$ s | Okay                                 | No                                              | No                                                     | Yes                          |
| Yes                       | +/- 16 ms       | No, CLK Sources must be GPS capable. | <b>Either</b> TPN Bridge (2 Redundant partners) | <b>Or</b> 2 GUS /APP Nodes with GPS capable LCNP4/GPSs | No                           |
| Yes                       | +/- 300 $\mu$ s | No, CLK Sources must be GPS capable  | <b>Either</b> TPN Bridge (2 Redundant partners) | <b>Or</b> 2 GUS /APP Nodes with GPS capable LCNP4/GPSs | Yes                          |

**Step 3—TPN Master / Segment Master Configuration Selections Based upon Results of Step 2**

**Understanding How Bridge Nodes know “Who is the Boss” when it comes to Time Messaging**

All bridge nodes automatically determine the synchronization reference to be used by detecting the assertion of specific signals within the GPS and Time Link cables. On a node-by-node basis, the 3 “system level synchronization references” translate to knowing if that particular node is to get its sync reference from the GPS / Time Link cable, from the Received Time Frame mailbox, or from neither.

Two (2) things need to be determined for each Bridge:

- What does this Bridge Node sync to?
  - Internal (Crystal)
  - LCN (LCN is the source of writes to Mailbox)
  - NT Driver (NT is the source of writes to Mailbox)
  - GPS / Time Link STP – Time Link Port (TLP)
- Does this Bridge Notify LCN Interface to TX Time Frame or Not? (i.e., Active Segment Master?)

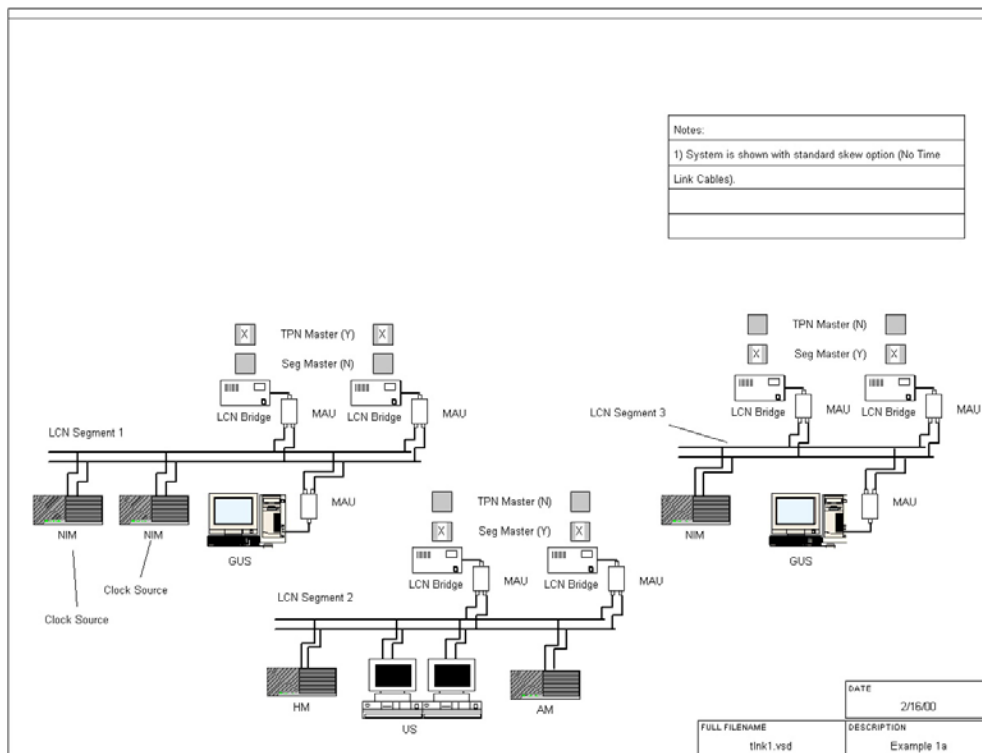
### 3. Configuring the TPN Bridge System Time

#### 3.1. TPN Bridge System Time Synchronization

##### Definition of TPN Master and Segment Master

- *TPN Master Role (TLMr)* is the node currently responsible for distribution of time data to all TPN Segments.
- *Segment Master Role (SMr)* is the node currently responsible for transmission of time data to the connected (one) Segment.
- *Primary/Secondary Bridge* state associated with Ethernet operation and activation of specific time synchronization Roles based upon configuration bits TLMc and TSMc.

Table 3-2 Clock Sources and Configuration Bits associates the clock source selection for a 3-segment system with the proper state of the TPN Master and Segment Master bits for each Bridge Pair. Redundant partners must always be configured in the same manner in regards to the TPN Master and Segment Master bits.



### 3. Configuring the TPN Bridge System Time

#### 3.1. TPN Bridge System Time Synchronization

---

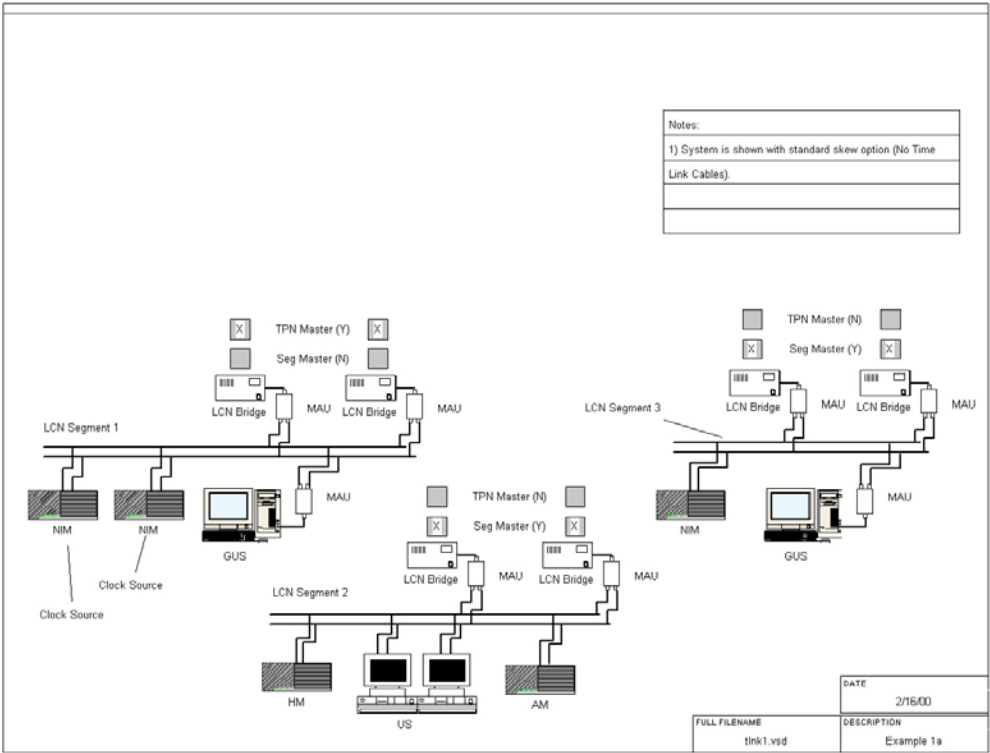
**Table 3-2 Clock Sources and Configuration Bits**

| Location of CLK Source |                                       | Clock TPN Master / Segment Master Configuration Bits |                |            |                |               |                |
|------------------------|---------------------------------------|------------------------------------------------------|----------------|------------|----------------|---------------|----------------|
|                        |                                       | Segment 1                                            |                | Segment 2  |                | Segment 3 ... |                |
| Segment #<br>(1,2,3)   | Type<br>Bridge Pair / Not Bridge Pair | TPN Master                                           | Segment Master | TPN Master | Segment Master | TPN Master    | Segment Master |
| 1                      | Not Bridge                            | Yes                                                  | No             | No         | Yes            | No            | Yes            |
| 1                      | Bridge                                | Yes                                                  | Yes            | No         | Yes            | No            | Yes            |
| 2                      | Not Bridge                            | No                                                   | Yes            | Yes        | No             | No            | Yes            |
| 2                      | Bridge                                | No                                                   | Yes            | Yes        | Yes            | No            | Yes            |
| 3                      | Not Bridge                            | No                                                   | Yes            | No         | Yes            | Yes           | No             |
| 3                      | Bridge                                | No                                                   | Yes            | No         | Yes            | Yes           | Yes            |

**3. Configuring the TPN Bridge System Time**  
**3.1. TPN Bridge System Time Synchronization**

**Example 1:**

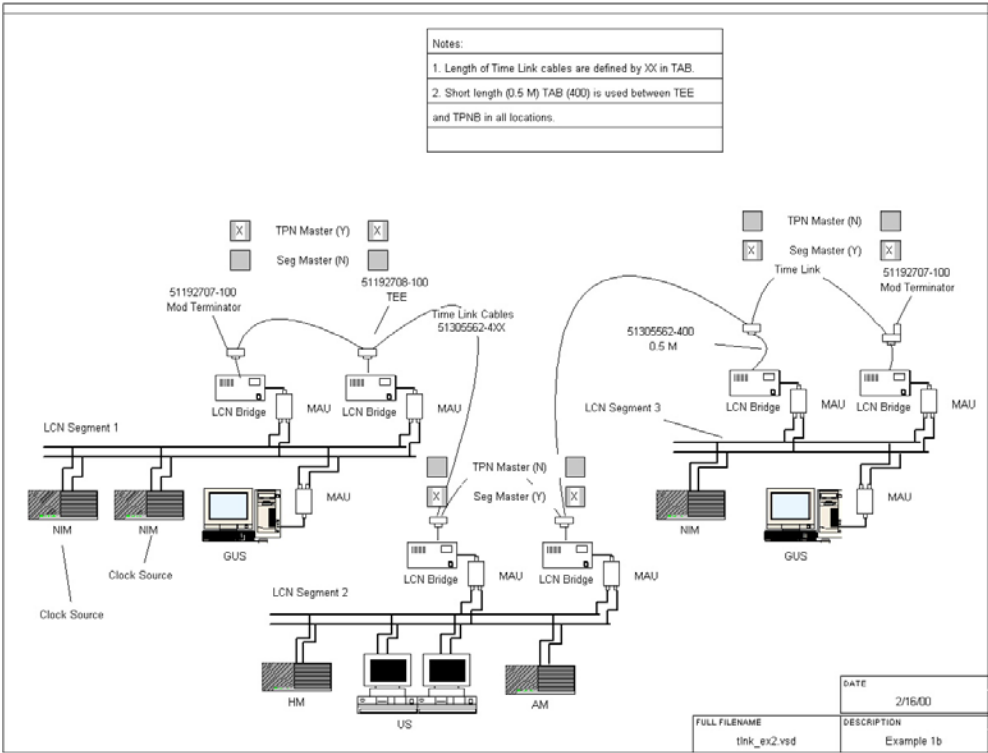
A system has 3 segments. The system clocks are two NIM's on the number 1 segment. The Bridge pair on the number 1 segment is configured with TPN Master (Yes), Segment Master (No). The Bridge pair on the number 2 and 3 segments is configured with TPN Master (No), Segment Master (Yes).



3. Configuring the TPN Bridge System Time  
3.1. TPN Bridge System Time Synchronization

Example 2:

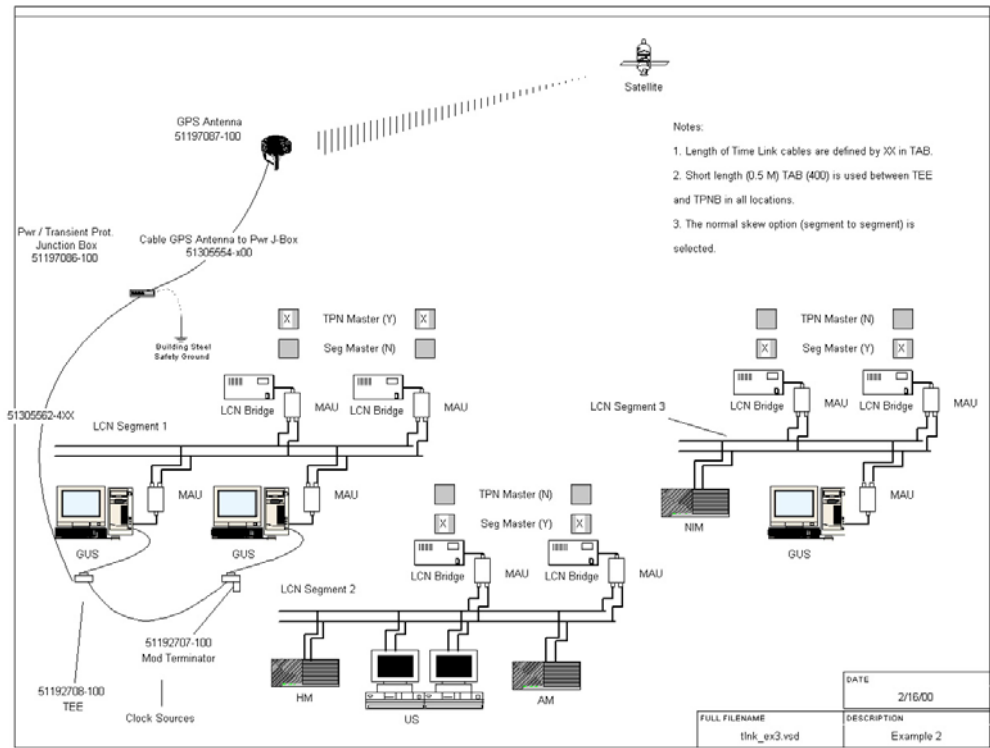
A system has 3 segments and utilizes GPS with connections to (2) GUS nodes on the number 1 segment. The Bridge pair on the number 1 segment is configured with TPN Master (Yes), Segment Master (No). The Bridge pair on the number 2 and 3 segments is configured with TPN Master (No), Segment Master (Yes).



3. Configuring the TPN Bridge System Time  
3.1. TPN Bridge System Time Synchronization

Example 3:

A system has 3 segments and utilizes GPS with connections to the TPN Bridge partners on the number 2 segment. The NCF is configured with addresses 95 and 96 as the clock sources (TPN Bridge Nodes). The Bridge pair on the number 1 segment is configured with TPN Master (No), Segment Master (Yes). The Bridge pair on the number 2 segment is configured with TPN Master (Yes), Segment Master (Yes) and the Bridge pair on the number 3 segment is configured with TPN Master (No), and Segment Master (Yes).



## 3.2 Time Link and GPS Options

### Options

The efficiency and accuracy of the system-wide time services can be increased through the use of GPS technology using a TPNB board that permits the installation of GPS based time sources. Accurate time reference is met by the optional addition of a GPS antenna that is connected to redundant TPN Bridge nodes by means of the shielded twisted pair cable. This GPS system is shown in illustration below.

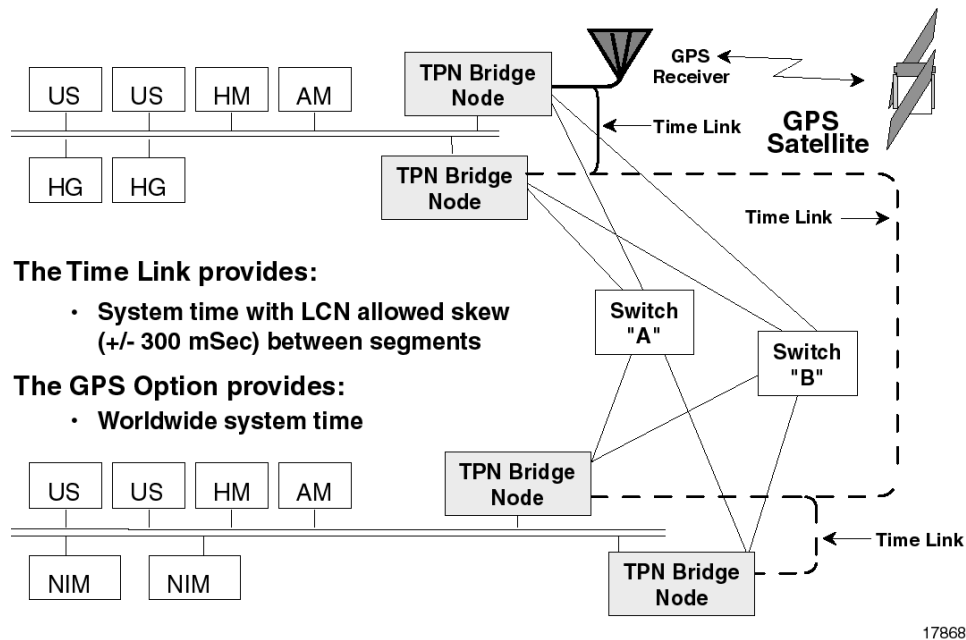


Figure 3-1 Time Links and GPS Option

### For further information

For further information on Time Link and GPS options, including installing Time Link without GPS, see the *Time Link and GPS Antenna User's Guide* (TP22).

### 3. Configuring the TPN Bridge System Time

#### 3.2. Time Link and GPS Options

---

##### Implementation of three segments with Time Link and GPS Options

A maximum of six segments of TPN nodes can be implemented using the TPN Bridge technology. A switch module is dedicated to TPN Cable A and to TPN Cable B. Each TPN Bridge node is cross-cabled to both switch module as shown below. The Time Link and GPS options are shown below.

3. Configuring the TPN Bridge System Time  
3.2. Time Link and GPS Options

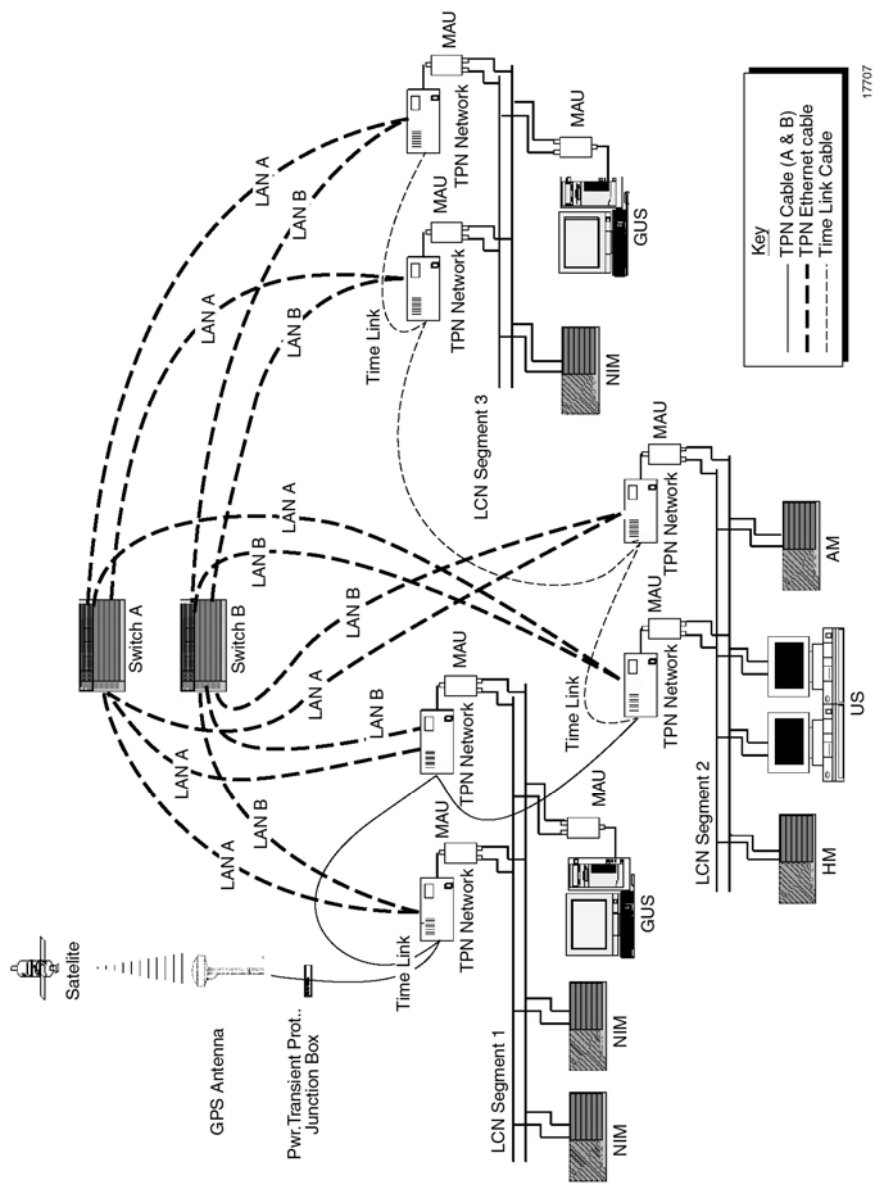


Figure 3-2 Three Segments with Time Link and GPS Options

### **3. Configuring the TPN Bridge System Time**

#### **3.2. Time Link and GPS Options**

---

## 4. Implementing the TPN Bridge System

### 4.1 Overview

#### Summary

Implementing a TPN Bridge system requires preliminary NT configuration of each Bridge node. The TPN Bridge system shipped from the factory is already initialized and the software installed. The TPN Bridge nodes must be configured to implement the keyfile option and added to the NCF. A customized boot disk will be created for each TPN Bridge node for error recovery. Various TPS and NT administrative tasks are performed on all TPN Bridge nodes. The Microsoft Management Console (MMC) is used to create a TPN Bridge View Console. Finally, other miscellaneous procedures are included in this section for implementing Time Sync options, adding a new segment to an existing TPN Bridge system, and replacing and restoring a TPN Bridge due to a failure.

#### Overview of procedures

Implementing the TPN Bridge system is a process that consists of the following sequence of procedures:

- Configuring the Nortel (BayStack) Switch
- TPN Bridge and NT Configuration
- Configuring the Bridge Segments Using MMC
- Setting TPN Bridge Time Sync Options
- Adding a New Segment to an Existing Bridge System
- Replacing the Network Card

This section includes information on a variety of displays that are used to monitor the activity of the Bridge nodes. This information is provided primarily for reference purposes (see "Using the Displays," "LCNP Status," and "Bridge Status" in this section). Also included is a series of tables with descriptions of the display fields that may be important when analyzing and troubleshooting the TPN Bridge.

Section 5, "TPN Bridge Hot Insertion," covers the physical insertion of the Bridge in a running system using a Hot Insertion procedure.

## 4. Implementing the TPN Bridge System

### 4.1. Overview

---

#### Software description

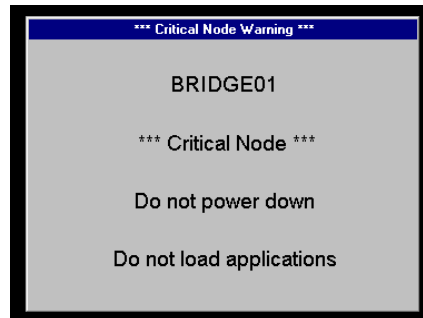
All required software is included on the TPN Bridge CD-ROM. The CD contains the following:

- **TPN Bridge** software – consisting of emulator software and special personality (TPN Bridge does not load standard LCN personality).
- **TPN Bridge Viewer** – provides detailed status of TPN Bridge node.
- **TPN Bridge Configuration/Status Snap-In** – this Microsoft Management Console (MMC) Snap-In configures TPN Bridge system and provides system status.



#### CAUTION

Once the TPN Bridge software has been loaded and configured, you must dedicate these nodes for Bridge use only. These nodes must not have other applications loaded and they must not be powered down. As a precaution, the following screen saver will be invoked soon after there is no human interface activity on the node.



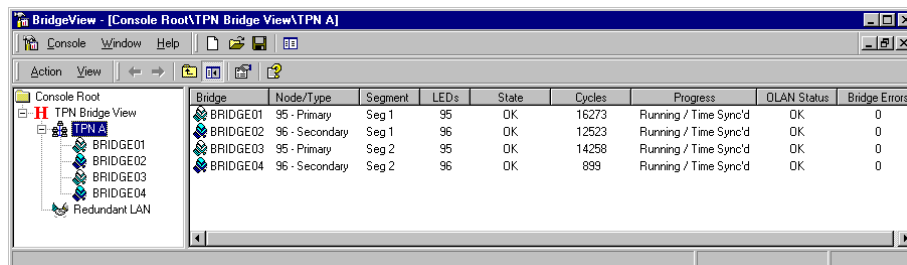
## 4.2 Using the Displays

There are displays installed on the Bridge node that are needed to ‘monitor’ the condition of the Bridge nodes and the passing of LCN messages between Bridge nodes during the Insertion Procedure.

- **Start\Programs\Honeywell TPS\LCNP Status:** This display shows the condition of the TPNB board and the state of the Bridge Personality. This applet also has a button to **Reset LCNP** board.



- **Start\Programs\My Administrative Tools:** This folder contains the MMC Snap-in created in Section 4.8, "Installing and Configuring the TPN System Software". In the following subsections this display will be referred to as the “**TPN Bridge Configuration/Status Snap-In.**”



4. Implementing the TPN Bridge System  
4.2. Using the Displays

- **TPN Bridge Viewer:** This display has been installed on the desktop. When invoked, choose the “View Bridge Status” button to get the “Bridge Status,” “TDC Status,” and “Node Status” displays.

The Bridge Status window displays a comprehensive overview of the bridge system. It includes sections for:

- Head, Tail, C1, C2, Add Data Section 1:** A table showing various data points and their corresponding addresses.
- Channel 1, Channel 2:** Detailed status for each channel, including input/output rates and buffer levels.
- System Status:** A summary of the overall system health, including notice status, cable status, and output status.
- Performance Metrics:** Real-time data on input/output rates, buffer usage, and error counts.
- Configuration Options:** Buttons for clearing counters, resetting LCNP, and updating the bridge version.

The TDC Status window provides a detailed view of the system's operational status. It is organized into several panels:

- System Status:** A list of system-wide indicators and their current values.
- Notice Status:** A section for monitoring and managing system notices.
- Output Status:** A panel for tracking output-related metrics and errors.
- Time Status:** A section for monitoring time-related parameters and synchronization.
- Configuration Options:** A series of checkboxes and buttons for configuring system behavior, such as enabling/disabling certain features and setting update rates.

The Node Status from SECONDARY window displays information specific to the secondary node. It includes:

- Node Information:** Fields for the node number, bridge status, and current transaction count.
- Performance Data:** Real-time data on new transaction counts, time spent on LCN, and TDC release numbers.
- Configuration and Control:** Buttons for refreshing the data and enabling/disabling updates.

## 4.3 LCNP Status

The LCNP Status applet displays the status of the LCNP / TPNB board. The display presents what were historically LED displays on board edges in an LCN module.

Slot information is available in the detailed display. The TPN Bridge utilizes the WSI slot for communication. Status that is more meaningful can be obtained using the Bridge Status Display.

## 4.4 Bridge Status

The *Bridge Status* display provides a view of the internal TPN Bridge state. Access to two major subsystems is provided. The primary display, shown by default when the Bridge Status display is invoked, shows the TPN Bridge personality state. A secondary display, invoked by pressing the WS Counters button, displays the Ethernet interface module state.

### TPN Bridge Personality State (Bridge Status Window)

The upper left corner of the Bridge Status window displays fields maintaining TPN Bridge message queues. During normal operation, the values in this section will change. The following table describes the contents of these fields.

**Table 4-1 Queue Status**

| Field                  | Description                                                                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NT->LCN Data           | Indicates reception of messages from the Ethernet module to the LCN.                                                                                                                                  |
| LCN->NT Data           | Indicates transmission of messages from the LCN to the Ethernet.                                                                                                                                      |
| LCN->NT Buffers        | Return of buffers once message is processed.<br><b>Note:</b> These buffers increment at a faster speed than the NT->LCN Buffers. See "Message buffering on TPN Bridge nodes" for further information. |
| NT->LCN Buffers        | Return of buffers once message is processed.<br><b>Note:</b> These buffers increment at a slower speed than the LCN->NT Buffers. See "Message buffering on TPN Bridge nodes" for further information. |
| Input Buffer Que Alloc | Percentage of output buffers allocated per channel.                                                                                                                                                   |
| Input Buffer Que Len   | Number of output buffers allocated per channel.                                                                                                                                                       |

## 4. Implementing the TPN Bridge System

### 4.4. Bridge Status

---

**Table 4-1      Queue Status**

| Field                          | Description                                                            |
|--------------------------------|------------------------------------------------------------------------|
| Input Msg Que Rate             | The rate at which LCN messages are received on the LCN.                |
| Input Buffers Processed        | Raw count of number of LCN messages received on the LCN and processed. |
| Count of Input Starts          | Number of channel starts executed.                                     |
| Total Buffers                  | Total number of buffers allocated for both input and output queues.    |
| Output Buffer Queue Allocation | Percentage of buffers allocated for output queue                       |

#### Message buffering on TPN Bridge nodes

You will notice that the LCN->NT buffers increment faster than the NT -> LCN buffers. This is normal behavior, and in fact there are several reasons why the buffer pointers move at different speeds.

First, there is a different quantity of traffic flowing "down" through a bridge than there is flowing "up" through a bridge, which can be seen by examining the rates in the "rate display". The traffic flow varies on a segment-by-segment basis with some segments originating a larger proportion of overall traffic than others. These higher traffic segments tend to have a greater upward-to-downward ratio of traffic.

Second, and more importantly, the buffers are sent upwards (LCN > NT) in very small groups, whereas the buffers are sent downwards (NT > LCN) in much larger groups. Upward buffers are serviced by the real-time LCN hardware and LCNP Bridge Personality. The LCNP Bridge Personality watches for incoming messages and queues them in immediately causing movement in the LCN-> NT buffer pointers.

In the downward direction, the NT side sends large batches of messages across the upper LAN, and when it receives them, the messages are all queued down to the LCN side in a single queuing operation. This means that the NT -> LCN buffer pointers in the display are only seen to change once, even though a large number of buffers was moved down.

#### Internal table addresses

The upper right corner of the Bridge Status display is filled with 10 address fields of tables that are allocated and maintained in TDC memory at the given location. The following table describes their content. To view table content, double-click the desired table address field.

**Table 4-2 Internal Table Addresses**

| Field              | Description                                                                                                                                                                                                                                                                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adr Data Section 1 | The location of the BRIDGE_QUEUES data structure. Used for communication between the Personality and the TPN Bridge Emulator.                                                                                                                                                                                                                       |
| Address NT LAT     | The location of the composite Logical Address Table (LAT) composed of the LAT of every node on remote segments. All tables are OR-ed together to create this table used for filtering Logically addressed messages from other segment coming into the Bridge via Ethernet. Valid if Use NT LAT flag is set.                                         |
| Address Seg LAT    | The location of the composite Logical Address Table (LAT) composed of the LAT of every node on the local segment. All tables are OR-ed together to create this table used for filtering Logically addressed messages bound for other segments.                                                                                                      |
| Address Indiv LAT  | A table containing all Logical Address Table (LAT) and Physical Address Table (PAT) of every node on the local segment. Each node's LAT and PAT may be viewed by entering the node number in the Current LAT is --- → field at the top of the display window.                                                                                       |
| Address Node Stat  | The location of the table containing the status of all nodes in this Bridge node's view. Primary bridge nodes contain only the nodes on the local segment. Secondary bridge nodes contain all nodes in the TPN. The contents of this table are automatically displayed during startup in the Node Status window in the Bridge's Node Status column. |
| Addr Cur Tx Cnt    | The location of the array containing the values displayed in the Node Status window Current Transact Count column. This value is the last node administrator transaction that was processed. This number reflects the number of times the LAT table has changed.                                                                                    |
| Addr New Tx Cnt    | The location of the array containing the values displayed in the Node Status window New Transact Count column. This value is the next node administrator transaction that is being processed. This number reflects the number of times the LAT table has changed.                                                                                   |
| Address Recency    | The location of the array containing the values displayed in the Node Status window Time Silent on LCN column. This value is used to derive the number of seconds since the given LCN node was heard from on the LCN.                                                                                                                               |
| Address Timeouts   | The location of the array containing the Resync timeout counter for each node. This value will change while nodes are loading and in the RESYNCH state. Normally this value will cycle 3 – 4 times while a node is completing the load sequence.                                                                                                    |

## 4. Implementing the TPN Bridge System

### 4.4. Bridge Status

---

#### Status registers

The following table describes the contents of the Status register fields and enumerates fields that are useful in troubleshooting. A complete enumeration of the bits in these fields is displayed in the TDC Status window.

**Table 4-3      Status Registers**

| Field         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System Status | Reflects the current state of the TPN Bridge personality. In a normal running mode, only the Physical Dirty (filtering on) and Looping flags will be checked. Other flags may occasionally flicker on and off.                                                                                                                                                                                                                                                                                                    |
| Notice Status | Errors requiring user notification are recorded in this status word. Any bit set in this register will cause the TPN Bridge to put itself into the TEST state on the TPN System Status Display.                                                                                                                                                                                                                                                                                                                   |
| Cable Status  | Reflects the state of personality LCN cable control. The In Use value indicates the LCN cable currently being listened to. The rest of the bits in this register are used for inter-segment cable swapping.                                                                                                                                                                                                                                                                                                       |
| Output Status | The output status register reflects the personalities need for special output messages to be sent. Values in this register are usually off or flicker.                                                                                                                                                                                                                                                                                                                                                            |
| Redund Status | This register contains the state of the Bridge redundancy subsystem. The Config Primary flag indicates that this bridge node is configured with an odd LCN address and is normally a primary bridge node. The Running Primary flag indicates the actual running state of the TPN Bridge. A secondary node running as primary would have Config Primary cleared and Running Primary set. Other bits of interest reflect the status of redundancy messages sent via Ethernet (Hearing NT) and via LCN (Heard LCNI). |
| Time Status   | The flags in the Status register reflect the state of the time synchronization subsystem. Bits of interest include the Ignore Synch bit – meaning time sync is disabled. Init Done – the time system has been initialized for the current time mastership mode. Sync Started - a time synchronization cycle has started. Sync Ended – a time synchronization cycle just ended.                                                                                                                                    |
| Time Master   | The bits in this register (Time Mastership) are used to control the time synchronization subsystem. Segment Master nodes broadcast diagnostic timeframe messages on their segments. TPN Master nodes source time to other segments.                                                                                                                                                                                                                                                                               |

### Individual Status Values

The right hand side, vertical center of the Bridge Status display contains several status values of interest. The following table describes each.

**Table 4-4 Individual Status Values**

| Field           | Description                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------|
| We Primary      | This bridge is running as primary if set                                                                                        |
| WSI LAT Changed | The composite LAT has changed                                                                                                   |
| Say OK          | A bridge redundancy message event is occurring.                                                                                 |
| Say Not OK      | A bridge redundancy broken message event is occurring                                                                           |
| Say Stun        | The bridge needs to be stunned                                                                                                  |
| Buddy OK        | The redundant bridge is OK                                                                                                      |
| Use NT LAT      | The remote composite LAT / PAT table is valid                                                                                   |
| OLAN OK         | Reflects the health of the Redundant Ethernet. Should be 3.                                                                     |
| Percent Idle    | Reflects CPU free time on the TPN Bridge                                                                                        |
| LATS Unknown    | The number of LATs that are not yet valid for this segment (nodes in Resync)                                                    |
| Hear OK         | Receiving bridge redundancy messages                                                                                            |
| Hear Not OK     | Receiving bridge redundancy broken messages                                                                                     |
| NT Die Request  | Another bridge has requested that this bridge kill itself                                                                       |
| He Moved        | A node moved from this segment to another.                                                                                      |
| Run Mode        | This value will be 0 if running as a normal TPN Bridge or 1 if running in special mode compatible with the LCNI diagnostic test |

### Raw / Rate Counters

In the center of the Bridge Status display, and below the individual status value fields, are counters presenting either raw counts or rates for their indicated values. The following table describes each.

**Table 4-5 Raw / Rate Counters**

## 4. Implementing the TPN Bridge System

### 4.4. Bridge Status

| Field             | Description                                                                                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LAT Recalcs       | The number of times this bridge has recalculated the composite segment LAT.                                                                                                                           |
| TX Errors         | The number of LCN transmit errors detected.                                                                                                                                                           |
| RX Errors         | The number of LCN receive errors detected. This number will frequently increment as nodes on the LCN are reset.                                                                                       |
| NT Discards       | The number of inbound messages that were discarded on their way from a remote segment to the local segment as a result of filtering. These messages were not bound for any node on the local segment. |
| NT Dis Rate       | The rate at which NT Discards are occurring. This is a measure of the effectiveness of inbound message filtering.                                                                                     |
| LCNP Discards     | The number of outbound messages that were discarded on their way from the local LCN to the Ethernet as the result of filtering. These are messages bound only for a node on the local segment.        |
| LCNP Dis Rate     | The rate at which LCNP Discards are occurring. This is a measure of the effectiveness of outbound message filtering.                                                                                  |
| Outputs Processed | The number of LCN messages that have been received from the local LCN.                                                                                                                                |
| Output Queue Rate | The rate at which outbound messages (from the local LCN) are processed.                                                                                                                               |

#### Network communication subsystem state

The TPN Bridge network communication subsystem state can be viewed by invoking the WS Counters display. Following are fields of interest and their description.

**Table 4-6 Network Communication Subsystem State**

| Field              | Description                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Configured Primary | 1 - This bridge is configured as a primary TPN Bridge node<br>0 – This bridge is configured as a secondary TPN Bridge node     |
| Running Primary    | 1 - This bridge is running as a primary TPN Bridge node<br>0 – This bridge is running as a secondary TPN Bridge node           |
| Buddy Bridge ID    | The ID assigned to the Buddy (Redundant partner) bridge during system configuration using the TPN Bridge config/status snapin. |

**Table 4-6 Network Communication Subsystem State**

| Field                 | Description                                                                                                                                                                                                       |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tx Total              | Total number of message transmitted                                                                                                                                                                               |
| Tx Failures           | Number of transmit failures. The most likely cause of this counter incrementing is failure to transmit Buddy messages (Ethernet redundancy messages). Check Redundant partner and Tx / Rx Buddy message counters. |
| Tx LCN Msg            | Number of LCN messages that have been transmitted.                                                                                                                                                                |
| Tx Buddy Msg          | Number of TCP/IP messages sent to Redundant partner bridge (Buddy).                                                                                                                                               |
| Tx PLAT Msg           | Number of LAT update messages sent from this bridge to remote bridges                                                                                                                                             |
| Rx Total              | Total number of messages received.                                                                                                                                                                                |
| Rx Failures           | Number of messages that experienced a failure during receipt.                                                                                                                                                     |
| Rx Out of Order       | Number of messages that have been received out of order. Each TPN Bridge and message class: LCN, Buddy, PLAT have their own sequence counter.                                                                     |
| Rx Duplicated         | Number of messages received for a Bridge and message class that had identical sequence numbers for consecutive messages.                                                                                          |
| Rx Lost               | Number of gaps in message sequencing that has been observed.                                                                                                                                                      |
| Rx LCN Lost           | Number of LCN messages that have been lost. See Rx Lost.                                                                                                                                                          |
| Rx LCN Out of Order   | Number of LCN messages that were received out of order. See Rx Out of Order.                                                                                                                                      |
| Rx Buddy Msg          | Number of Buddy messages received from Redundant partner bridge.                                                                                                                                                  |
| Rx Buddy Lost         | Number of Buddy messages that have been lost. See Rx Lost.                                                                                                                                                        |
| Rx Buddy Out of Order | Number of Buddy messages that were received out of order. See Rx Out of Order.                                                                                                                                    |
| Rx PLAT Msg           | Number of LAT update messages received from remote bridges.                                                                                                                                                       |
| Rx PLAT Lost          | Number of LAT update messages that have been lost. See Rx Lost.                                                                                                                                                   |
| Rx PLAT Out of Order  | Number of LAT update messages that were received out of order.                                                                                                                                                    |

## 4. Implementing the TPN Bridge System

### 4.4. Bridge Status

---

**Table 4-6      Network Communication Subsystem State**

| Field                | Description                                                                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | See Rx Out of Order.                                                                                                                                          |
| Tx Max PDU size      | Maximum size of network message that has been sent.                                                                                                           |
| Tx Min PDU size      | Minimum size of network message that has been sent.                                                                                                           |
| Tx Ave PDU size      | Average size of network messages that have been sent.                                                                                                         |
| Remote Bridges Known | Binary map of remote bridges that have sent PLAT updates. Bit 31 indicates this bridge is alone. Bit 0 through 30 indicate Bridge 1 through 29, respectively. |

### Bridge Operation Checklists

The following items should be checked using the *LCN* displays to verify that the TPN Bridge is working correctly. If the expected value is not observed proceed with corrective action until the condition has been resolved. If condition is corrected, do not continue with additional corrective action steps.

#### Check for the following values using the LCN Displays

**Table 4-7 Bridge Operation Checklist – LCN Display**

| Display                      | Expected Value                                                                                                                                                       | Corrective Action (expected value not observed)                                                                                                                                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System Status                | Bridge Node State should display: OK-BRG                                                                                                                             | Bridges in TEST indicate a problem in other Bridge nodes. Use the remaining entries in this checklist to identify bridges that may be causing this condition.<br><br>Reset Bridge nodes that are showing FAIL or OFF mode.                                       |
| CLOKSTAT                     | If the clock source is not a bridge, one node will display <i>Master</i> , one node will display <i>Slave</i> , and all other nodes will be in <i>Listener</i> mode. | If <i>Master</i> and <i>Slave</i> state are not displayed, reload master and/or slave node. If nodes across segment boundaries are not in <i>Listener</i> mode, ensure bridge time sync has been configured correctly according to Table 4-11 TPN Clock Sources. |
| NET MEDIA DEADS              | Net media dead values should be zero.                                                                                                                                | Reset all counters. If counts increment, locate LCN cable fault or defective node.                                                                                                                                                                               |
| CROSSED CABLE MESSAGE TOTALS | Cable master (lowest numbered US) should display a count of one greater than all other node counts. Values should increment every 2-minutes.                         | Reset lowest numbered US. Verify cable swaps occur at 2-minute intervals.                                                                                                                                                                                        |
| LOCAL DRIVER DATA            | Lowest US node number displays MASTER and Swap Ticks Left changes.                                                                                                   | Shutdown node and reload.                                                                                                                                                                                                                                        |

## 4. Implementing the TPN Bridge System

### 4.5. Bridge Operation Checklists

---

## 4.5 Bridge Operation Checklists

### Bridge Status display

The following items should be checked using the **Bridge Status** display to verify that the TPN Bridge is working correctly. If the expected value is not observed proceed with corrective action until the condition has been resolved. If condition is corrected, do not continue with additional corrective action steps.

**Table 4-8 Bridge Operation Checklist – Status Display**

| Field                          | Expected Value                      | Corrective Action (expected value not observed)                                                                                                                                                                                                                      |
|--------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bridge Error Code              | 0                                   | Place cursor over the Bridge Error Code field to decode the value. Reset TPN Bridge to reload the bridge node.                                                                                                                                                       |
| Cycle Counter                  | Incrementing                        | Check the Bridge Error Code. If the error code is 0, check the TPNB status using the LCNP status applet. Most likely, the bridge node has been stunned by another bridge.                                                                                            |
| Seconds Since Msg Received NT  | Never reaches timeout of 61 seconds | No messages from remote TPN Bridge nodes are being received. Check all Ethernet communication routes. Shutdown TPN Bridge computer, power off, pull AC plug, wait 10 seconds, restore to running state.                                                              |
| Seconds Since Msg Receive LCNI | Never reaches timeout of 61 seconds | No messages from local LCN are being received. Check LCN connections. Verify that LCN nodes are powered on and ring is established (3 or more nodes, reset has been performed). Shutdown TPN Bridge computer, power off, pull AC plug, wait 10 seconds, restore all. |
| OLAN Status                    | 3                                   | A value other than 3 indicates a problem somewhere in the Redundant Ethernet system. Use the TPN Bridge config/status snapin to get a detailed view of Redundant LAN status.<br>2 = Cable LAN A Suspect; 1 = Cable LAN B Suspect                                     |

#### 4. Implementing the TPN Bridge System

##### 4.5. Bridge Operation Checklists

**Table 4-8 Bridge Operation Checklist – Status Display**

| Field         | Expected Value | Corrective Action (expected value not observed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Load Progress | 254 or 255     | A TPN Bridge with time sync enabled in a normal loaded state will have a Load Progress value of 255. If time sync is disabled, the running Load Progress value will be 254. A node that does not proceed past state 10 is waiting to hear an input on its LCNI channels. If the node hangs at state 105, the Redundant LAN system is not healthy. Check the OLAN OK status for a value of 3 – any other value indicates a Redundant Ethernet problem. A node in a state 127 or greater is running and in a valid time synchronization state and should eventually reach state 254 or 255. |

## 4. Implementing the TPN Bridge System

### 4.5. Bridge Operation Checklists

---

**Table 4-8 Bridge Operation Checklist – Status Display**

| Field                        | Expected Value                                                                                                         | Corrective Action (expected value not observed)                                                                                                                                                                                                                                |
|------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Node Status from XXXX window | XXXX = Primary displays all nodes on the local segment.<br><br>XXXX = Secondary displays all nodes on all TPN segments | If all other values are correct, this list should be correct. Use the list to verify the bridge's view of the TPN.                                                                                                                                                             |
| WS Counters – Tx Buddy Msg   | Incrementing at following rates:<br>Primary - 2 per second.<br>Secondary - 1 every 10 seconds                          | TCP/IP connection is broken between Redundant partners. Check status of Redundant bridge. Reboot both bridges (one at a time – ensuring partner is in a running state before rebooting each). Shutdown, power down, remove AC plug, wait 10 seconds, restore to running state. |
| WS Counters – Rx Buddy Msg   | Incrementing at following rates:<br>Primary - 1 every 10 seconds.<br>Secondary - 2 per second.                         | TCP/IP connection is broken between Redundant partners. Check status of Redundant bridge. Reboot both bridges (one at a time – ensuring partner is in a running state before rebooting each). Shutdown, power down, remove AC plug, wait 10 seconds, restore to running state. |

### LCNP Status applet display

The following items should be checked using the **LCNP Status Applet** display to verify that the TPN Bridge is working correctly. If the expected value is not observed proceed with corrective action until the condition has been resolved. If condition is corrected, do not continue with additional corrective action steps.

| Field            | Expected Value | Corrective Action (expected value not observed)                                                                      |
|------------------|----------------|----------------------------------------------------------------------------------------------------------------------|
| Passed Self Test | Green          | Simulates the processor board Passed LED                                                                             |
| Board Failed     | Empty          | Simulates the processor board Self-Test / Failed LED                                                                 |
| LEDs             | Node Address   | Simulates the processor board, 3 digit LED display. This field will display an error code – XXX if a failure occurs. |
| State            | OK             | Reflects the board load / running state.                                                                             |
| Substate         | Blank          | In failure condition, this field may decode the failure cause.                                                       |

All values listed in the paragraphs above are also available from the **TPN Bridge Configuration/Status Snap-In** console window. The snap-in uses DCOM to communicate with remote nodes for status updates. If the display is not updating, the connection is broken and the Bridge Status display must be used for troubleshooting the problem. Online Help is provided for the snap-in.

## 4.6 Capturing Failure Status for TAC Analysis

### Overview

The following procedure will be used when a TPN Bridge failure has occurred. This will provide preliminary information that can be presented to the TAC for failure analysis.



#### ATTENTION

The following procedure will capture the status environment that exists when the failure occurred. It must be performed on [each](#) TPN Bridge computer.

---

### Save the System and Application Events Logs

| Step | Action                                                                                                                                                                  |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Start the Event Viewer by clicking <b>Start → Programs → Administrative Tools (Common) → Event Viewer</b> .                                                             |
| 2    | Select <b>Log</b> and <b>System</b> . Click <b>Save As...</b> and save the file in the format of<br><br>SYSTEM_dd-mm-yy.EVT<br><br>to a selected drive and directory.   |
| 3    | Select <b>Log</b> and <b>Application</b> . Click <b>Save As...</b> and save the file in the format of<br><br>APP_dd-mm-yy.EVT<br><br>to a selected drive and directory. |

---

### Screen Capture of LCNP Status Applet

|   |                                                                                                                                                                                |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | Select the LCNP Status.                                                                                                                                                        |
| 5 | Press <b>Alt + Print Screen</b> to capture the active window on the screen (LCNP Status).                                                                                      |
| 6 | Invoke the Microsoft Paint application and paste ( <b>Ctrl + V</b> ) the captured screen information.                                                                          |
| 7 | Select <b>File</b> and click <b>Save As...</b> and save the file in the format of<br><br>LCNP_dd-mm-yy.BMP<br><br>(save as 16 color bitmap) to a selected drive and directory. |

---

***Screen Capture of TPN Bridge View - TDC Status Window***

- 8**      Select the TDC Status window.
  - 9**      Press **Alt + Print Screen** to capture the active window on the screen (TDC Status).
  - 10**     Invoke the Microsoft Paint application and paste (**Ctrl + V**) the captured screen information.
  - 11**     Select **File** and click **Save As...** and save the file in the format of  
  
                                         TDCSTATUS\_dd-mm-yy.BMP  
  
                                         (save as 16 color bitmap) to a selected drive and directory.
- 

***Documenting other Failure Input***

- 12**     Document the events or a detailed step-by-step procedure that seemed to have caused the problem.
  - 13**     From a US or GUS station, print the `SYSTEM STATUS` display from each TPS LCN Segment.
  - 14**     Please send all information to TAC for review and analysis.
-

## 4.7 Configuring the Nortel (BayStack) Switch

### Purpose

---



#### ATTENTION

To minimize the opportunity for IP Multicast problems, ALL switches on ALL systems, regardless of software revision, should be configured to disable snooping.

---

### Prerequisites

Before beginning the procedures in this section, verify the following:

- You have a UL-listed RS-232 cable configured as required by the switch vendor to connect the computer's serial port to the switch's comm port.
- Have HyperTerminal configured on the computer that will be used as the interface to the Switch Configuration menu.
- Have reviewed the specific vendor switch user guide.
- Have checked MMC to ensure the modification will take place on the NON-ACTIVE SWITCH

### Disable "snooping" on Nortel (BayStack) switch

Use the following procedure to disable IGMP Snooping on a Nortel (BayStack) Switch. Refer also to your BayStack Series Switch manual for details on the switch menus.

| Step | Action                                                                                                                                         |
|------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Connect the RS-232 cable to the Switch's Comm Port and the computer's serial port.                                                             |
| 2    | Click Start, point to Programs, point to Accessories, point to Communications, and then click HyperTerminal.                                   |
| 3    | From the <b>Connection Description</b> dialog box, type a name that describes the connection in the <b>Name</b> box and then click <b>OK</b> . |
| 4    | In the <b>Icon</b> box, click the appropriate icon, and then click <b>OK</b> .                                                                 |
| 5    | From the <b>Connect To</b> dialog box, select the serial port being used by the computer in Connect Using box and then click <b>OK</b> .       |

| Step                                                                              | Action                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6                                                                                 | From the <b>Connect To</b> dialog, select the serial port being used by the computer and click <b>OK</b> .                                                                                                                                        |
| 7                                                                                 | From the <b>Properties</b> page make the following Port Settings: <ul style="list-style-type: none"> <li>• Bits per second: 9600</li> <li>• Data Bits: 8</li> <li>• Parity: NONE</li> <li>• Stop bits: 1</li> <li>• Flow control: None</li> </ul> |
| 8                                                                                 | Verify the connection between the computer and the switch.                                                                                                                                                                                        |
|  | <p><b>TIP</b></p> <p>See the section on <a href="#">Using the Console Interface in your Nortel Switch manual for additional information on using the Console Interface menus.</a></p>                                                             |
| 9                                                                                 | Press CTRL+Y to display the switch's control interface main menu.                                                                                                                                                                                 |
| 10                                                                                | Select <b>Switch Configuration</b> from the main menu.                                                                                                                                                                                            |
| 11                                                                                | Select <b>IGMP Configuration</b> from the <b>Switch Configuration Menu</b> .                                                                                                                                                                      |
| 12                                                                                | Select the <b>Snooping</b> option from the <b>IGMP Configuration</b> .                                                                                                                                                                            |
| 13                                                                                | Highlight <b>Enabled</b> and press the space bar to set the <b>Snooping</b> option to <b>Disabled</b> .                                                                                                                                           |
| 14                                                                                | Press Ctrl –C to return to the <b>Main Menu</b> .                                                                                                                                                                                                 |
| 15                                                                                | Check the system to ensure all communications are working correctly. If not, correct the problem before continuing.                                                                                                                               |
| 16                                                                                | <b>Save configuration changes and repeat the procedure for the second switch.</b>                                                                                                                                                                 |

## 4.8 Installing and Configuring the TPN System Software

### Installing "Greater than 64 Node" keyfile option

The following procedure will install the software option to exceed the 64-node maximum using TPS R610.4 or greater.

| Step | Action                                                                                                                                                          |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Mount the TPS R610.4 or greater System "SOFTWARE OPTIONS" media into the local drive.                                                                           |
| 2    | Go to the <b>Engineering Main Menu</b> and select <b>Support Utilities</b> .                                                                                    |
| 3    | Select <b>Modify Volume Paths</b> :<br><br>Set path for KEY FILE VOLUME to \$Fs>&KFO><br>Set path for NCF BACKUP PATH to \$Fd>&ASY><br><br>Press <b>Enter</b> . |
| 4    | Select MAIN MENU<br>Select SYSTEM WIDE VALUES target<br>Select Software OPTIONS target.                                                                         |
| 5    | Press <CTRL> <F10> simultaneously (to allow reading of the new OPTIONS).                                                                                        |
| 6    | Ensure the GREATER Than 64 Nodes is displayed in CYAN.<br><br>Press <b>Enter</b> .                                                                              |
| 7    | Press <CTRL> <F1> (to RUN the Checker)<br><br>Press <CTRL> <F2> (to INSTALL the Changes)<br><br>Press <CTRL> HELP (to RETURN to the MAIN MENU)                  |
| 8    | Reload <u>all</u> nodes.                                                                                                                                        |

### Adding the TPN Bridge to the NCF

The following procedure will configure the TPN Bridge on the TPN system using TPS R610.



#### ATTENTION

If TPN Bridge node numbers greater than 64 are to be used, you must first perform the Installing "Greater than 64 Node" keyfile option procedure.

---

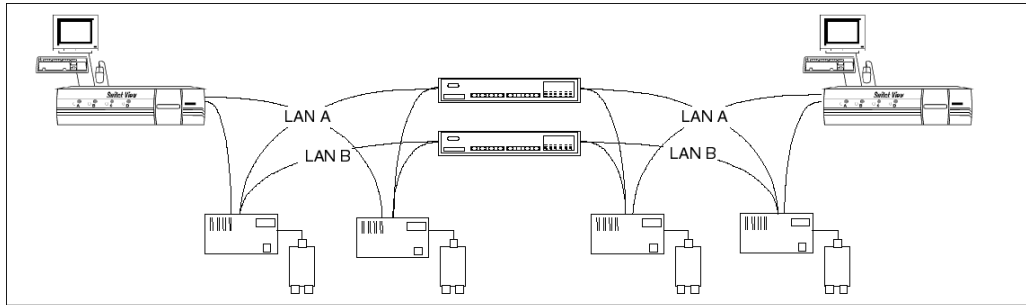
| Step | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Go to the <b>Engineering Main Menu</b> and select <b>Support Utilities</b> .                                                                                                                                                                                                                                                                                                                                                                                         |
| 2    | Select <b>Modify Volume Paths</b> :<br><br>Edit the entry for NCF BACKUP PATH to contain<br><br>\$Fs>&ASY where s = a removable media or emulated disk<br>containing the BACKUP NCF files.<br><br>Press <b>Enter</b> .                                                                                                                                                                                                                                               |
| 3    | Select MAIN MENU<br>Select LCN NODES                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 4    | Select the node number for the Primary Bridge. This must be an odd address and the next consecutive number must be available for the Secondary Bridge.                                                                                                                                                                                                                                                                                                               |
| 5    | Select NETWORK GATEWAY.                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 6    | Enter the PIN LOCAL ADDRESS..                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|      |  <b>WARNING</b><br><br>When adding a TPN Bridge to the NCF (LCN NODES), the PIN LOCAL ADDRESS specified must be unique on both Local and Remote LCNs. If there is a duplication of the PIN address, the Network Gateways that are already configured will stop communicating. Confirm there are no duplicate PIN LOCAL ADDRESS by checking the existing NETWORK GATEWAY LCN NODE. |
| 7    | Press F1 to check the NCF change.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8    | Press F2 to install the NCF.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 9    | Repeat step 5 through 9 for the Secondary Bridge node.                                                                                                                                                                                                                                                                                                                                                                                                               |
| 10   | Reload all US stations.                                                                                                                                                                                                                                                                                                                                                                                                                                              |

---

#### 4.9. TPN Bridge and NT Configuration

## Overview

- Monitor, keyboard and mouse connected to a peripheral switch box.
- Two Bridge computers (A and B) with monitor, keyboard and mouse ports connected to the peripheral switch box.



- Do not connect Bridge computers to Ethernet Switch (or power on) until computer names and IP addresses have been set and computers have been rebooted.
- Do not connect MAUs to the LCN cable until instructed to do so during Hot Insertion procedure.

## Initializing the node

---



### ATTENTION

It is critical that you perform the following task on **each** newly arrived TPN Bridge computers in preparation for initializing the node.

---

## Overview

Perform **all** procedures in this section on **every** TPN Bridge node and archive the diskette. The diskette will be used when a TPN Bridge node must be recovered. The TPN Bridge computers being initialized **must not** be connected to the LCN. They should be disconnected from the Ethernet switches or optionally (if no TPN Bridges are running) the switches may be powered down.

## Create a Customized Boot Disk

---



### ATTENTION

The following procedure **must be repeated** for **all** TPN Bridge nodes.

---

Perform the following on each TPN Bridge node, label diskette with the TPN Bridge computer name, and archive the diskette.

| Step | Action                                                                                    |
|------|-------------------------------------------------------------------------------------------|
| 1    | Insert the <b>TPN Bridge</b> software CD.                                                 |
| 2    | Navigate to the following folder on the <b>TPN Bridge</b> software CD:<br><b>dosimage</b> |
| 3    | Insert a blank diskette into the A drive.                                                 |
| 4    | Double-click the following file:<br><b>Makedisk.bat</b>                                   |
| 5    | A console window will be displayed as the diskette is formatted.                          |
| 6    | The diskette image will be created.                                                       |
| 7    | The diskette image will be verified.                                                      |

---

## 4. Implementing the TPN Bridge System

### 4.9. TPN Bridge and NT Configuration

---

| Step | Action                                                                                                                                                                                                                                  |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8    | Continue to the next step if the diskette verification completes without error.<br><br>If verification errors are reported, insert a different blank diskette into the A drive, and then return to step 1.                              |
| 9    | Copy the "options.bat" and "customer.udb" files from the root of the node's hard drive to the diskette.                                                                                                                                 |
| 10   | Remove the diskette from the drive and label it as follows:<br><br><b>Customized Boot Diskette</b> (with Bridge Computer Name).<br><br>Refer to the latest SCN to verify the TPSSOFTREL is correct for the TPN Bridge Software version. |

---



#### ATTENTION

The diskette may only be used to recover the node that was used to create the diskette because the options batch file and the customer database files are unique to the node.

---

### Change the Local Administrator Password (Optional)

---



#### ATTENTION

The following procedure must be repeated for all Bridge nodes.

---

The following procedure changes the password. A newly initialized node does not have an Administrator password.

| Step | Action                                                                                                                                                                                                                                                                 |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Press <Ctrl> <Alt> <Delete> keys and press <Enter> to allow logon as Administrator.                                                                                                                                                                                    |
| 2    | Once logged on, press <Ctrl> <Alt> <Delete> and click <b>Change Password</b> .                                                                                                                                                                                         |
| 3    | Enter the old password into <b>Old Password</b> field and enter new password into the <b>New Password</b> and <b>Confirm New Password</b> fields and click <b>OK</b> .<br><br>Continue on to the next section after you have changed the Local Administrator password. |

---



#### WARNING

All TPN Bridge nodes must have the same Local Administrator password. Record and store the Local Administrator password in a secure place. If you forget the password, you will have to re-perform the Node Initialization procedure in order to recover.

---

#### Set the Date and Time



#### ATTENTION

The following procedure must be repeated for all Bridge nodes.

---

| Step | Action                                                                                                                                                                                                                                                            |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Identify the physical computer to be configured as BRIDGE01. Using the Cybex SwitchView box, select this computer for viewing on the monitor. This same node will be used to configure <u>all</u> the TPN Bridge nodes.<br><br>Log on as the local Administrator. |
| 2    | Open the <b>Control Panel</b> program as follows:<br><br><b>Start &gt; Settings &gt; Control Panel</b>                                                                                                                                                            |
| 3    | Double-click the <b>Date/Time</b> icon.                                                                                                                                                                                                                           |
| 4    | Click on the <b>Time Zone</b> tab and then click the down arrow next to the highlighted time zone entry to bring up the list of time zones.                                                                                                                       |
| 5    | Select the local time zone from the list of time zones.                                                                                                                                                                                                           |
| 6    | Select <b>Automatically adjust for Daylight Savings changes</b> <u>ONLY</u> if it is the local custom and you are not using time sync from the LCN.                                                                                                               |
| 7    | Click <b>Apply</b> .                                                                                                                                                                                                                                              |
| 8    | Click on <b>Date &amp; Time</b> tab.                                                                                                                                                                                                                              |
| 9    | Enter the correct date.                                                                                                                                                                                                                                           |
| 10   | Enter the correct time.                                                                                                                                                                                                                                           |
| 11   | Click <b>OK</b> .                                                                                                                                                                                                                                                 |
| 12   | Close the <b>Control Panel</b> window.                                                                                                                                                                                                                            |
| 13   | Continue on to the next section.                                                                                                                                                                                                                                  |

---

## 4. Implementing the TPN Bridge System

### 4.9. TPN Bridge and NT Configuration

---

#### ***Set the Computer Name for TPN Bridge Identification***

Nodes that are part of a TPN Bridge network differ from standard TPS nodes in only two respects:

- TPS domain name is pre-assigned and set to **WORKGROUP**
- TPN Bridge IDs (Computer Names), IP addresses, and Dual NIC port identification of each TPN Bridge node are assigned according to the following table.

**Table 4-9 TPN Bridge Node Identification**

| TPN Bridge ID<br>(Computer Name) | IP Address<br>Assignment | TPN Bridge ID<br>(Computer Name) | IP Address<br>Assignment |
|----------------------------------|--------------------------|----------------------------------|--------------------------|
| BRIDGE01                         | 192.168.1.216            | BRIDGE07                         | 192.168.1.222            |
| BRIDGE02                         | 192.168.1.217            | BRIDGE08                         | 192.168.1.223            |
| BRIDGE03                         | 192.168.1.218            | BRIDGE09                         | 192.168.1.224            |
| BRIDGE04                         | 192.168.1.219            | BRIDGE10                         | 192.168.1.225            |
| BRIDGE05                         | 192.168.1.220            | BRIDGE11                         | 192.168.1.226            |
| BRIDGE06                         | 192.168.1.221            | BRIDGE12                         | 192.168.1.227            |

**Table 4-10 Ethernet Switch IP Addressing (Maintenance Use Only  
Not Required)**

| Ethernet Switch ID<br>(Computer Name) | IP Address<br>Assignment | Ethernet Switch ID<br>(Computer Name) | IP Address<br>Assignment |
|---------------------------------------|--------------------------|---------------------------------------|--------------------------|
| SWITCH01                              | 192.168.1.200            | SWITCH06                              | 192.168.1.205            |
| SWITCH02                              | 192.168.1.201            | SWITCH07                              | 192.168.1.206            |
| SWITCH03                              | 192.168.1.202            | SWITCH08                              | 192.168.1.207            |
| SWITCH04                              | 192.168.1.203            | SWITCH09                              | 192.168.1.208            |
| SWITCH05                              | 192.168.1.204            | SWITCH10                              | 192.168.1.209            |

To implement the TPN Bridge nodes, continue with this procedure for configuring the Node Names and IP address. The procedure will indicate when you should use the information in the above table to configure your TPN Bridge node.



### ATTENTION

The following procedure must be repeated for all Bridge nodes.

---

| Step | Action                                                                                                                                                                                                                    |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Double-click the <b>Network</b> program icon                                                                                                                                                                              |
| 2    | The <b>Network</b> page appears showing the factory default <b>Computer Name:</b> (which is set to <b>TPNBRIDGE01</b> ) and <b>WORKGROUP:</b> on the <b>Identification</b> tab. <div data-bbox="703 777 1148 1102"></div> |
| 3    | Select <b>Change...</b> button to change the <b>Computer Name:</b> .                                                                                                                                                      |

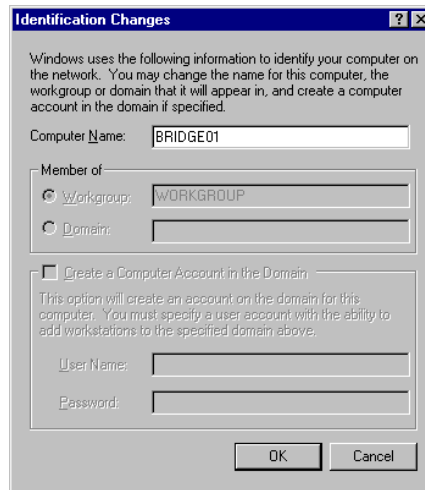
---

## 4. Implementing the TPN Bridge System

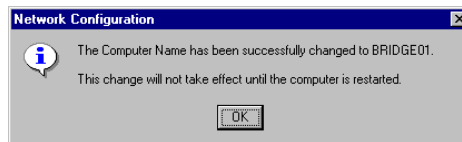
### 4.9. TPN Bridge and NT Configuration

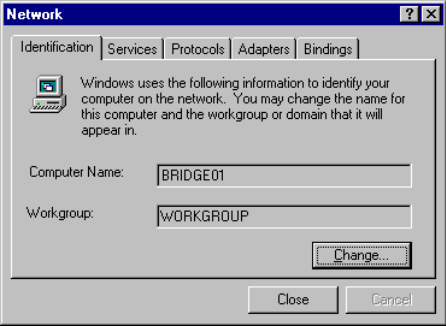
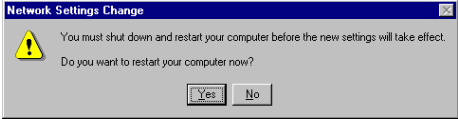
---

| Step | Action                                                                                                                                                                                                                                                                                                                                      |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4    | The <b>Identification Changes</b> window appears. Change the <b>Computer Name:</b> to identify the Bridge node on the Bridge network. Reference the TPN Bridge Node Ethernet Identification chart showing the Computer Name and IP Address for the Bridge node being configured. This example node uses <b>BRIDGE01</b> . Click <b>OK</b> . |



- 5 A **Network Configuration** message window now appears. Click **OK**.



| Step | Action                                                                                                                |
|------|-----------------------------------------------------------------------------------------------------------------------|
| 6    | The <b>Network</b> page now shows the new <b>Computer Name:</b> for the Bridge node. Click <b>Close</b> .             |
|      |                                     |
| 7    | A network message appears prompting you to reboot the computer. Do not restart (reboot). Click <b>No</b> to continue. |
|      |                                    |

---

## 4. Implementing the TPN Bridge System

### 4.9. TPN Bridge and NT Configuration

---

#### Set the Network Protocol and IP Address

---

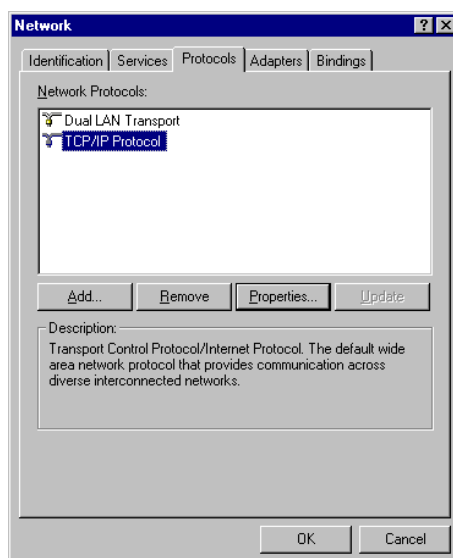


#### ATTENTION

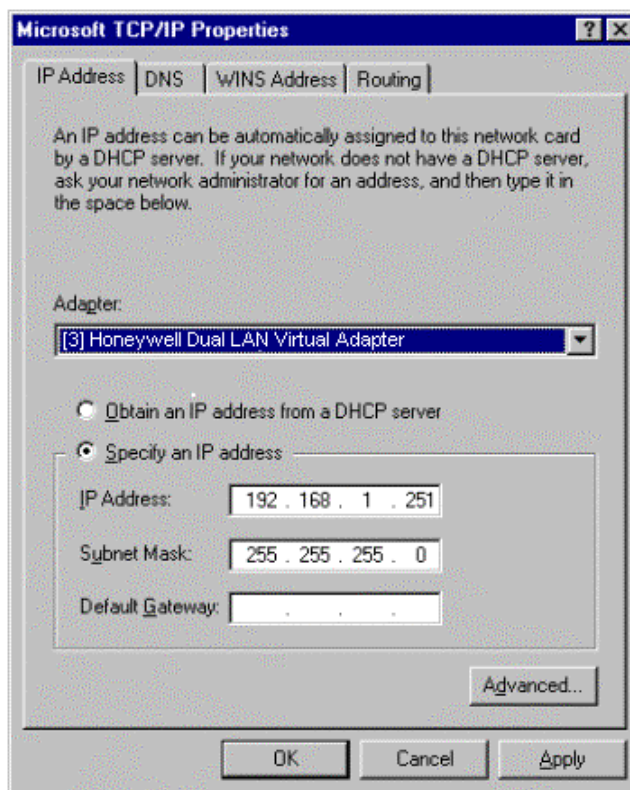
The following procedure must be repeated for all Bridge nodes.

---

| Step | Action                                                                                                                                                                                |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8    | Double-click the <b>Network</b> program icon                                                                                                                                          |
| 9    | Select the <b>Protocols</b> tab on the <b>Network</b> page. A list of installed <b>Network Protocols:</b> appears.<br><br>Select <b>TCP/IP Protocol</b> and click <b>Properties</b> . |



| Step | Action                                                                                                                                                                                                                                                                                                          |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10   | <p>The <b>Microsoft TCP/IP Properties</b> window appears. Select the <b>Specify an IP address</b> radio button. The default IP Address (octet equal to 251) is shown below.</p> <p><b>Note:</b> Only Adapter: <b>[3] Honeywell Dual LAN Virtual Adapter</b> from the drop down window should be configured.</p> |



Enter the following data using Table 4-10 Ethernet Switch IP Addressing (Maintenance Use Only Not Required for the TPN Bridge node name and IP address following the dotted quad notation shown above.

- IP Address for the appropriate TPN Bridge node name (change least significant octet).
- Subnet Mask (use the default of 255.255.255.0 octets)
- Default Gateway (leave blank for TPN Bridge node).

## 4. Implementing the TPN Bridge System

### 4.9. TPN Bridge and NT Configuration

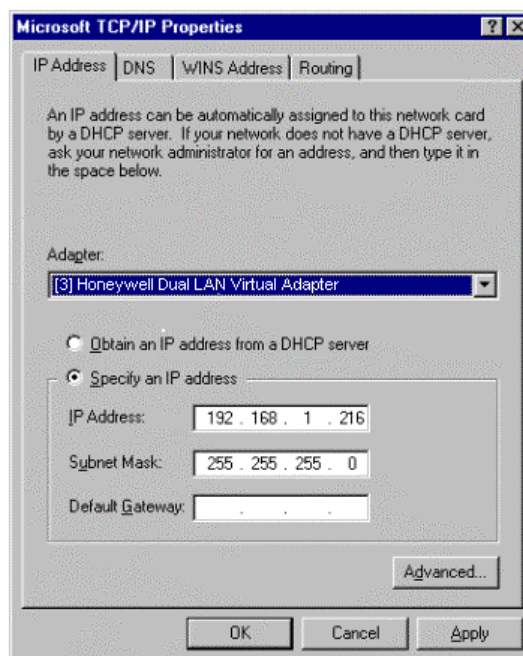
---

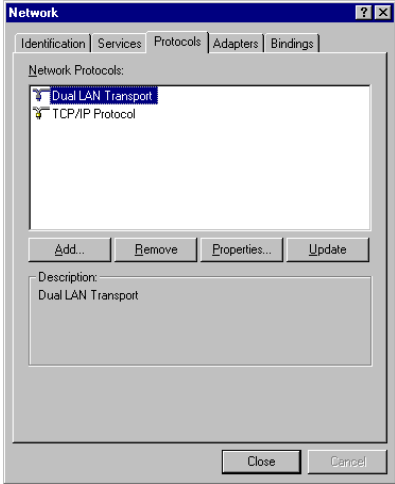
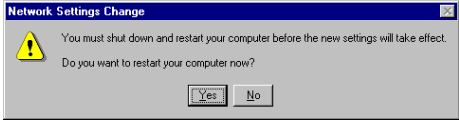
| Step | Action |
|------|--------|
|------|--------|

---

- |    |                                    |
|----|------------------------------------|
| 11 | Click <b>Apply</b> and <b>OK</b> . |
|----|------------------------------------|

**Note:** In this example only the least significant octet of the IP address (216) was changed to match the computer name of BRIDGE01 (See Table 4-10).



| Step | Action                                                                                                                                                                                                                                                               |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12   | <p>The Network window appears.</p>                                                                                                                                                |
| 13   | <p>A network message appears prompting you to reboot the computer. <b>DO NOT restart (reboot) the computer on <u>this</u> node at this time.</b> Continue to the next step.</p>  |
| 14   | <p>Repeat steps 2 through 13 for <u>each</u> remaining TPN Bridge node to set the Computer Name and IP Address until <u>all</u> nodes on the Bridge network have been configured. When finished, continue to next step.</p>                                          |
| 15   | <p>Click <b>Yes</b> in the <b>Network Settings Change</b> message box. Perform a computer shut down and restart.</p> <p>Connect the TPN Bridge computers to the Ethernet switches while the computers reboot.</p>                                                    |
| 16   | <p>Logon to each Bridge node as "Administrator."</p>                                                                                                                                                                                                                 |
| 17   | <p>After rebooting, verify the Computer Name and IP Address is correct on each Bridge node by entering <b>Command Prompt</b> (MS-DOS) and executing the command: <code>ipconfig -all   more</code>.</p>                                                              |

## 4.10 Configuring the Bridge Segments Using MMC

### Overview

The MMC offers a common environment for Snap-ins. When the MMC tool is loaded one or more Snap-Ins is initialized. These Snap-In's are integrated to create the tool's *namespace*—the ordered collection of nodes that appears in the tree view in the scope pane (left pane). The *namespace* is a master tree that represents what the tool can do. It appears similar to a tree view of the files and folders on a hard drive. The *namespace* can include all manageable aspects of a network. In this case, the network that results contains these TPN Bridge nodes.

### Configuration using TPN Bridge configuration/status snap-In

---



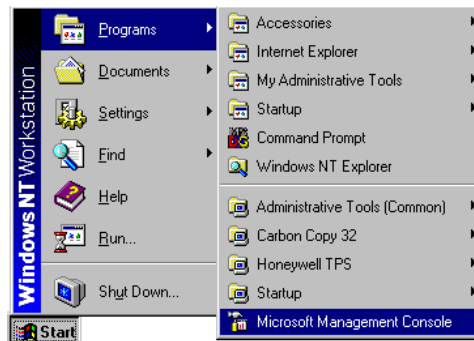
#### ATTENTION

The following steps will be used to configure ALL TPN Bridge nodes. All TPN Bridge nodes must be configured using the same node as a user interface and running the Microsoft Management Console program. All Bridge nodes need to be connected to the Redundant Ethernet.

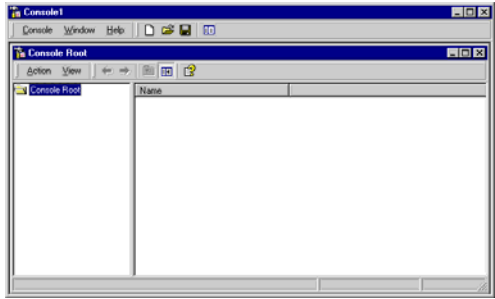
---

### Creating an MMC TPN Bridge View Console

| Step | Action                                                           |
|------|------------------------------------------------------------------|
| 1    | Open the <b>Microsoft Management Console</b> program as follows: |



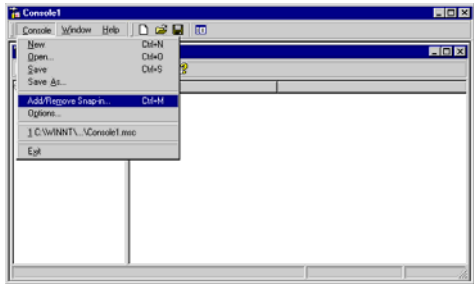
| Step | Action                                                                                                                                                                                                                                                                                                                                                                 |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | The MMC parent frame appears with a master menu and toolbar. Also, a child window appears that includes a command bar, a scope pane (left pane) and a result pane (right pane). The scope pane is a tree control displaying the tool's <i>namespace</i> , the tree-formatted listing of all visible nodes, each of which represent a manageable object, task, or view. |



---

**Add Snap-in to Child Frame**

- 3 Click **Console** on the parent frame command bar. Select and click **Add/Remove Snap-in**.

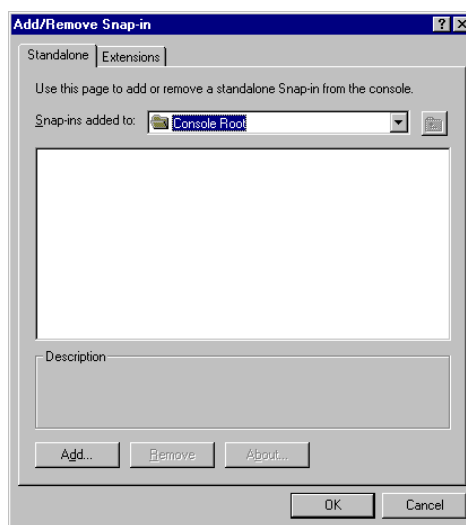


## 4. Implementing the TPN Bridge System

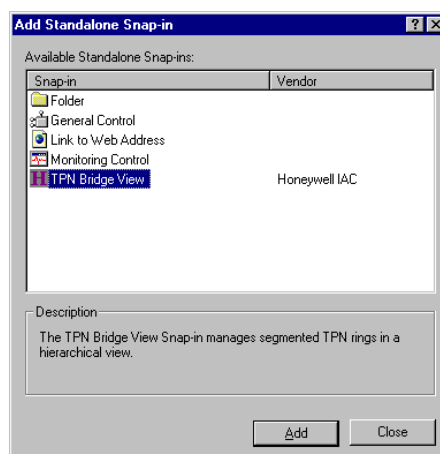
### 4.10. Configuring the Bridge Segments Using MMC

---

- 4 The **Add/Remove Snap-in** page appears. Click **Add**.



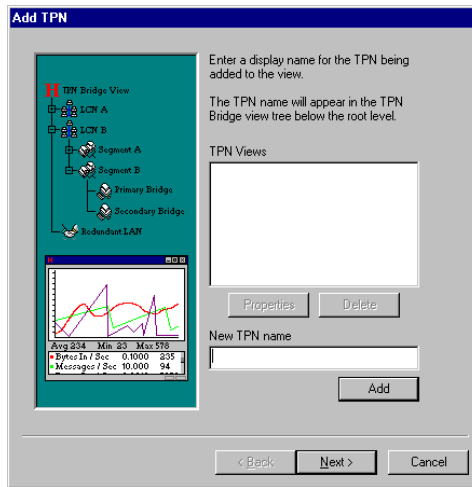
- 5 A list of available Snap-in appears. Select **TPN Bridge View** Snap-in and click **Add**.



**Add a TPN to the Bridge View**

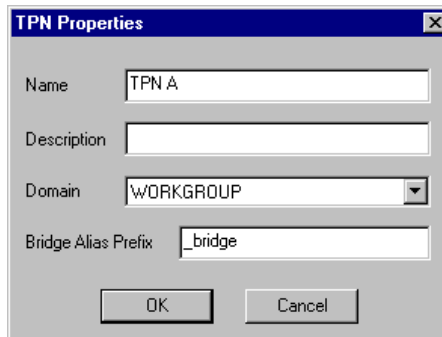
- 6 The **Add TPN** window appears. Enter the **New TPN name** and click **Add**.

**Note:** A **TPN name** is a short title (32-character limit) that will be displayed in the scope pane (tree view) of the MMC console.



- 7 The **TPN Properties** window appears. Enter the desired **Description** and click **OK**.

**Note:** This description is optional and is displayed from the **Bridge View** level of the scope pane in the MMC console.

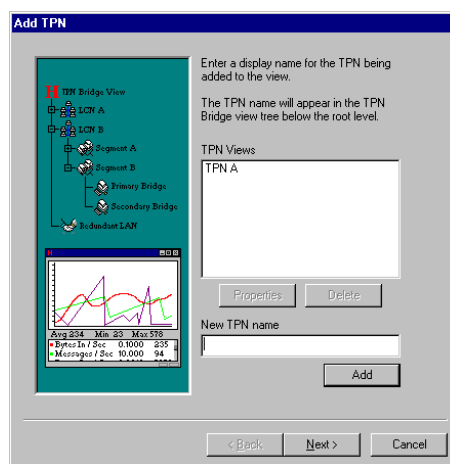


## 4. Implementing the TPN Bridge System

### 4.10. Configuring the Bridge Segments Using MMC

---

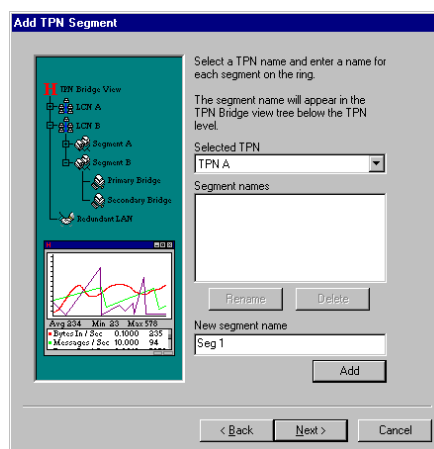
- 8 The **Add TPN** window appears with the **TPN Properties** name appearing in the **TPN Views**. Click **Next >**.



#### Add TPN Segments to TPN

- 9 The **Add TPN Segment** window appears. Enter the **New segment name** (this example uses **Seg 1**) and click **Add**.

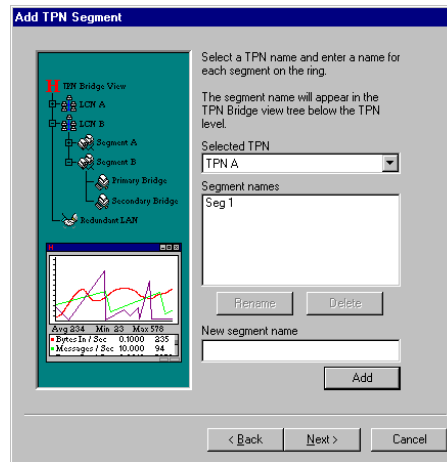
**Note:** The **segment name** is a short title (31-character limit) that describes the TPN segment being bridged (e.g., North Boiler).



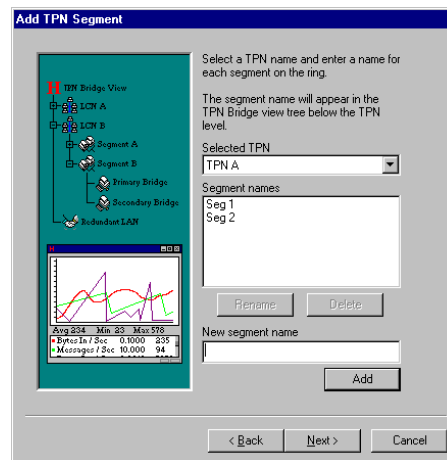
## 4. Implementing the TPN Bridge System

### 4.10. Configuring the Bridge Segments Using MMC

- 10 The name just entered appears in the **Segment names** list.



- 11 Repeat the previous two steps for the remaining segments.



Click **Next >**.

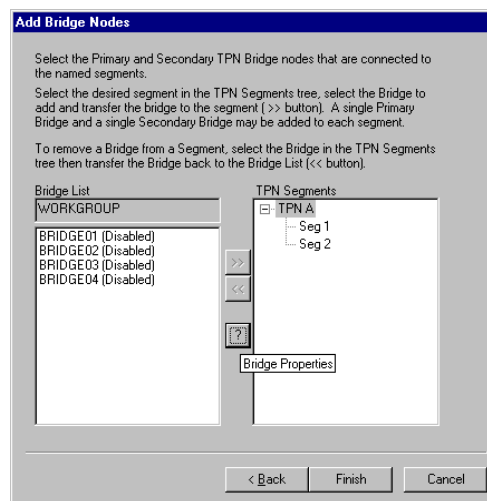
## 4. Implementing the TPN Bridge System

### 4.10. Configuring the Bridge Segments Using MMC

---

#### Transfer Bridge Nodes to TPN

| Step | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12   | <p><b>Note:</b> The wizard will identify all computers on the local area network that are installed as TPN Bridges. The following message appears in a pop-up window:</p> <p><b>Searching WORKGROUP for Bridge...</b></p> <p>The <b>Add Bridge Nodes</b> window appears. Initially, all Bridge entries in the <b>Bridge List</b> are set as (Disabled) nodes. Select each Bridge and set the bridge LCN address using the <b>Bridge Properties</b> page. Primary Bridges will be set to a single odd LCN address. Secondary Bridges will be set to the next consecutive even address (e.g., 95 (Primary), 96 (Secondary)).</p> <p>Select <b>BRIDGE01 (Disabled)</b> from <b>Bridge List</b> and click ? button (<b>Bridge Properties</b>).</p> |



#### ATTENTION

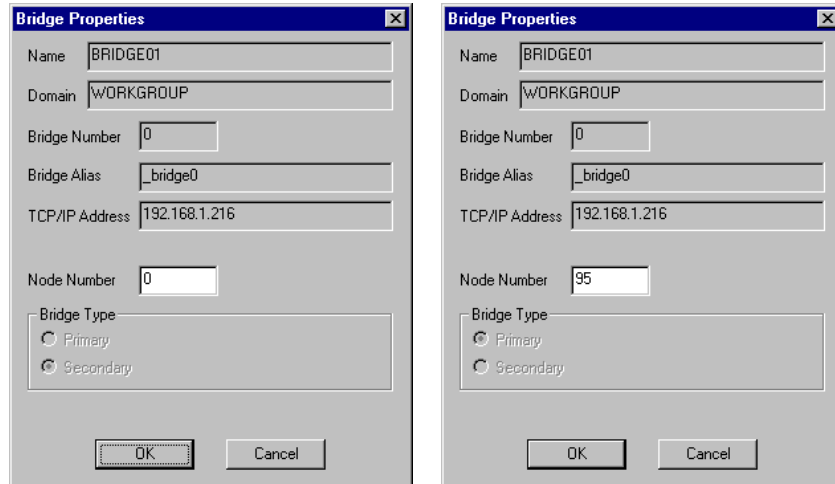
The following steps configure four (4) Bridge nodes as Primary/Secondary pairs at addresses 95 and 96, respectively.

---

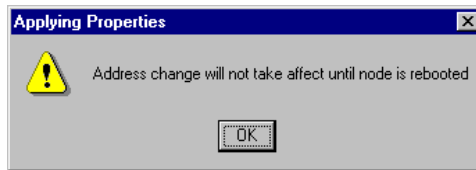
**4. Implementing the TPN Bridge System**  
**4.10. Configuring the Bridge Segments Using MMC**

---

- 13** The **Bridge Properties** page appears. Change the **Node Number** from 0 to 95 (Primary) and click **OK**.



- 14** The **Applying Properties** pop-up screen appears. Click **OK**.



- 15** Repeat previous three steps (12 through 14) for BRIDGE02 (Secondary and **Node Number** 96), BRIDGE03 (Primary and **Node Number** 95), and BRIDGE04 (Secondary and **Node Number** 96) or until all **Bridge Properties** are configured.

When completed, continue to next step.

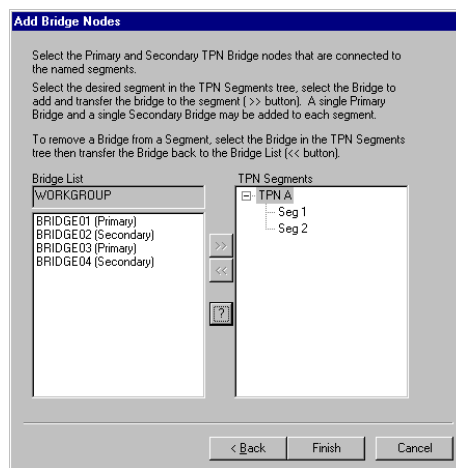
---

## 4. Implementing the TPN Bridge System

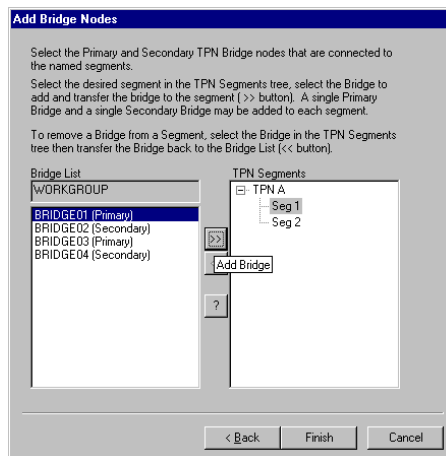
### 4.10. Configuring the Bridge Segments Using MMC

---

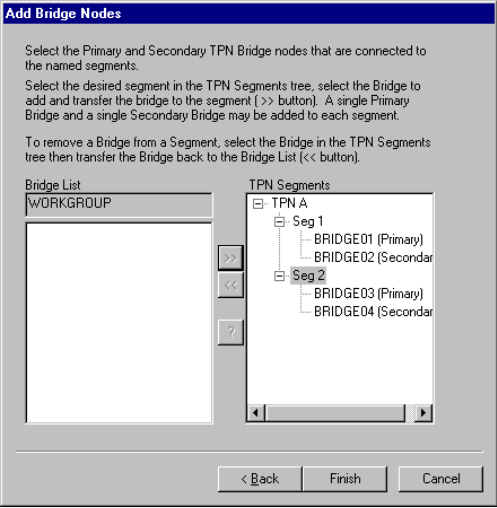
- 16** The **Bridge List** should now be set to the Primary and Secondary configuration as shown below.



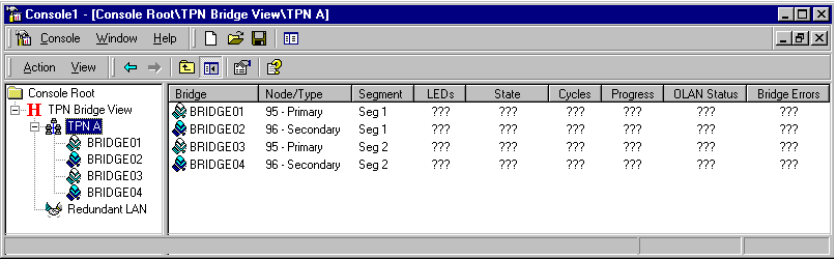
- 17** Select Seg 1 in the right pane. Select **BRIDGE01 (Primary)** in the left pane. Click the >> (Add Bridge) button to transfer the selected Bridge to **Seg 1**.



- 18 Repeat previous step until all Bridges in the **Bridge List** (left pane) are transferred to **Seg 1** and **Seg 2** (right pane) as shown below.
- Click **Finish**. Click **Close**. Click **OK**.



- 19 Click **TPN Bridge View** and double-click **TPN A**. The parent and child window appears. Maximize the child console window, expand the Bridge View tree, and resize the console window so that all Bridge status is displayed. This will be the default view for subsequent sessions.



## 4. Implementing the TPN Bridge System

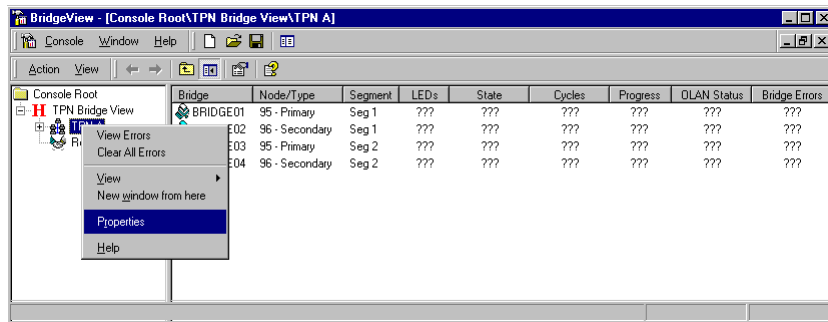
### 4.10. Configuring the Bridge Segments Using MMC

---

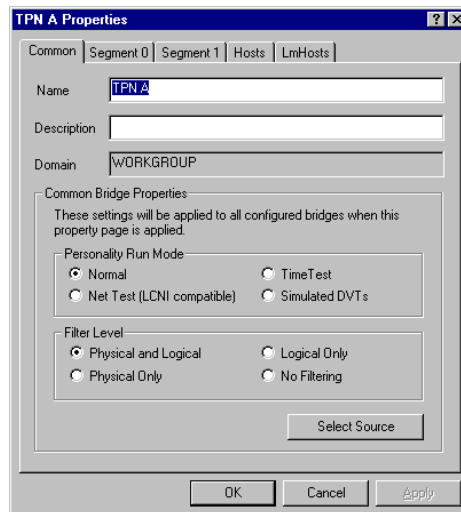
#### Set the TPN Properties

- 20 Right-click **TPN A** in the scope view (left pane).

A pop-up menu appears. Select and click **Properties**.

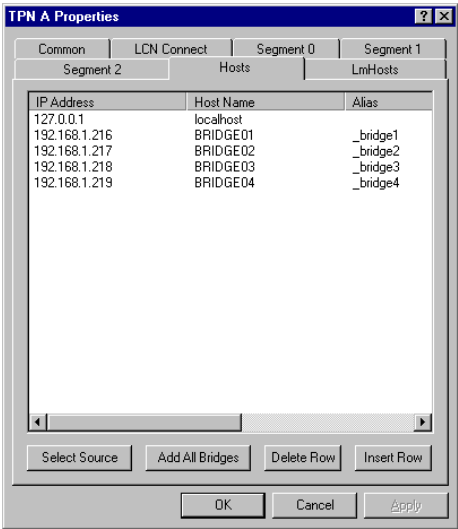


- 21 The **TPN A Properties** page appears. Select and click **Hosts** tab.

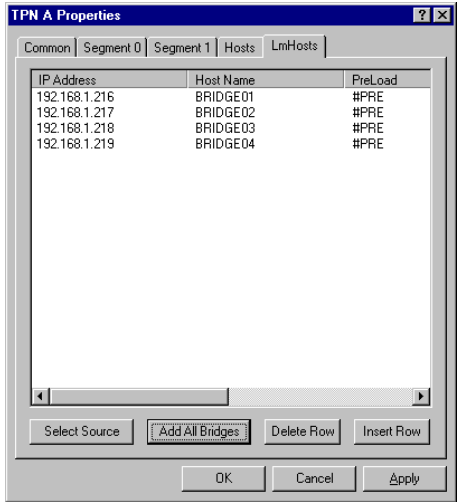


4. Implementing the TPN Bridge System  
4.10. Configuring the Bridge Segments Using MMC

- 22 The **Hosts** window appears. Click **Add All Bridges**. Click **Apply**. The **Host Name** data is transferred for all bridges.



- 23 Select and click **LmHosts** tab. The **LmHosts** window appears. Click **Add All Bridges**. Click **Apply** and **OK**. The **Host Name** data is transferred for all bridges.



## 4. Implementing the TPN Bridge System

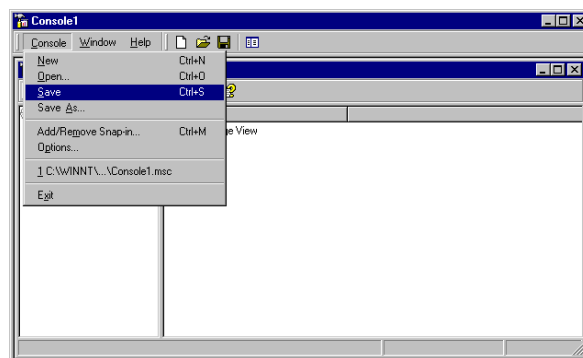
### 4.10. Configuring the Bridge Segments Using MMC

---

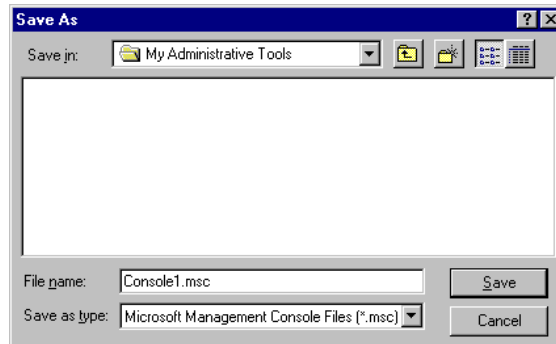
#### *Saving the MMC Bridge View Console File*

- 24 The newly configured Snap-ins must now be saved. Click **Console** on the parent window command bar.

A drop-down menu appears. Select and click **Save** or **Save As....**

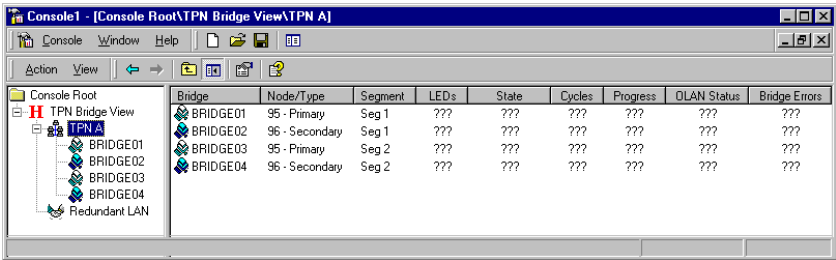


- 25 The **Save As** window appears. Enter a file name for the console file. This example uses **BridgeView**. Click **Save**.



**26** The final **BridgeView** console window should look similar to the screen below.

Exit the MMC program. This completes the configuration of the TPN Bridge nodes.



**ATTENTION**

All TPN Bridge nodes are now configured. You must now reboot all Bridge nodes before all configuration changes will take affect.

**27** The MSC console file may be copied to the other Bridge nodes at this time.

Copy the

C:\WINNT\Profiles\Administrator\Start Menu\Programs\  
My Administrative Tools

folder to the

C:\WINNT\Profiles\Administrator\Start Menu\Programs  
folder of the desired destination computer.

## 4.11 Setting TPN Bridge Time Sync Options

### Overview

Each TPN Bridge segment will have the system TPN Time Master and Segment Time Master bits set or cleared based on TPN clock source configuration.



### ATTENTION

DO NOT perform this procedure until after hot insertion of all TPN Bridges is complete.

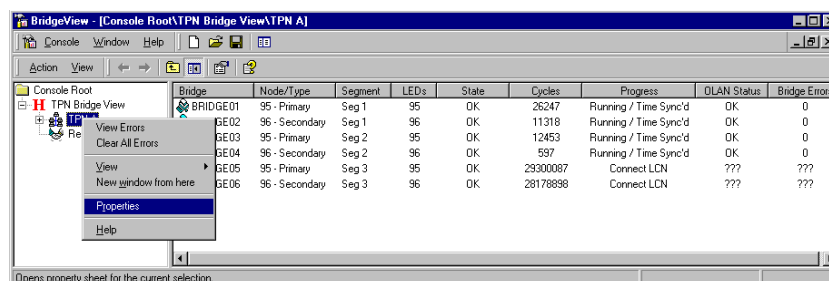
### Setting the time synchronization option

This procedure sets the TPN Bridge time synchronization option.

The TPN Clock Source bit selection can be summarized as follows:

- If the Bridge should synchronize to an existing clock source on this segment (LCN Clock source or GPS source), the TPN Master bit will be set.
- If the Bridge should transmit diagnostic timeframes on this segment (the bridge is acting as the segment clock source or repeater) the Segment Master bit will be set. No other clock source can be present on this segment if this bit is set.
- The clock sources must be on the same segment and not across bridge segments.

| Step | Action                                                                         |
|------|--------------------------------------------------------------------------------|
| 1    | Complete Table 4-11 based on determination of TPN Clock Source settings using. |
| 2    | Open the TPN Bridge View console.                                              |
| 3    | Right-click the TPN level. Select <b>Properties</b> .                          |



4. Implementing the TPN Bridge System  
4.11. Setting TPN Bridge Time Sync Options

| Step | Action                                 |
|------|----------------------------------------|
| 4    | Select the segment page for a segment. |

The image shows the 'TPN A Properties' dialog box with the 'Segment 3' tab selected. The 'Name' field is 'Seg 3'. Under 'Primary Bridge', the 'Name' is 'BRIDGE05', 'IP Address' is '192.168.1.220', and 'Node' is '95'. Under 'Secondary Bridge', the 'Name' is 'BRIDGE06', 'IP Address' is '192.168.1.121', and 'Node' is '96'. The 'Bridge Redundancy' section has 'Use Redundant TPN Bridge' checked and 'LAN heartbeat period' set to '5' seconds. The 'LCN Clock Source' section has 'System Time Master' unchecked and 'Segment Time Master' unchecked. A 'Select Source' button is present. At the bottom are 'OK', 'Cancel', and 'Apply' buttons.

|   |                                                                                     |                   |
|---|-------------------------------------------------------------------------------------|-------------------|
| 5 | Set the <b>LCN Clock Source</b> based on Table 4-11 in step 1. Click <b>Apply</b> . | TPN Clock Sources |
|---|-------------------------------------------------------------------------------------|-------------------|

This image is identical to the previous one, but the 'Segment Time Master' checkbox in the 'LCN Clock Source' section is now checked.

|   |                                           |
|---|-------------------------------------------|
| 6 | Repeat previous 2 steps for all segments. |
|---|-------------------------------------------|

#### 4. Implementing the TPN Bridge System

##### 4.11. Setting TPN Bridge Time Sync Options

---

| Step | Action                                                                                                                                                                                         |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | If the TPN Bridge nodes are connected to a Universal Station, use the TPN CLOKSTAT schematic to verify that all TPN nodes are in the appropriate time sync state (Master, Slave, or Listener). |

#### Turning off the time synchronization option

Use the following procedure to turn off the time sync option.

| Step | Action                                                                                                                                                                                                                |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Open the TPN Bridge View console.                                                                                                                                                                                     |
| 2    | Right-click the TPN level. Select <b>Properties</b> .                                                                                                                                                                 |
| 3    | Select the segment page for a segment.                                                                                                                                                                                |
| 4    | Review the System Time Master and Segment Time Master options selected in the <b>LCN Clock Source</b> area. Record the time sync values in Table 4-11 for re-application when time synchronization is turned back on. |
| 5    | Clear all checkboxes in the <b>LCN Clock Source</b> area and click OK.                                                                                                                                                |
| 6    | Repeat previous 3 steps for all segments.                                                                                                                                                                             |

#### Recording TPN clock sources

Use the following procedure to record the time sync values.

**Table 4-11 TPN Clock Sources**

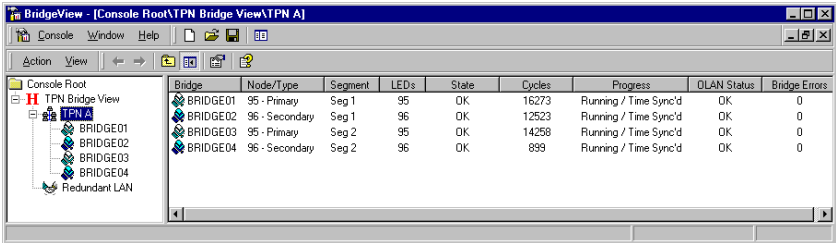
| Segment | Segment Name | TPN Master | Segment Master |
|---------|--------------|------------|----------------|
| 1       |              |            |                |
| 2       |              |            |                |
| 3       |              |            |                |
| 4       |              |            |                |
| 5       |              |            |                |
| 6       |              |            |                |

## 4.12 Adding a New Segment to an Existing Bridge System

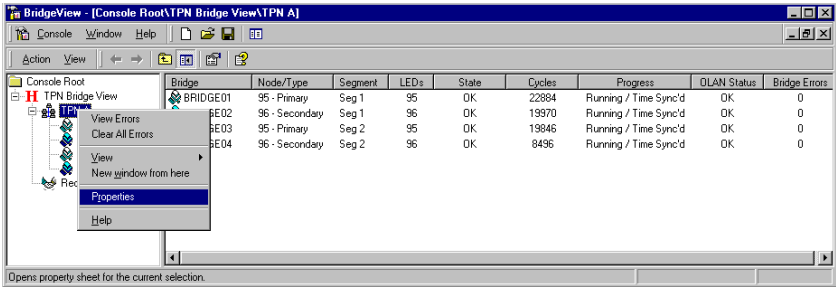
### Overview

The following procedure is used to configure new segments that are added to a running, Bridge system.

| Step | Action                                                                                                                                                                      |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Initialize new Bridges with Computer Name and network IP address by performing procedures in Section 4.9, "TPN Bridge and NT Configuration." Shutdown only the new bridges. |
| 2    | Connect the new bridges to the existing LAN switch box.<br>DO NOT CONNECT to the LCN!                                                                                       |
| 3    | Using an existing Bridge, close all MMC console sessions. Bring up a single MMC session using the current MSC file.                                                         |



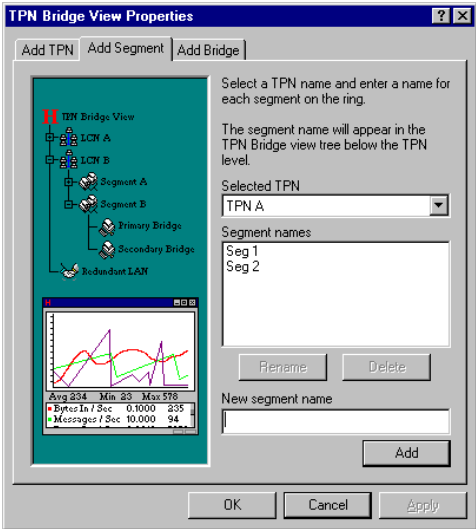
- 4 Select the **Redundant LAN** icon and ensure that Cable A is the active LAN cable. If not, force the LAN cable to A.
- 5 Right-click **TPN Bridge View** and select **Properties**.



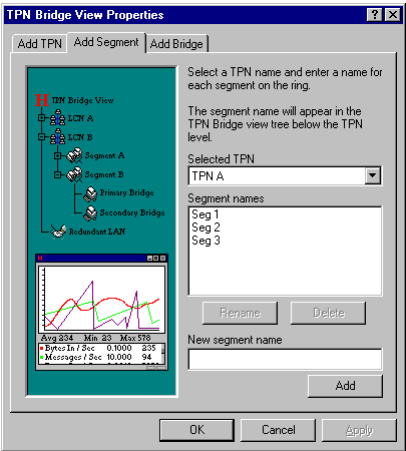
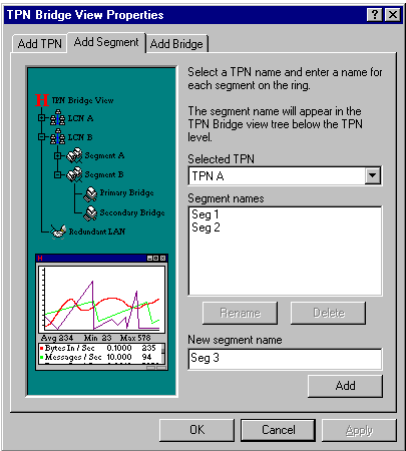
- 6 The **TPN Bridge View Properties** page appears. Select the **Add Segment** tab.

4. Implementing the TPN Bridge System  
4.12. Adding a New Segment to an Existing Bridge System

Step Action



- 7 Add a new segment. This example adds **Seg 3**. Enter the segment name and click **Add** then click **Apply**.



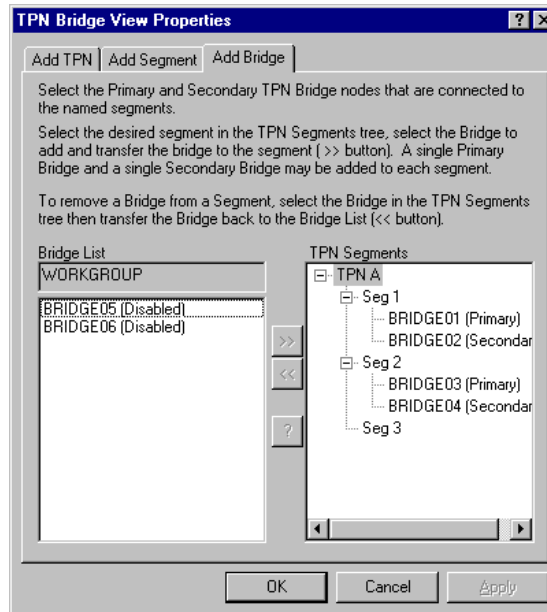
- 8 Select **Add Bridge** tab. The Bridge View snap-in will discover the new bridges and add the Bridges to the **Bridge List**. The current TPN configured displays in the **TPN Segments** tree.

## 4. Implementing the TPN Bridge System

### 4.12. Adding a New Segment to an Existing Bridge System

#### Step

#### Action



- 9 Select the Bridge node to add as Primary TPN Bridge in the **Bridge List** and click ? button. A **Bridge Properties** dialog appears.
- 10 Set the new Bridges' address to the correct Primary LCN address. This example uses address 95. Click **OK**.

## 4. Implementing the TPN Bridge System

### 4.12. Adding a New Segment to an Existing Bridge System

---

#### Step

#### Action

Bridge Properties

Name: BRIDGE05

Domain: WORKGROUP

Bridge Number: 0

Bridge Alias: \_bridge0

TCP/IP Address: 192.168.1.220

Node Number: 1

Bridge Type:  
☐ Primary  
☒ Secondary

OK Cancel

Bridge Properties

Name: BRIDGE05

Domain: WORKGROUP

Bridge Number: 0

Bridge Alias: \_bridge0

TCP/IP Address: 192.168.1.220

Node Number: 96

Bridge Type:  
☒ Primary  
☐ Secondary

OK Cancel

- 11** Repeat previous 2 steps to set the LCN address for the Secondary TPN Bridge. Results are shown below. This example uses address 96.

Bridge Properties

Name: BRIDGE06

Domain: WORKGROUP

Bridge Number: 0

Bridge Alias: \_bridge0

TCP/IP Address: 192.168.1.121

Node Number: 1

Bridge Type:  
☐ Primary  
☒ Secondary

OK Cancel

Bridge Properties

Name: BRIDGE06

Domain: WORKGROUP

Bridge Number: 0

Bridge Alias: \_bridge0

TCP/IP Address: 192.168.1.121

Node Number: 96

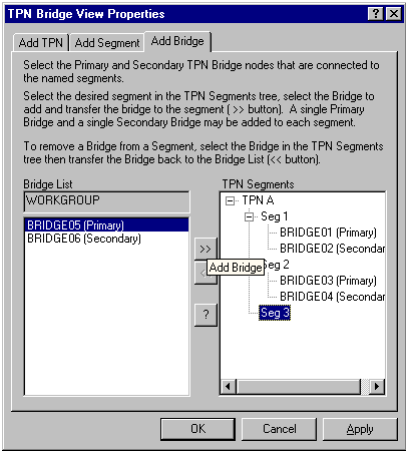
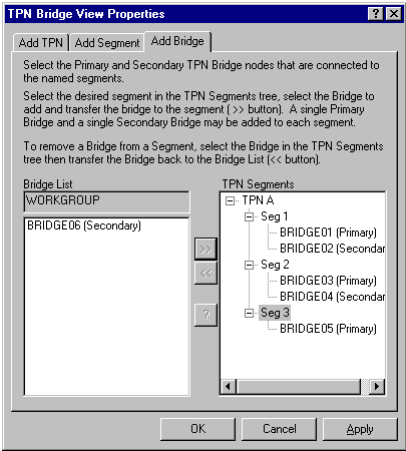
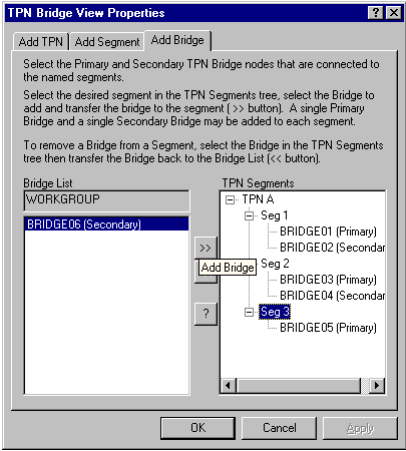
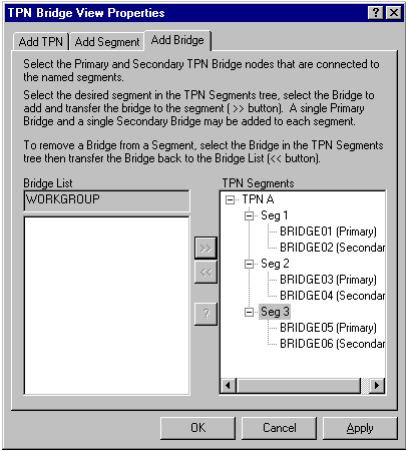
Bridge Type:  
☐ Primary  
☒ Secondary

OK Cancel

- 12** Select the Primary Bridge, the new segment, and then click the >> (Add Bridge) button. Click the **OK** button in the TPN Bridge View Properties dialog box. The selected bridge will be transferred to the selected segment. Click **Apply**. The workgroup will be searched again to refresh the bridge list.
-

4. Implementing the TPN Bridge System

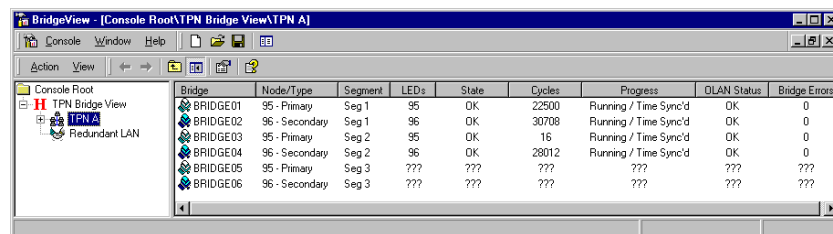
4.12. Adding a New Segment to an Existing Bridge System

| Step | Action                                                                                                                                                                                                   |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <div><div></div><div></div></div>     |
| 13   | Repeat previous step for the Secondary Bridge.                                                                                                                                                           |
|      | <div><div></div><div></div></div> |
| 14   | Click <b>OK</b> in the <b>TPN Bridge View Properties</b> dialog box.                                                                                                                                     |
| 15   | The new Bridges for Segment 3 now appear in the MMC scope pane.                                                                                                                                          |

## 4. Implementing the TPN Bridge System

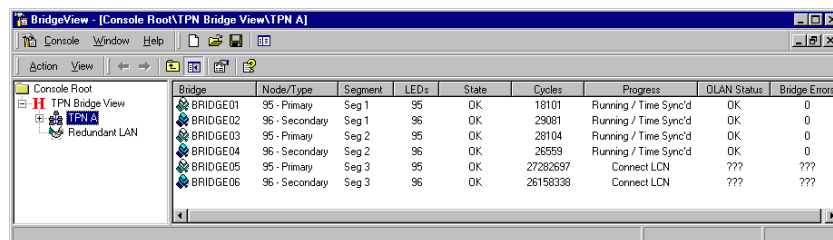
### 4.12. Adding a New Segment to an Existing Bridge System

| Step | Action |
|------|--------|
|------|--------|



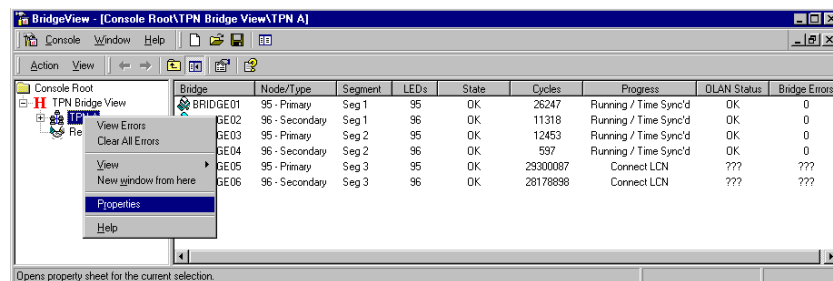
| Bridge   | Node/Type      | Segment | LEDs | State | Cycles | Progress              | OLAN Status | Bridge Errors |
|----------|----------------|---------|------|-------|--------|-----------------------|-------------|---------------|
| BRIDGE01 | 95 - Primary   | Seg 1   | 95   | OK    | 22500  | Running / Time Sync'd | OK          | 0             |
| BRIDGE02 | 96 - Secondary | Seg 1   | 96   | OK    | 30708  | Running / Time Sync'd | OK          | 0             |
| BRIDGE03 | 95 - Primary   | Seg 2   | 95   | OK    | 16     | Running / Time Sync'd | OK          | 0             |
| BRIDGE04 | 96 - Secondary | Seg 2   | 96   | OK    | 28012  | Running / Time Sync'd | OK          | 0             |
| BRIDGE05 | 95 - Primary   | Seg 3   | ???  | ???   | ???    | ???                   | ???         | ???           |
| BRIDGE06 | 96 - Secondary | Seg 3   | ???  | ???   | ???    | ???                   | ???         | ???           |

- 16 Save the MSC file and **Close the MMC session.**
- 17 Shutdown and restart the new bridges. When reboot is complete, continue to next step.
- 18 Restart the MMC session. The new bridges will be displayed with their status updating.



| Bridge   | Node/Type      | Segment | LEDs | State | Cycles   | Progress              | OLAN Status | Bridge Errors |
|----------|----------------|---------|------|-------|----------|-----------------------|-------------|---------------|
| BRIDGE01 | 95 - Primary   | Seg 1   | 95   | OK    | 18101    | Running / Time Sync'd | OK          | 0             |
| BRIDGE02 | 96 - Secondary | Seg 1   | 96   | OK    | 29081    | Running / Time Sync'd | OK          | 0             |
| BRIDGE03 | 95 - Primary   | Seg 2   | 95   | OK    | 28104    | Running / Time Sync'd | OK          | 0             |
| BRIDGE04 | 96 - Secondary | Seg 2   | 96   | OK    | 26559    | Running / Time Sync'd | OK          | 0             |
| BRIDGE05 | 95 - Primary   | Seg 3   | 95   | OK    | 27282697 | Connect LCN           | ???         | ???           |
| BRIDGE06 | 96 - Secondary | Seg 3   | 96   | OK    | 26158338 | Connect LCN           | ???         | ???           |

- 19 Right-click the TPN level and select **Properties.**



| Bridge   | Node/Type      | Segment | LEDs | State | Cycles   | Progress              | OLAN Status | Bridge Errors |
|----------|----------------|---------|------|-------|----------|-----------------------|-------------|---------------|
| BRIDGE01 | 95 - Primary   | Seg 1   | 95   | OK    | 26247    | Running / Time Sync'd | OK          | 0             |
| GE02     | 96 - Secondary | Seg 1   | 96   | OK    | 11318    | Running / Time Sync'd | OK          | 0             |
| GE03     | 95 - Primary   | Seg 2   | 95   | OK    | 12453    | Running / Time Sync'd | OK          | 0             |
| GE04     | 96 - Secondary | Seg 2   | 96   | OK    | 597      | Running / Time Sync'd | OK          | 0             |
| GE05     | 95 - Primary   | Seg 3   | 95   | OK    | 29300087 | Connect LCN           | ???         | ???           |
| GE06     | 96 - Secondary | Seg 3   | 96   | OK    | 28178898 | Connect LCN           | ???         | ???           |

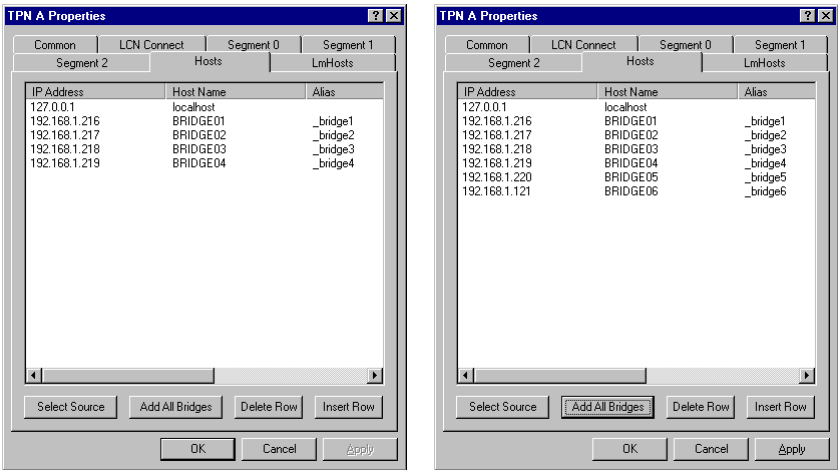
- 20 Select Hosts page and Add All Bridges. Click Apply.

4. Implementing the TPN Bridge System

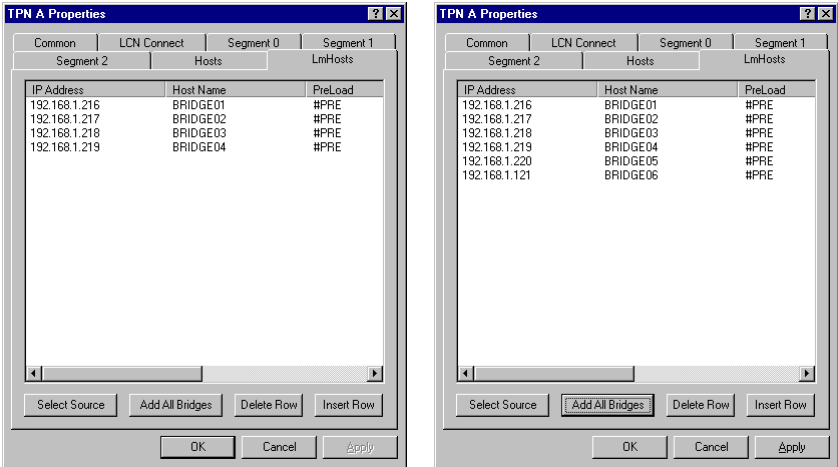
4.12. Adding a New Segment to an Existing Bridge System

Step

Action



21 Select LmHosts page and Add All Bridges. Click Apply.



22 Close the MMC session. The configuration of all TPN Bridges is now complete.

23 Shutdown and reboot the new bridges.

## **4. Implementing the TPN Bridge System**

### **4.12. Adding a New Segment to an Existing Bridge System**

---

## 5. TPN Bridge Hot Insertion

### 5.1 Overview

#### Summary

When dividing a TPN (LCN) into multiple segments, bridges are inserted one redundant pair at a time rather than all at once. Prior to doing anything, the LCN must be running error free. There must be no crossed cables. All configuration rules must be adhered to. Redundant LCN nodes must be on the same LCN segment. Each LCN segment must have a Human Interface (US, GUS) such that if the segment is isolated, there is a view to the process.

After each redundant pair is inserted, proper operation of the bridge pair and the LCN network is confirmed. Only after the proper operation of each bridge pair and the LCN network is confirmed, is the next pair inserted. In addition, between insertions of bridge pairs, enough time must elapse to enable any transient LCN errors to clear. The stepwise insertion means that an LCN that is intended to be divided into 4 segments, for example, goes up through the steps of 2 segments, 3 segments, and finally 4 segments. It does NOT go directly from the un-segmented condition to 4 segments.

The LCN A & B cable system is designed to have each “coax cable segment’s shield” tied to logic ground at only one spot. The existing cabling system needs to be inspected to locate the current grounding spot, and to locate the NEW grounding spot. When the cable is broken to create segments, a cable shield ground point must be added. This is done with grounded terminators (51305492-100) as shown later in this procedure.



#### CAUTION

Removing a TPN Bridge node and connecting the LCN segments back together may violate LCN configuration rules. The LCN must be shut down and the LCN configuration re-engineered whenever a Bridge is removed and the segments are to be rejoined.

---

## 5. TPN Bridge Hot Insertion

### 5.1. Overview

---

#### Hot insertion concepts

All Hot Insertion scenarios fall into the following two categories:

- The First Segmentation of a TPN (LCN) - Use this procedure if installing TPN Bridge on an unbridged LCN.
- Connecting Additional Segments - Use this procedure if adding a new bridge segment to an existing bridged LCN.

The hot insertion process consists of the following sequence:

- Configuring the NCF for the TPN Bridge.
- Initializing the TPN Bridge node.
- Configuring the TPN Bridge node software.
- Inserting Hot Insertion Switchbox Tool.
- Offline Testing of Bridge operation.
- Checking the Bridge Node Communications on the LCN.
- Breaking the LCN.
- Removal of the Hot Insertion Switchbox Tool.
- Setting the TPN Bridge Time Sync Options.

## 5.2 Special Considerations

### TPN Bridge shut-down

---



#### CAUTION

Shutting down (or the crash) of a TPN Bridge node may hang the TPN Bridge Config/Status Snap-in for extended periods. When shutting down a TPN Bridge, close all MMC sessions first to avoid this lengthy timeout. The TPN Bridge Config/Status snap-in is primarily intended for use in configuration and installation of healthy systems. Use the local TPN Bridge Viewer to diagnose system problems. Recovery of MMC status updates may be possible by closing the MMC session, waiting 2 minutes per TPN Bridge in the system, then restarting the MMC console session

---

### TPN Bridge node failover

---



#### CAUTION

Messages may be lost during a TPN Bridge Node failover. The TPN recovers from these lost messages with the following anomalies:

- There may be short gaps in data trends of displays during the failure or recovery (up to 15 seconds).
  - Data Access between nodes may be delayed in some cases, but recover normally after a few seconds.
  - File operations between HMs, consoles and HMs, or across NGs may see a delay of 5 minutes.
  - Node loads in progress during a bridge failure or recovery may fail or be significantly delayed.
-

## 5.3 About TPN Bridge Cables

### Cable breaks



#### CAUTION

LCN cable breaks in both cables in any segment (breaking cable A and B at the same time, regardless of location) should be avoided as they may cause node isolation. Inhibit cable swapping while performing cable maintenance to reduce the risk of failing the primary TPN Bridge nodes. Maintenance and troubleshooting activities requiring cable breaks or removal should be confined to a single cable on a single segment

### Cable error interpretation

The TPN Bridge segments an LCN into physically separate LCNs. Because of this, error indications on bridged LCNs are different than on traditional LCNs. Each TPN Bridge segment has its own token. This means that *media deads* caused by cable problems on one bridge segment are not reported by nodes on another bridge segment.

Cable problems on a bridged LCN will be reported by suspecting the problematic cable the same way that traditional LCNs do, with the exception that problems on non-LCN cable master segments will not cause the cable to suspect immediately. Three (3) automatic swaps to the affected cable will be performed before the cable will go suspect.

Cable errors will be confined to a TPN Bridge segment. Cable Suspect Status will still be displayed by all System Status displays on the network. However, only those nodes on the TPN Bridge segment that is experiencing errors will report cable errors in the LCN cable diagnostic displays (\$LNMENU).

Because media deads are not reported to the cable master node (lowest numbered US), cable breaks on remote (non-cable master) segments will not increment the cable master media dead count. Nodes on the broken cable may or may not log media deads or other errors. A bridged LCN with suspect cables and no cable errors on any nodes will indicate a cable break on a non-cable master segment. Since this condition has traditionally been reported by only the cable master reporting the error, this new symptom will aid in eliminating bridge segments as suspects.

The HM will not contain journal information about cable errors when the cable is broken on a segment that does not contain the cable master.

### Cable error symptoms and causes

The following chart summarizes symptoms and causes of cable errors. Cable errors include media deads, alone-in- rings, stolen tokens and IOCB errors including FCS, framing, oversize frame, buffer overruns, and manchester coding errors.

**Table 5-1 Cable Error Symptoms and Causes**

| Cable Symptom                                                         | Description of Possible Cause                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LCN Cable Suspect<br>Cable master segment nodes have cable errors     | A cable problem exists on the cable master segment. Troubleshoot the cable master bridge segment as though it is a traditional LCN.                                                                                                                                                                                                                                                                                                                              |
| LCN Cable Suspect<br>Non-cable master segment nodes have cable errors | A cable problem exists on the segment logging the errors. Troubleshoot the non-cable master segment as though it is a traditional LCN.                                                                                                                                                                                                                                                                                                                           |
| LCN Cable Suspect<br>No errors detected                               | This symptom indicates an error on a non-cable master segment. It is typically a complete cable break on the non-cable master segment. This problem would traditionally be logged by the cable master node only and an entire LCN cable examination would be required. This is a rare condition on a heavily loaded LCN. Testing has shown that generally nodes on the affected bridge segment will log cable errors.                                            |
| LCN Cable Suspect<br>A TPN Bridge has failed                          | A primary TPN Bridge that cannot communicate with the secondary bridge due to LCN cable problems will be stunned by the secondary TPN Bridge node. The secondary TPN Bridge node will remain running and communicate if cable conditions allow. The cable error is located on the segment that has stunned the primary TPN Bridge. Repair the LCN cable and reset the primary TPN Bridge TPNB (LCNP reset). If cable errors persist, the primary may be stunned. |

## 5. TPN Bridge Hot Insertion

### 5.3. About TPN Bridge Cables

---

**Table 5-1      Cable Error Symptoms and Causes**

| Cable Symptom                         | Description of Possible Cause                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A TPN Bridge is logging LCN Rx Errors | A TPN Bridge will log ALL LCN errors that have been identified on its local segment. A bridge does not log errors occurring on remote bridge segments. Rx Errors are no longer logged when LCN nodes load. Clear TPN Bridge Rx Errors by double clicking Rx Errors in the TPN Bridge View LCN Status display, or by resetting Bridge Error Counters using the TPN Bridge Config/Status Snap-in. Rx Errors include FCS, framing, oversize frame, buffer overruns, manchester decoder, media dead, alone-in-ring and stolen token errors. The Rx Error count is the combined count of both cable A and cable B errors. Detail of these errors can be viewed by right clicking the TPN Bridge Viewer – Bridge Status - Rx Errors edit box and selecting Display Error Details... in the context menu or by selecting the Detailed Status menu option using the TPN Bridge Config/Status Snap-in. There may be additional information logged to the Windows NT Event Log that could be useful in troubleshooting LCN cable problems. |

## 5.4 First Segmentation of a TPN (LCN)

You will perform this procedure on the initial (or first) installation of a TPN Bridge. For systems with TPN Bridges already installed, see Section 5.5, "Creating Additional Segments."



### ATTENTION

For these procedures, it is assumed that the TPN Bridge is being inserted into a system of greater than 64 nodes and it is assumed that the "Keyfile Option" has been installed. If the system is less than 64 node and the "Keyfile Option" has not been installed use two consecutive LCN Node addresses up to 64 for the Bridge LCN address instead of address 95 and 96. The lower LCN address must be an odd node number.

If the existing LCN is configured for 64 nodes or less and the Bridge is going to be used to allow expansion to greater than 64 nodes the "Keyfile Option" must be purchased and installed.

---

### Configuring the NCF for the TPN Bridge

The TPN Bridge nodes must be added to the system NCF. Refer to Section 4.8, "Installing and Configuring the TPN System Software".

### Configuring NT on the TPN Bridge nodes

Each TPN Bridge must be initialized. Refer to the procedure in Section 4.9, "TPN Bridge and NT Configuration."



### ATTENTION

At this point the Bridge hardware must have been installed in cabinets, or desktop, and set up according to the Configuration Rules for TPN Bridge (see *TPN Bridge Planning, Installation and Service*).

---



### CAUTION

The four Bridge PCs for the first segmentation are configured and **disconnected** from the LCN and **connected** to the Redundant Ethernet Bridge LAN. All LCN MAUs should be labeled to ensure no mix-ups.

All cables should be run to the Bridges and all necessary Tees should be inserted.

**The system Time Sync must be disabled. For additional information, refer to the "Turning off the time synchronization option" in Section 4.11, "Setting TPN Bridge Time Sync Options."**

---

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)



#### CAUTION

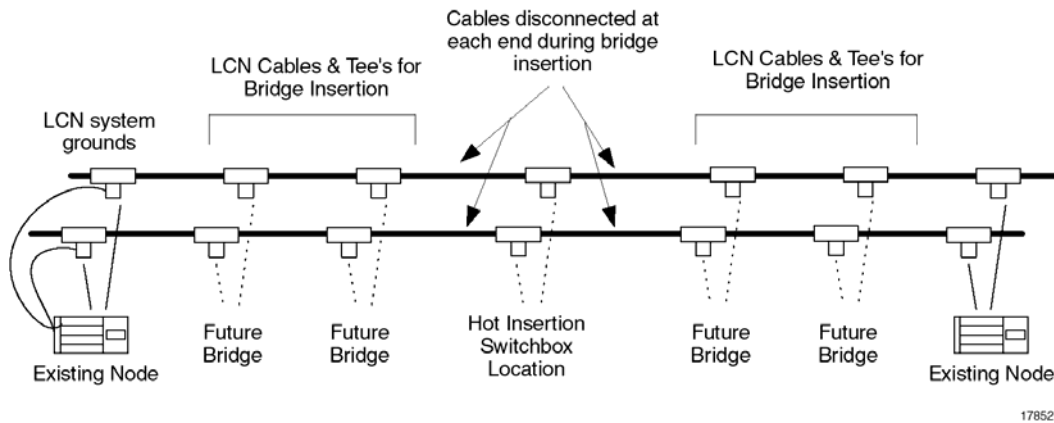
Do not connect the TPN Bridge MAUs to the LCN until directed to do so.

#### Configuring the TPN Bridge node software

All TPN Bridge nodes must be configured to reflect the new TPN topology. Refer to Section 4.10, "Configuring the Bridge Segments Using MMC."

#### Preparing the LCN for TPN Bridge insertion

The TPN cable must be modified to provide taps for insertion of the Hot Insertion Switchbox Tool. The following illustration shows the location of the LCN cables, the Tee's and the Switchbox Tool insertion point.



**Figure 5-1 Location of Switch Box, Cables and Tees**

The following section outlines a procedure for safely installing cables, Tees and Hot Insertion Switchbox tool.

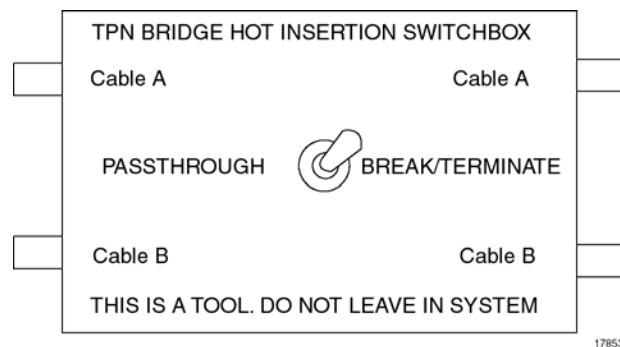
### Inserting hot insertion switchbox tool

The TPN Bridge Hot Insertion Switchbox is a tool used to break and terminate both LCN cables at the same time to prevent node Isolation and Media Dead counts. The switchbox also allows the reconnection of the LCN if a failure occurs during the insertion of the Bridges.



#### ATTENTION

The TPN Bridge nodes must not be inserted if there are any LCN cable problems. Verify there are no cross-cable errors using the schematic \$LNMENU on a TDC Universal Station or GUS. If cable errors exist, make repairs before continuing.



**Figure 5-2 Hot Insertion Switchbox Tool**

The following procedure installs the Hot Insertion Switchbox tool. It outlines the procedure for inserting cable segments and T-connectors for installing the TPN Bridges.

| Step | Action                                                             |
|------|--------------------------------------------------------------------|
| 1    | Bring up the schematic \$LNMENU on a TDC Universal Station or GUS. |
| 2    | Choose RESET TIMES.                                                |
| 3    | Choose SELECT TO UPDATE DISPLAY.                                   |
| 5    | Choose RESET ALL COUNTERS.                                         |
| 6    | Choose MAIN MENU.                                                  |

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)

---

| Step | Action                                                                                                                                                       |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | Choose NET MEDIA DEADS.<br><br>Observe that the Media Dead counts on all the nodes for several minutes to determine that there are no cable problems.        |
| 8    | Locate the LCN break point and label the cables.                                                                                                             |
| 9    | Bring up the SYSTEM STATUS display on a US or GUS.                                                                                                           |
| 10   | Force the LCN to Cable B.                                                                                                                                    |
| 11   | Bring up the schematic \$LNMENU on a TDC Universal Station or GUS.                                                                                           |
| 12   | Choose CABLE SWAP INH.                                                                                                                                       |
| 13   | Choose INHIBIT SWAPPING.                                                                                                                                     |
| 14   | Make sure the Hot Insertion Switchbox tool is in the <b>PASSTHROUGH</b> position.                                                                            |
| 15   | At the LCN break point, break the A cable of the LCN and connect both ends to the BNC connectors labeled <b>Cable A</b> on the Hot Insertion Switchbox tool. |
| 16   | At the INHIBIT SWAPPING display, choose ENABLE SWAPPING.                                                                                                     |
| 17   | Bring up the SYSTEM STATUS display.                                                                                                                          |
| 18   | Wait until both cable A and cable B are OK.                                                                                                                  |
| 19   | Force the LCN to Cable A.                                                                                                                                    |
| 20   | Bring up the schematic \$LNMENU on a TDC Universal Station or GUS.                                                                                           |
| 21   | Choose CABLE SWAP INH.                                                                                                                                       |
| 22   | Choose INHIBIT SWAPPING.                                                                                                                                     |
| 23   | At the LCN break point, break the B cable of the LCN and connect both ends to the BNC connectors labeled <b>Cable B</b> on the Hot Insertion Switchbox tool. |
| 24   | At the INHIBIT SWAPPING display, choose ENABLE SWAPPING.                                                                                                     |
| 25   | Bring up the SYSTEM STATUS display.                                                                                                                          |
| 26   | Wait until both cable A and cable B are OK.                                                                                                                  |

---

### Offline testing of Bridge operation

The next process is to bring up the TPN Bridge nodes so that they are communicating with one another over the Ethernet and then connect them to a private LCN to verify LCN interface functionality.

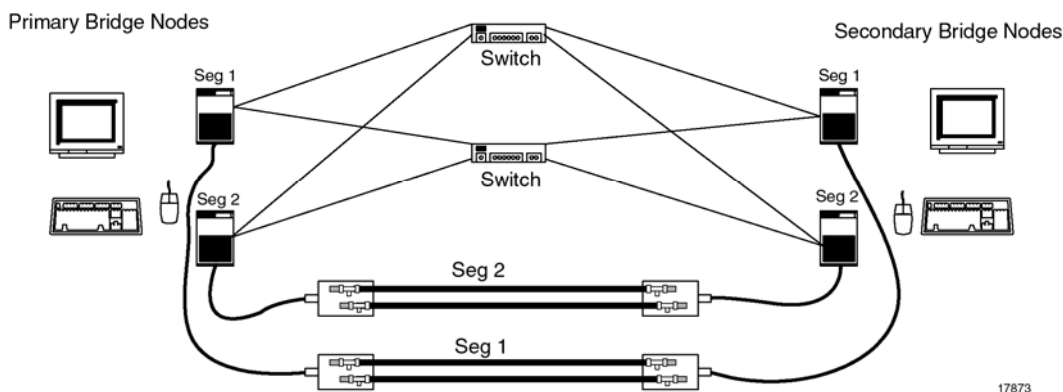


Figure 5-3 Private LCN Setup

| Step | Action                                                                                                                                                                                                                   |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Using the <b>TPN Bridge Configuration/Status Snap-In</b> make sure that Time Sync has been turned off for all segments.<br><br>Refer to Section 4.11, "Setting TPN Bridge Time Sync Options" for additional information. |
| 2    | Connect each Bridge Primary to its Bridge Secondary with a private LCN.                                                                                                                                                  |
| 3    | Reset each Primary TPN Bridge TPNB.<br>If the Bridge fails reset the Primary TPN Bridge TPNB again.<br><br><b>Note:</b> This procedure is establishing a two-node LCN ring.                                              |
| 4    | Using the <b>TPN Bridge Configuration/Status Snap-In</b> and selecting <b>TPN Properties</b> make sure that all the segments are in the <b>Connected to LCN</b> mode.                                                    |

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)

#### Step

#### Action

- 5 Bring up the **Bridge Status**, **TDC Status** and **Node Status** displays, verify the expected values using the "Private LCN Health Check" chart below.

Private LCN Health Check

| Value Name                                                        | Primary Expected Value          | Secondary Expected Value                 |
|-------------------------------------------------------------------|---------------------------------|------------------------------------------|
| <b>TPN Bridge Viewer - Bridge Status</b>                          |                                 |                                          |
| Cycle Counter                                                     | Changing                        | Changing                                 |
| Load Progress                                                     | 254                             | 254                                      |
| OLAN OK                                                           | 3                               | 3                                        |
| Buddy OK                                                          | 1                               | 1                                        |
| Seconds Since Msg Recived (NT)                                    | Cycles 0 – 10                   | Cycles 0 – 1                             |
| Seconds Since Msg Recived (LCNI)                                  | Cycles 0 – 10                   | Cycles 0 – 1                             |
| <b>TPN Bridge Viewer - Node Status</b>                            |                                 |                                          |
| Node [primary node number]                                        | ME-BRIDGE prim                  | BUDDY prim                               |
| Node [secondary node number]                                      | BUDDY sec                       | ME-BRIDGE sec                            |
| <b>TPN Bridge Viewer – TDC Status</b>                             |                                 |                                          |
| System Status – Physical Dirty                                    | Set                             | Set                                      |
| System Status – Looping                                           | Set                             | Set                                      |
| System Status – Disconnected                                      | Cleared                         | Cleared                                  |
| Notice Status – Multi Alone in Ring                               | Cleared                         | May be Set                               |
| Notice Status – All other bits                                    | Cleared                         | Cleared                                  |
| Time Mastership – All bits                                        | Cleared                         | Cleared                                  |
| <b>TPN Bridge Viewer – WS Counters</b>                            |                                 |                                          |
| <b>Note: Clear all counters before proceeding with this check</b> |                                 |                                          |
| Configured Primary                                                | 1                               | 0                                        |
| Running Primary                                                   | 1                               | 0                                        |
| Tx Total                                                          | Incrementing 2 per second       | Incrementing at least 1 every 10 seconds |
| Tx Failures                                                       | 0                               | 0                                        |
| Tx LCN Msg                                                        | 0                               | 0                                        |
| Tx Buddy Msg                                                      | Incrementing 2 per second       | Incrementing at least 1 every 10 seconds |
| Tx PLAT Msg                                                       | Incrementing 1 every 10 seconds | May increment                            |
| Rx Total                                                          | Incrementing 1 every 10 seconds | Incrementing 2 per second                |
| Rx Failures                                                       | 0                               | 0                                        |

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)

| Step | Action                    |                                          |
|------|---------------------------|------------------------------------------|
|      | Rx Lost                   | 0                                        |
|      | Rx Out of Order           | 0                                        |
|      | Rx Duplicated             | 0                                        |
|      | Rx LCN Msg                | 0                                        |
|      | Rx LCN Lost               | 0                                        |
|      | Rx LCN Out of Order       | 0                                        |
|      | Rx Buddy Msg              | Incrementing at least 1 every 10 seconds |
|      | Rx Buddy Msg Lost         | 0                                        |
|      | Rx Buddy Msg Out of Order | 0                                        |
|      | Rx PLAT Msg               | May increment                            |
|      | Rx PLAT Msg Lost          | 0                                        |
|      | Rx PLAT Msg Out of Order  | 0                                        |

Refer to Table 4-8 for additional information.

- 6 Call-up MMC. Right-click TPN Properties for LCN Connection page.
- 7 Using the **TPN Bridge Configuration/Status Snap-In** place all Bridge segments into the **Disconnected from LCN** mode. Verify the expected values using the "Disconnected State Health Check" table below.

| Disconnected State Health Check          |                        |                          |
|------------------------------------------|------------------------|--------------------------|
| Value Name                               | Primary Expected Value | Secondary Expected Value |
| <b>TPN Bridge Viewer - Bridge Status</b> |                        |                          |
| Cycle Counter                            | Changing               | Changing                 |
| Load Progress                            | 254                    | 254                      |
| OLAN OK                                  | 3                      | 3                        |
| Buddy OK                                 | 0                      | 1                        |
| <b>TPN Bridge Viewer - Node Status</b>   |                        |                          |
| Node [primary node number]               | ME-BRIDGE prim         | Not displayed            |
| Node [secondary node number]             | Not Displayed          | ME-BRIDGE sec            |
| <b>TPN Bridge Viewer - TDC Status</b>    |                        |                          |
| System Status – Physical Dirty           | Set                    | Set                      |
| System Status – Looping                  | Set                    | Set                      |
| System Status – Disconnected             | Set                    | Set                      |
| System Status – LCNI Media Dead          | Cleared                | May be Set – this is OK  |
| Notice Status – Multi Alone In Ring      | Cleared                | May be Set – this is OK  |
| Notice Status – All Other Bits           | Cleared                | Cleared                  |

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)

---

#### Step

#### Action

---

| TPN Bridge Viewer – WS Counters<br>Note: Clear all counters before proceeding with this check |                                 |                                          |
|-----------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------|
| Configured Primary                                                                            | 1                               | 0                                        |
| Running Primary                                                                               | 1                               | 0                                        |
| Tx Total                                                                                      | Incrementing 1 every 10 seconds | 0                                        |
| Tx Failures                                                                                   | 0                               | 0                                        |
| Tx LCN Msg                                                                                    | 0                               | 0                                        |
| Tx Buddy Msg                                                                                  | 0                               | 0                                        |
| Tx PLAT Msg                                                                                   | Incrementing 1 every 10 seconds | 0                                        |
| Rx Total                                                                                      | May increment                   | Incrementing at least 1 every 10 seconds |
| Rx Failures                                                                                   | 0                               | 0                                        |
| Rx Lost                                                                                       | 0                               | 0                                        |
| Rx Out of Order                                                                               | 0                               | 0                                        |
| Rx Duplicated                                                                                 | 0                               | 0                                        |
| Rx LCN Msg                                                                                    | 0                               | 0                                        |
| Rx LCN Lost                                                                                   | 0                               | 0                                        |
| Rx LCN Out of Order                                                                           | 0                               | 0                                        |
| Rx Buddy Msg                                                                                  | 0                               | 0                                        |
| Rx Buddy Msg Lost                                                                             | 0                               | 0                                        |
| Rx Buddy Msg Out of Order                                                                     | 0                               | 0                                        |
| Rx PLAT Msg                                                                                   | May increment                   | Incrementing at least 1 every 10 seconds |
| Rx PLAT Msg Lost                                                                              | 0                               | 0                                        |
| Rx PLAT Msg Out of Order                                                                      | 0                               | 0                                        |

#### 8 Remove the private LCNs.

---

### Checking the Bridge node communications on the LCN

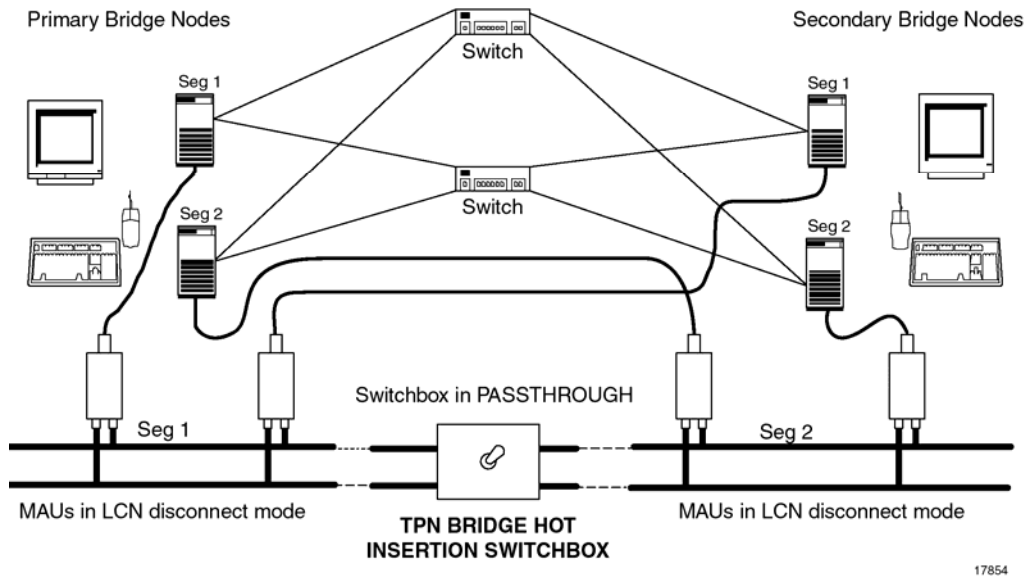


Figure 5-4 Connection of the MAUs to Primary & Secondary Bridge Nodes



#### ATTENTION

During this communication check, placing BOTH segments in **Connected Mode** at the same time WILL result in LCN node to go **ISOLATED** and **FAILURE**.

| Step | Action                                                                                        |
|------|-----------------------------------------------------------------------------------------------|
| 1    | Connect all of the Bridge MAUs to the LCN cables.                                             |
| 2    | Check the SYSTEM STATUS display for LCN cable status to ensure no cable problems.             |
| 3    | Check the LCN OVERVIEW FROM ALL NODES display to ensure no nodes indicate status of ISOLATED. |

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)

| Step | Action                                                                                                                                                                                                                              |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4    | Using the <b>TPN Bridge Configuration/Status Snap-In</b> place the <u>first (1<sup>st</sup>)</u> segment in <b>Connected Mode</b> .                                                                                                 |
| 5    | Verify the <b>SYSTEM STATUS</b> display indicates a status OK-BRG for the Primary Bridge. The Secondary Bridge may indicate OK-BRG or TEST.<br><br>Refer to Table 5-2 for additional information.                                   |
| 6    | Using the <b>LOCAL DRIVER DATA</b> display, verify that the lowest US is still functioning as Cable Master and its Swap Ticks Left counter is changing. Reload the node if it is not functioning as a cable Master.                 |
| 7    | Bring up the <b>Bridge Status</b> , <b>TDC Status</b> and <b>Node Status</b> displays on the connected Bridge nodes. Verify the expected values using the "Connected to LCN (Test Mode - Single Segment) Health Check" chart below. |

Connected to LCN (Test Mode – Single Segment) Health Check

| Value Name                               | Primary Expected Value | Secondary Expected Value |
|------------------------------------------|------------------------|--------------------------|
| <b>TPN Bridge Viewer - Bridge Status</b> |                        |                          |
| Cycle Counter                            | Changing               | Changing                 |
| Load Progress                            | 254                    | 254                      |
| OLAN OK                                  | 3                      | 3                        |
| Buddy OK                                 | 1                      | 1                        |
| LCN → NT Data (C1)                       | Changing               | Changing                 |
| NT → LCN Buffers (C1)                    | Changing               | Changing                 |
| Use NT LAT                               | 0                      | 0                        |
| Seconds Since Msg Received (NT)          | Cycling from 0 to 10   | Cycling 0 to 1           |
| Seconds Since Msg Received (LCNI)        | <11 seconds            | <11 seconds              |
| <b>TPN Bridge Viewer - Node Status</b>   |                        |                          |
| Node [primary node number]               | ME-BRIDGE prim         | BUDDY prim               |
| Node [secondary node number]             | BUDDY sec              | ME-BRIDGE sec            |
| All other LCN Node numbers ...           | Appear in list         | Appear in list           |
| <b>TPN Bridge Viewer – TDC Status</b>    |                        |                          |
| System Status – Physical Dirty           | Set                    | Set                      |
| System Status – Looping                  | Set                    | Set                      |
| System Status – Disconnected             | Cleared                | Cleared                  |
| Notice Status – Multi Alone In Ring      | Cleared                | May be Set – this is OK  |
| Notice Status – All Other Bits           | Cleared                | Cleared                  |
| Time Mastership – all Bits               | Cleared                | Cleared                  |

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)

Step

Action

| TPN Bridge Viewer – WS Counters                            |                                 |                                 |
|------------------------------------------------------------|---------------------------------|---------------------------------|
| Note: Clear all counters before proceeding with this check |                                 |                                 |
| Configured Primary                                         | 1                               | 0                               |
| Running Primary                                            | 1                               | 0                               |
| Tx Total                                                   | Changing rapidly                | Incrementing 1 every 10 seconds |
| Tx Failures                                                | 0                               | 0                               |
| Tx LCN Msg                                                 | Changing rapidly                | 0                               |
| Tx Buddy Msg                                               | Incrementing 2 per second       | Incrementing 1 every 10 seconds |
| Tx PLAT Msg                                                | Changing – within 10 seconds    | 0                               |
| Rx Total                                                   | Incrementing 1 every 10 seconds | Changing rapidly                |
| Rx Failures                                                | 0                               | 0                               |
| Rx Lost                                                    | 0                               | 0                               |
| Rx Out of Order                                            | 0                               | 0                               |
| Rx Duplicated                                              | 0                               | 0                               |
| Rx LCN Msg                                                 | 0                               | Changing rapidly                |
| Rx LCN Lost                                                | 0                               | 0                               |
| Rx LCN Out of Order                                        | 0                               | 0                               |
| Rx Buddy Msg                                               | Incrementing 1 every 10 seconds | Incrementing 2 per second       |
| Rx Buddy Msg Lost                                          | 0                               | 0                               |
| Rx Buddy Msg Out of Order                                  | 0                               | 0                               |
| Rx PLAT Msg                                                | <11 seconds                     | <11 seconds                     |
| Rx PLAT Msg Lost                                           | 0                               | 0                               |
| Rx PLAT Msg Out of Order                                   | 0                               | 0                               |

Refer to Table 4-7 for additional information.

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)

#### Step

#### Action

- 8 Bring up the **Bridge Status**, **TDC Status**, and **Node Status** displays on the disconnected Bridge nodes. Verify the expected values using the "Disconnected State Health Check" chart below.

**Disconnected State Health Check**

| Value Name                                                        | Primary Expected Value          | Secondary Expected Value |
|-------------------------------------------------------------------|---------------------------------|--------------------------|
| <b>TPN Bridge Viewer - Bridge Status</b>                          |                                 |                          |
| Cycle Counter                                                     | Changing                        | Changing                 |
| Load Progress                                                     | 254                             | 254                      |
| OLAN OK                                                           | 3                               | 3                        |
| Buddy OK                                                          | 0                               | 1                        |
| <b>TPN Bridge Viewer - Node Status</b>                            |                                 |                          |
| Node [primary node number]                                        | ME-BRIDGE prim                  | Not displayed            |
| Node [secondary node number]                                      | Not Displayed                   | ME-BRIDGE sec            |
| <b>TPN Bridge Viewer - TDC Status</b>                             |                                 |                          |
| System Status – Physical Dirty                                    | Set                             | Set                      |
| System Status – Looping                                           | Set                             | Set                      |
| System Status – Disconnected                                      | Set                             | Set                      |
| System Status – LCNI Media Dead                                   | Cleared                         | May be Set – this is OK  |
| Notice Status – Multi Alone In Ring                               | Cleared                         | May be Set – this is OK  |
| Notice Status – All Other Bits                                    | Cleared                         | Cleared                  |
| <b>TPN Bridge Viewer - WS Counters</b>                            |                                 |                          |
| <i>Note: Clear all counters before proceeding with this check</i> |                                 |                          |
| Configured Primary                                                | 1                               | 0                        |
| Running Primary                                                   | 1                               | 0                        |
| Tx Total                                                          | Incrementing 1 every 10 seconds | 0                        |
| Tx Failures                                                       | 0                               | 0                        |
| Tx LCN Msg                                                        | 0                               | 0                        |
| Tx Buddy Msg                                                      | 0                               | 0                        |
| Tx PLAT Msg                                                       | Incrementing 1 every 10 seconds | 0                        |
| Rx Total                                                          | Incrementing rapidly            | Incrementing rapidly     |
| Rx Failures                                                       | 0                               | 0                        |
| Rx Lost                                                           | 0                               | 0                        |
| Rx Out of Order                                                   | 0                               | 0                        |
| Rx Duplicated                                                     | 0                               | 0                        |

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)

| Step | Action                    |                                          |
|------|---------------------------|------------------------------------------|
|      | Rx LCN Msg                | Incrementing rapidly                     |
|      | Rx LCN Lost               | 0                                        |
|      | Rx LCN Out of Order       | 0                                        |
|      | Rx Buddy Msg              | 0                                        |
|      | Rx Buddy Msg Lost         | 0                                        |
|      | Rx Buddy Msg Out of Order | 0                                        |
|      | Rx PLAT Msg               | Incrementing at least 1 every 10 seconds |
|      | Rx PLAT Msg Lost          | 0                                        |
|      | Rx PLAT Msg Out of Order  | 0                                        |

- 9** Using the **TPN Bridge Configuration/Status Snap-In** place the segment just tested in **Disconnected Mode**.

Both segments should be in the **Disconnected Mode** at this time.

- 10** Verify the **SYSTEM STATUS** display indicates **OFF** for both Bridge LCN addresses.
- 11** Using the **TPN Bridge Configuration/Status Snap-In** place the second (2<sup>nd</sup>) segment in **Connected Mode**.
- 12** Verify the **SYSTEM STATUS** displays the Bridge nodes with **OK-BRG** status.
- 13** Bring up the **Bridge Status**, **TDC Status** and **Node Status** displays on all Bridge nodes. Verify the expected values using the "Connected to LCN (Test Mode – Single Segment) Health Check" chart below.

Connected to LCN (Test Mode – Single Segment) Health Check

| Value Name                               | Primary Expected Value | Secondary Expected Value |
|------------------------------------------|------------------------|--------------------------|
| <b>TPN Bridge Viewer - Bridge Status</b> |                        |                          |
| Cycle Counter                            | Changing               | Changing                 |
| Load Progress                            | 254                    | 254                      |
| OLAN OK                                  | 3                      | 3                        |
| Buddy OK                                 | 1                      | 1                        |
| LCN → NT Data (C1)                       | Changing               | Changing                 |
| NT → LCN Buffers (C1)                    | Changing               | Changing                 |
| Use NT LAT                               | 0                      | 0                        |

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)

| Step | Action                                                            |                                 |
|------|-------------------------------------------------------------------|---------------------------------|
|      | Seconds Since Msg Received (NT)                                   | Cycling from 0 to 10            |
|      | Seconds Since Msg Received (LCNI)                                 | Cycling 0 to 1                  |
|      | <b>TPN Bridge Viewer - Node Status</b>                            |                                 |
|      | Node [primary node number]                                        | ME-BRIDGE prim                  |
|      | Node [secondary node number]                                      | BUDDY prim                      |
|      | Node [secondary node number]                                      | BUDDY sec                       |
|      | Node [secondary node number]                                      | ME-BRIDGE sec                   |
|      | All other LCN Node numbers ...                                    | Appear in list                  |
|      | <b>TPN Bridge Viewer - TDC Status</b>                             |                                 |
|      | System Status – Physical Dirty                                    | Set                             |
|      | System Status – Looping                                           | Set                             |
|      | System Status – Disconnected                                      | Cleared                         |
|      | Notice Status – Multi Alone In Ring                               | Cleared                         |
|      | Notice Status – All Other Bits                                    | May be Set – this is OK         |
|      | Time Mastership – all Bits                                        | Cleared                         |
|      | <b>TPN Bridge Viewer – WS Counters</b>                            |                                 |
|      | <b>Note: Clear all counters before proceeding with this check</b> |                                 |
|      | Configured Primary                                                | 1                               |
|      | Running Primary                                                   | 0                               |
|      | Tx Total                                                          | Changing rapidly                |
|      | Tx Failures                                                       | Incrementing 1 every 10 seconds |
|      | Tx LCN Msg                                                        | 0                               |
|      | Tx Buddy Msg                                                      | Changing rapidly                |
|      | Tx PLAT Msg                                                       | Incrementing 2 per second       |
|      | Rx Total                                                          | Incrementing 1 every 10 seconds |
|      | Rx Failures                                                       | Incrementing 1 every 10 seconds |
|      | Rx Lost                                                           | 0                               |
|      | Rx Out of Order                                                   | 0                               |
|      | Rx Duplicated                                                     | 0                               |
|      | Rx LCN Msg                                                        | 0                               |
|      | Rx LCN Lost                                                       | Changing rapidly                |
|      | Rx LCN Out of Order                                               | 0                               |

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)

| Step | Action                    |                                          |
|------|---------------------------|------------------------------------------|
|      | Rx Buddy Msg              | Incrementing 1 every 10 seconds          |
|      | Rx Buddy Msg Lost         | Incrementing 2 per second                |
|      | Rx Buddy Msg Out of Order | 0                                        |
|      | Rx PLAT Msg               | Incrementing at least 1 every 10 seconds |
|      | Rx PLAT Msg Lost          | Incrementing at least 1 every 10 seconds |
|      | Rx PLAT Msg Out of Order  | 0                                        |

- 14** Bring up the **Bridge Status**, **TDC Status** and **Node Status** displays on the disconnected Bridge nodes. Verify the expected values using the "Disconnected State Health Check" chart below.

Disconnected State Health Check

| Value Name                               | Primary Expected Value      | Secondary Expected Value    |
|------------------------------------------|-----------------------------|-----------------------------|
| <b>TPN Bridge Viewer - Bridge Status</b> |                             |                             |
| Cycle Counter                            | Changing                    | Changing                    |
| Load Progress                            | 254                         | 254                         |
| OLAN OK                                  | 3                           | 3                           |
| Buddy OK                                 | 0                           | 0 or 1                      |
| <b>TPN Bridge Viewer - Node Status</b>   |                             |                             |
| Node [primary node number]               | ME-BRIDGE prim              | Not displayed               |
| Node [secondary node number]             | Not Displayed               | ME-BRIDGE sec               |
| All other nodes will disappear           | Wait for 120 second timeout | Wait for 120 second timeout |
| <b>TPN Bridge Viewer - TDC Status</b>    |                             |                             |
| System Status – Physical Dirty           | Set                         | Set                         |
| System Status – Looping                  | Set                         | Set                         |
| System Status – Disconnected             | Set                         | Set                         |
| System Status – LCNI Media Dead          | Cleared                     | May be Set – this is OK     |
| Notice Status – Multi Alone In Ring      | Cleared                     | May be Set – this is OK     |
| Notice Status – All Other Bits           | Cleared                     | Cleared                     |

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)

| Step      | Action                                                                                                                     |                                                                                   |
|-----------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|           | <b>TPN Bridge Viewer – WS Counters</b><br><b>Note: Clear all counters before proceeding with this check</b>                |                                                                                   |
|           | Configured Primary                                                                                                         | 1 0                                                                               |
|           | Running Primary                                                                                                            | 1 0                                                                               |
|           | Tx Total                                                                                                                   | Incrementing 1 every 10 seconds 0                                                 |
|           | Tx Failures                                                                                                                | 0 0                                                                               |
|           | Tx LCN Msg                                                                                                                 | 0 0                                                                               |
|           | Tx Buddy Msg                                                                                                               | 0 0                                                                               |
|           | Tx PLAT Msg                                                                                                                | Incrementing 1 every 10 seconds 0                                                 |
|           | Rx Total                                                                                                                   | Incrementing rapidly Incrementing rapidly                                         |
|           | Rx Failures                                                                                                                | 0 0                                                                               |
|           | Rx Lost                                                                                                                    | 0 0                                                                               |
|           | Rx Out of Order                                                                                                            | 0 0                                                                               |
|           | Rx Duplicated                                                                                                              | 0 0                                                                               |
|           | Rx LCN Msg                                                                                                                 | Incrementing rapidly Incrementing rapidly                                         |
|           | Rx LCN Lost                                                                                                                | 0 0                                                                               |
|           | Rx LCN Out of Order                                                                                                        | 0 0                                                                               |
|           | Rx Buddy Msg                                                                                                               | 0 0                                                                               |
|           | Rx Buddy Msg Lost                                                                                                          | 0 0                                                                               |
|           | Rx Buddy Msg Out of Order                                                                                                  | 0 0                                                                               |
|           | Rx PLAT Msg                                                                                                                | Incrementing at least 1 every 10 seconds Incrementing at least 1 every 10 seconds |
|           | Rx PLAT Msg Lost                                                                                                           | 0 0                                                                               |
|           | Rx PLAT Msg Out of Order                                                                                                   | 0 0                                                                               |
| <b>15</b> | Using the <b>TPN Bridge Configuration/Status Snap-In</b> place the segment just tested into the <b>Disconnected Mode</b> . |                                                                                   |
| <b>16</b> | Verify the <b>SYSTEM STATUS</b> display indicates <b>OFF</b> for both Bridge LCN addresses.                                |                                                                                   |

**Table 5-2 TDC Status Display**

| <i>In this situation . . .</i>                                                                                                                                               | <i>TDC Status Display for Bridge nodes . . .</i>                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| When everything is correct.                                                                                                                                                  | Primary Bridge and Redundant Partner Bridge will show OK-BRG                                                                    |
| When there is something wrong with the redundant partner Bridge                                                                                                              | Bridge state will go to TEST<br><br>Failed Bridge will show ISOLATED once the 2 minute communication timeout period has elapsed |
| When a Bridge node is systematically shutdown.                                                                                                                               | Bridge will show FAIL or OFF                                                                                                    |
| When a Bridge node is not systematically shutdown.                                                                                                                           | Bridge will show ISOLATED after 2 minutes and cause the System Status Display to show NETWORK <> (shattered network)            |
| If ANY Bridge node has failed.                                                                                                                                               | ALL secondary TPN Bridge nodes will show TEST                                                                                   |
| If any Redundant Ethernet LAN fault is detected.                                                                                                                             | ALL TPN Bridge nodes will show TEST                                                                                             |
| When loading a TPN Bridge node from a TPN console, the load will appear to fail due to a transition to an incorrect state; however, the TPN Bridge node will load correctly. | Bridge will show OK-BRG status                                                                                                  |



**ATTENTION**

Bridge failures that occur on segments other than the System HM segment will be indicated in the System Journal with a TEST status instead of FAIL. A TEST status on a bridge node at any time other than during node load should be considered an indication of a severe problem.

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)

#### Breaking the LCN



#### WARNING

Failure to do the following steps in order and in a timely manner can, and will, cause LCN nodes to go **ISOLATED**. All remaining steps in this procedure must be completed within 2 minutes!

| Step | Action                                                                                                                                                                                                                                                                                                     |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Place the switch on the TPN Bridge Hot Insertion Switchbox in the <b>BREAK/TERMINATE</b> position. The cable may go <b>SUSPECT</b> at this point. This is expected.                                                                                                                                        |
| 2    | Using the <b>TPN Bridge Configuration/Status Snap-In</b> place all segments in <b>Connected Mode</b> .                                                                                                                                                                                                     |
| 3    | From a US console, refresh the <b>SYSTEM STATUS</b> on each Bridge segment.<br><br>If the Status display responds very slowly (does not respond or redraws only a portion of the <b>SYSTEM STATUS</b> display) the system is going isolated causing timeouts in status gathering. Go to step 5 to resolve. |
| 4    | Bring up the <b>Bridge Status</b> , <b>TDC Status</b> and <b>Node Status</b> displays on all Bridge nodes. Verify the proper state using the "Connected to LCN (Initial Insertion – 2 Segments) Health Check" chart below.                                                                                 |

Connected to LCN (Initial Insertion – 2 Segments) Health Check

| Value Name                               | Primary Expected Value | Secondary Expected Value |
|------------------------------------------|------------------------|--------------------------|
| <b>TPN Bridge Viewer - Bridge Status</b> |                        |                          |
| Cycle Counter                            | Changing               | Changing                 |
| Load Progress                            | 254                    | 254                      |
| OLAN OK                                  | 3                      | 3                        |
| Buddy OK                                 | 1                      | 1                        |
| NT → LCN Data (C1)                       | Changing               | Changing                 |
| LCN → NT Data (C1)                       | Changing               | Changing                 |
| LCN → NT Buffers (C1)                    | Changing slowly        | Changing slowly          |
| NT → LCN Buffers (C1)                    | Changing               | Changing                 |
| Use NT LAT                               | 1                      | 1                        |
| Seconds Since Msg Received (NT)          | <11 seconds            | <11 seconds              |
| Seconds Since Msg Received (LCNI)        | <11 seconds            | <11 seconds              |

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)

| Step | Action                                                                                                      |                                                           |
|------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
|      | <b>TPN Bridge Viewer - Node Status</b>                                                                      |                                                           |
|      | Node [primary node number]                                                                                  | ME-BRIDGE prim BUDDY prim                                 |
|      | Node [secondary node number]                                                                                | BUDDY sec ME-BRIDGE sec                                   |
|      | All other LCN Node numbers ...                                                                              | Appear in list Appear in list                             |
|      | <b>TPN Bridge Viewer – TDC Status</b>                                                                       |                                                           |
|      | System Status – Recompute LAT                                                                               | Cleared May be Set                                        |
|      | System Status – Physical Dirty                                                                              | Set Set                                                   |
|      | System Status – Looping                                                                                     | Set Set                                                   |
|      | System Status – Disconnected                                                                                | Cleared Cleared                                           |
|      | Notice Status – Multi Alone In Ring                                                                         | Cleared Cleared                                           |
|      | Notice Status – All Other Bits                                                                              | Cleared Cleared                                           |
|      | Time Mastership – all Bits                                                                                  | Cleared Cleared                                           |
|      | <b>TPN Bridge Viewer – WS Counters</b><br><b>Note: Clear all counters before proceeding with this check</b> |                                                           |
|      | Configured Primary                                                                                          | 1 0                                                       |
|      | Running Primary                                                                                             | 1 0                                                       |
|      | Tx Total                                                                                                    | Changing rapidly Incrementing 1 every 10 seconds          |
|      | Tx Failures                                                                                                 | 0 0                                                       |
|      | Tx LCN Msg                                                                                                  | Changing rapidly 0                                        |
|      | Tx Buddy Msg                                                                                                | Incrementing 2 per second Incrementing 1 every 10 seconds |
|      | Tx PLAT Msg                                                                                                 | Incrementing within 10 seconds 0                          |
|      | Rx Total                                                                                                    | Changing rapidly Changing rapidly                         |
|      | Rx Failures                                                                                                 | 0 0                                                       |
|      | Rx Lost                                                                                                     | 0 0                                                       |
|      | Rx Out of Order                                                                                             | 0 0                                                       |
|      | Rx Duplicated                                                                                               | 0 0                                                       |
|      | Rx LCN Msg                                                                                                  | Changing rapidly Changing rapidly                         |
|      | Rx LCN Lost                                                                                                 | 0 0                                                       |
|      | Rx LCN Out of Order                                                                                         | 0 0                                                       |
|      | Rx Buddy Msg                                                                                                | Incrementing 1 every 10 seconds Incrementing 2 per second |
|      | Rx Buddy Msg Lost                                                                                           | 0 0                                                       |

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)

---

| Step | Action                    |                                          |
|------|---------------------------|------------------------------------------|
|      | Rx Buddy Msg Out of Order | 0                                        |
|      | Rx PLAT Msg               | Incrementing at least 1 every 10 seconds |
|      | Rx PLAT Msg Lost          | 0                                        |
|      | Rx PLAT Msg Out of Order  | 0                                        |

If all Bridge nodes are OK, go to step 6.

#### 5 IF THE BRIDGES ARE NOT FUNCTIONING:

- Using the **TPN Bridge Configuration/Status Snap-In** place all segments in **Disconnected Mode**.
- Any errors during Disconnect – pull the power on all Bridges.
- Place the switch on the Insertion Switchbox in the **PASSTHROUGH** position.
- Check for crossed cables.
- Refer to "Table 4-7" for additional information on troubleshooting the problem.

**6** Check the **LOCAL DRIVER DATA** display for the lowest US. If it displays **SLAVE** or Swap Ticks Left counter is not changing, shut it down. The next lowest US will become the **MASTER**.

**7** While monitoring the **TPN Bridge Config/Status Snap-in**, right-click the TPN level, select **Clear All Errors** in the context menu.

**8** If Bridge errors exist, right-click the Bridge, select **All Tasks** in the context menu and select **Reset Bridge Error Counters** to clear the counters. Do this for each Bridge.

**9** Using the **TPN Bridge Configuration/Status Snap-In** enable the **Time Sync** pre-defined function for each of the two segments.

**10** Use the schematic **CLOKSTAT** to verify that all the nodes on both segments transition to **LISTENER** mode within a few minutes. If the **CLOKSTAT** displays two nodes as **MASTER** (no **SLAVE**) shutdown one of the **MASTER** nodes and reload.

---

### Removing the TPN Bridge hot insertion switchbox

---



#### WARNING

The TPN Bridge Hot Insertion Switchbox is a tool used during Bridge insertion and **is not** to be left connected in the LCN.

One of the LCN segments will not be grounded after the Switchbox is removed. That segment **must** be grounded either before removal of the Switchbox with a chassis ground in a cabinet or grounded as the Switchbox is removed using a terminator with a grounding wire.

---

| Step | Action                                                                                                                                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | From the <b>Bridge Status</b> display, select <b>Clear All Counters</b> .                                                                             |
| 2    | Verify the <b>TX Errors</b> and <b>RX Errors</b> remain at 0.                                                                                         |
| 3    | Bring up the schematic \$LNMENU on a TDC Universal Station or GUS.                                                                                    |
| 4    | Choose RESET TIMES.                                                                                                                                   |
| 5    | Choose SELECT TO UPDATE DISPLAY.                                                                                                                      |
| 6    | Choose RESET ALL COUNTERS.                                                                                                                            |
| 7    | Choose MAIN MENU.                                                                                                                                     |
| 8    | Choose NET MEDIA DEADS.<br><br>Observe that the Media Dead counts on all the nodes for several minutes to determine that there are no cable problems. |
| 9    | Choose MAIN MENU.                                                                                                                                     |
| 10   | Bring up the SYSTEM STATUS display on a US or GUS.                                                                                                    |
| 11   | Force the LCN to Cable B.                                                                                                                             |
| 12   | Bring up the schematic \$LNMENU on a TDC Universal Station or GUS.                                                                                    |
| 13   | Choose CABLE SWAP INH.                                                                                                                                |
| 14   | Choose INHIBIT SWAPPING.                                                                                                                              |
| 15   | Determine where the LCN A Cable is to be terminated on each segment.                                                                                  |

---

## 5. TPN Bridge Hot Insertion

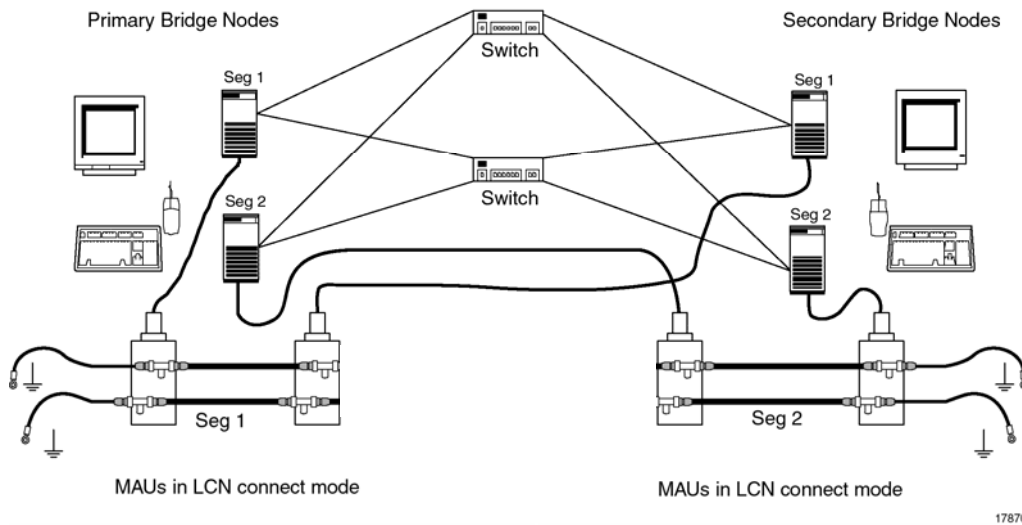
### 5.4. First Segmentation of a TPN (LCN)

---

| Step | Action                                                                                                                                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16   | Determine which segment needs the LCN A Cable grounded.                                                                                               |
| 17   | Disconnect the LCN A Cable from the first segment and terminate with a regular terminator or terminator with a grounding wire (part # 51305492-100).  |
| 18   | Disconnect the LCN A Cable from the second segment and terminate with a regular terminator or terminator with a grounding wire (part # 51305492-100). |
| 19   | At the INHIBIT SWAPPING display, choose ENABLE SWAPPING.                                                                                              |
| 20   | Bring up the SYSTEM STATUS display.                                                                                                                   |
| 21   | Allow the system to swap to cable A and back to cable B and verify no cable SUSPECT status before going to the next step.                             |
| 22   | Force the LCN to Cable A.                                                                                                                             |
| 23   | Bring up the schematic \$LNMENU on a TDC Universal Station or GUS.                                                                                    |
| 24   | Choose CABLE SWAP INH.                                                                                                                                |
| 25   | Choose INHIBIT SWAPPING.                                                                                                                              |
| 26   | Determine where the LCN B Cable is to be terminated on each segment.                                                                                  |
| 27   | Determine which segment needs the LCN B Cable grounded.                                                                                               |
| 28   | Disconnect the LCN B Cable from the first segment and terminate with a regular terminator or terminator with a grounding wire (part # 51305492-100).  |
| 29   | Disconnect the LCN B Cable from the second segment and terminate with a regular terminator or terminator with a grounding wire (part # 51305492-100). |
| 30   | At the INHIBIT SWAPPING display, choose ENABLE SWAPPING.                                                                                              |
| 31   | Remove the TPN Bridge Hot Insertion Switchbox and any excess LCN cable.                                                                               |

## 5. TPN Bridge Hot Insertion

### 5.4. First Segmentation of a TPN (LCN)



**Figure 5-5 Hot Insertion Switchbox Removed and Cables Terminated**

#### Setting Time Sync options

Set all segment time sync options according to the procedure in Section 4.11, "Setting TPN Bridge Time Sync Options." Also refer to "Table 3-2" for additional information.

## 5.5 Creating Additional Segments

You will perform this procedure if TPN Bridge segments already exist on your TPN system.

To install the initial TPN Bridge on your TPN system, see Section 5.4, "First Segmentation of a TPN (LCN)."

### Configuring the NCF for the TPN Bridge

No NCF changes are required for installing additional TPN Bridge segments unless the LCN is to be moved from a "Less than or equal to 64 Node" to a "Greater than 64 Node." The "Greater than 64 Node" Keyfile option must be purchased and installed. Refer to Section 4.8, "Installing and Configuring the TPN System Software" for further information.

### Configuring the NT on TPN Bridge nodes

Each TPN Bridge must be initialized. Refer to the procedure in Section 4.9, "TPN Bridge and NT Configuration."



#### ATTENTION

At this point the Bridge hardware must have been installed in cabinets, console or desktop, and set up according to the Configuration Rules for TPN Bridge (see *TPN Bridge Planning, Installation and Service*).

The two new Bridge PCs for the additional segment are configured and **disconnected** from the LCN and **connected** to the Redundant Ethernet Bridge LAN. All LCN MAUs should be labeled to ensure no mix-ups.

All cables should be run to the Bridges and all necessary Tees should be inserted.

The system Time Sync should be disabled. For additional information, refer to Section 4.11, "Setting TPN Bridge Time Sync Options".



#### CAUTION

Do not connect the TPN Bridge MAUs to the LCN until directed to do so.

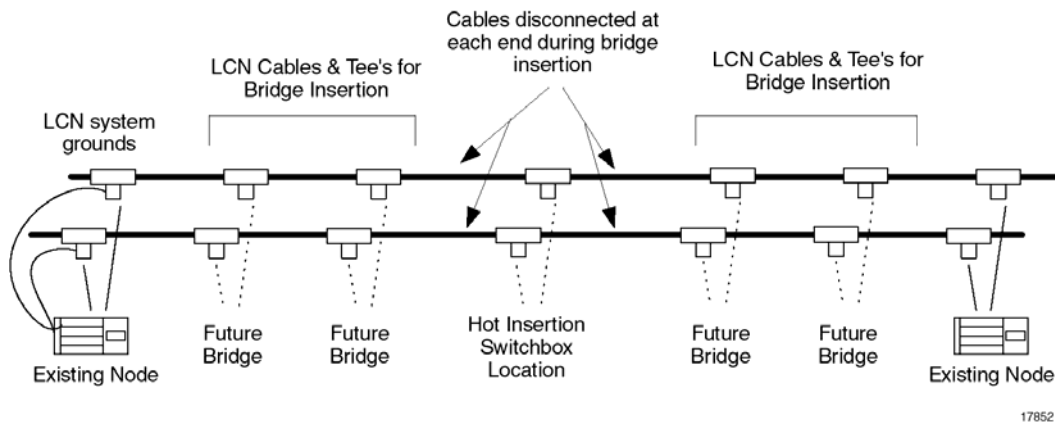
---

### Configuring the TPN Bridge Node software

All TPN Bridge nodes must be configured to reflect the new TPN topology. Refer to Section 4.12, "Adding a New Segment to an Existing Bridge System."

### Preparing the LCN for TPN Bridge insertion

The TPN cable must be modified to provide taps for insertion of the Hot Insertion Switchbox Tool. The following illustration shows the location of the LCN cables, the Tee's and the Switchbox Tool insertion point.



**Figure 5-6 Location of Switch Box, Cables and Tees**

The following section outlines a procedure for safely installing cables, Tees and Hot Insertion Switchbox tool.

### Inserting Hot Insertion switchbox tool

The TPN Bridge Hot Insertion Switchbox is a tool used to break and terminate both LCN cables at the same time to prevent node Isolation and Media Dead counts. The switchbox also allows the reconnection of the LCN if a failure occurs during the insertion of the Bridges.



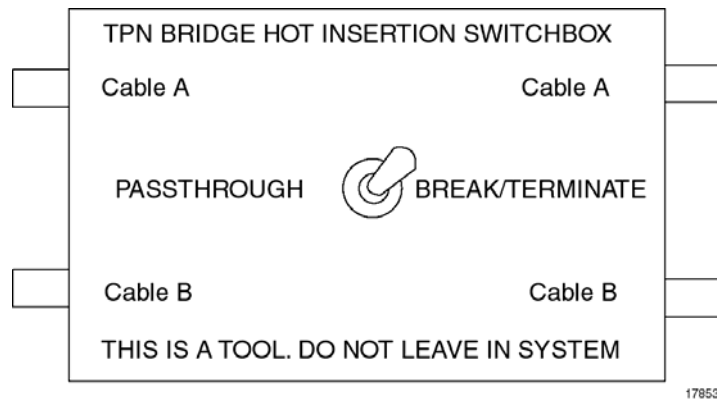
#### ATTENTION

The TPN Bridge nodes must not be inserted if there are any LCN cable problems. If cable errors exist, make repairs before continuing.

## 5. TPN Bridge Hot Insertion

### 5.5. Creating Additional Segments

---



**Figure 5-7 Hot Insertion Switchbox Tool**

The following procedure installs the Hot Insertion Switchbox tool. It outlines the procedure for inserting cable segments and T-connectors for installing the TPN Bridges.

| Step | Action                                                                                                                                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Bring up the schematic \$LNMENU on a TDC Universal Station or GUS.                                                                                    |
| 2    | Choose RESET TIMES.                                                                                                                                   |
| 3    | Choose SELECT TO UPDATE DISPLAY.                                                                                                                      |
| 5    | Choose RESET ALL COUNTERS.                                                                                                                            |
| 6    | Choose MAIN MENU.                                                                                                                                     |
| 7    | Choose NET MEDIA DEADS.<br><br>Observe that the Media Dead counts on all the nodes for several minutes to determine that there are no cable problems. |
| 8    | Locate the LCN break point and label the cables.                                                                                                      |
| 9    | Bring up the SYSTEM STATUS display on a US or GUS.                                                                                                    |
| 10   | Force the LCN to Cable B.                                                                                                                             |
| 11   | Bring up the schematic \$LNMENU on a TDC Universal Station or GUS.                                                                                    |

| Step | Action                                                                                                                                                       |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12   | Choose CABLE SWAP INH.                                                                                                                                       |
| 13   | Choose INHIBIT SWAPPING.                                                                                                                                     |
| 14   | Make sure the Hot Insertion Switchbox tool is in the <b>PASSTHROUGH</b> position.                                                                            |
| 15   | At the LCN break point, break the A cable of the LCN and connect both ends to the BNC connectors labeled <b>Cable A</b> on the Hot Insertion Switchbox tool. |
| 16   | At the INHIBIT SWAPPING display, choose ENABLE SWAPPING.                                                                                                     |
| 17   | Bring up the SYSTEM STATUS display.                                                                                                                          |
| 18   | Wait until both cable A and cable B are OK.                                                                                                                  |
| 19   | Force the LCN to Cable A.                                                                                                                                    |
| 20   | Bring up the schematic \$LNMENU on a TDC Universal Station or GUS.                                                                                           |
| 21   | Choose CABLE SWAP INH.                                                                                                                                       |
| 22   | Choose INHIBIT SWAPPING.                                                                                                                                     |
| 23   | At the LCN break point, break the B cable of the LCN and connect both ends to the BNC connectors labeled <b>Cable B</b> on the Hot Insertion Switchbox tool. |
| 24   | At the INHIBIT SWAPPING display, choose ENABLE SWAPPING.                                                                                                     |
| 25   | Bring up the SYSTEM STATUS display.                                                                                                                          |
| 26   | Wait until both cable A and cable B are OK.                                                                                                                  |

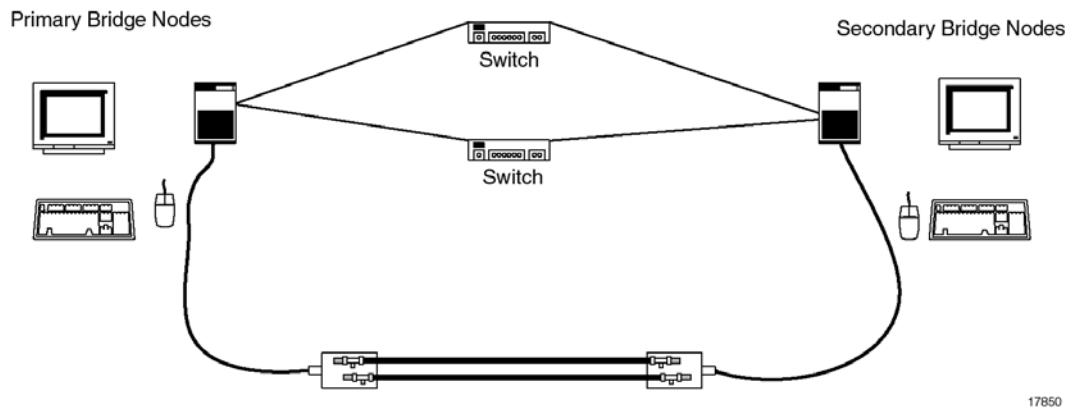
## 5. TPN Bridge Hot Insertion

### 5.5. Creating Additional Segments

---

#### Offline testing of Bridge operation

The next process is to bring up the TPN Bridge nodes so that they are communicating with one another over the Ethernet and then connect them to a private LCN to verify LCN interface functionality.



**Figure 5-8 Private LCN Setup**

| Step | Action                                                                                                                                                                     |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Using the <b>TPN Bridge Configuration/Status Snap-In</b> make sure that <b>Time Sync</b> has been turned off for all segments.                                             |
| 2    | Connect the new Bridge Primary to its Bridge Secondary with a private LCN.                                                                                                 |
| 3    | Reset the new Primary TPN Bridge TPNB.<br>If the Bridge fails reset the Primary TPN Bridge TPNB again.<br><b>Note:</b> This procedure is establishing a two-node LCN ring. |
| 4    | Using the <b>TPN Bridge Configuration/Status Snap-In</b> and selecting <b>TPN Properties</b> make sure that the new segment is in the <b>Connected Mode</b> .              |

## 5. TPN Bridge Hot Insertion

### 5.5. Creating Additional Segments

| Step | Action                                                                                                                                                                                                             |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5    | Bring up the <b>Bridge Status</b> , <b>TDC Status</b> and <b>Node Status</b> displays on both new Bridge nodes. Verify the expected values using the "Private LCN (Additional Segments) Health Check" chart below. |

**Private LCN (Additional Segments) Health Check**

| Value Name                                                        | Primary Expected Value                   | Secondary Expected Value           |
|-------------------------------------------------------------------|------------------------------------------|------------------------------------|
| <b>TPN Bridge Viewer - Bridge Status</b>                          |                                          |                                    |
| Cycle Counter                                                     | Changing                                 | Changing                           |
| Load Progress                                                     | 254                                      | 254                                |
| OLAN OK                                                           | 3                                        | 3                                  |
| Buddy OK                                                          | 1                                        | 1                                  |
| Seconds Since Msg Recived (NT)                                    | <11 seconds                              | Cycles 0 – 1                       |
| Seconds Since Msg Recived (LCNI)                                  | Cycles 0 – 10                            | Cycles 0 – 1                       |
| <b>TPN Bridge Viewer - Node Status</b>                            |                                          |                                    |
| Node [primary node number]                                        | ME-BRIDGE prim                           | BUDDY prim                         |
| Node [secondary node number]                                      | BUDDY sec                                | ME-BRIDGE sec                      |
| All other node numbers                                            | None                                     | May be present                     |
| <b>TPN Bridge Viewer – TDC Status</b>                             |                                          |                                    |
| System Status – Physical Dirty                                    | Set                                      | Set                                |
| System Status – Looping                                           | Set                                      | Set                                |
| System Status – Disconnected                                      | Cleared                                  | Cleared                            |
| Notice Status – Multi alone in Ring                               | Cleared                                  | Cleared                            |
| Notice Status – All other bits                                    | Cleared                                  | Cleared                            |
| Time Mastership – All bits                                        | Cleared                                  | Cleared                            |
| <b>TPN Bridge Viewer – WS Counters</b>                            |                                          |                                    |
| <i>Note: Clear all counters before proceeding with this check</i> |                                          |                                    |
| Configured Primary                                                | 1                                        | 0                                  |
| Running Primary                                                   | 1                                        | 0                                  |
| Tx Total                                                          | Incrementing 2 per second                | Incrementing 1 every 10 seconds    |
| Tx Failures                                                       | 0                                        | 0                                  |
| Tx LCN Msg                                                        | 0                                        | 0                                  |
| Tx Buddy Msg                                                      | Incrementing 2 per second                | Incrementing 1 every 10 seconds    |
| Tx PLAT Msg                                                       | Incrementing 1 every 10 seconds          | 0                                  |
| Rx Total                                                          | Incrementing at least 1 every 10 seconds | Incrementing at least 2 per second |

## 5. TPN Bridge Hot Insertion

### 5.5. Creating Additional Segments

---

| Step | Action                    |                                 |
|------|---------------------------|---------------------------------|
|      | Rx Failures               | 0                               |
|      | Rx Lost                   | 0                               |
|      | Rx Out of Order           | 0                               |
|      | Rx Duplicated             | 0                               |
|      | Rx LCN Msg                | Increasing                      |
|      | Rx LCN Lost               | 0                               |
|      | Rx LCN Out of Order       | 0                               |
|      | Rx Buddy Msg              | Incrementing 1 every 10 seconds |
|      | Rx Buddy Msg Lost         | 0                               |
|      | Rx Buddy Msg Out of Order | 0                               |
|      | Rx PLAT Msg               | May increase                    |
|      | Rx PLAT Msg Lost          | 0                               |
|      | Rx PLAT Msg Out of Order  | 0                               |

- 6 Place the new Bridge segment into the **Disconnected Mode**. Verify the expected values using the "Disconnected State (Additional Segments) Health Check" chart below.

| Disconnected State (Additional Segments) Health Check |                        |                          |
|-------------------------------------------------------|------------------------|--------------------------|
| Value Name                                            | Primary Expected Value | Secondary Expected Value |
| <b>TPN Bridge Viewer - Bridge Status</b>              |                        |                          |
| Cycle Counter                                         | Changing               | Changing                 |
| Load Progress                                         | 254                    | 254                      |
| OLAN OK                                               | 3                      | 3                        |
| Buddy OK                                              | 0                      | 0 or 1                   |
| <b>TPN Bridge Viewer - Node Status</b>                |                        |                          |
| Node [primary node number]                            | ME-BRIDGE prim         | Not displayed            |
| Node [secondary node number]                          | Not Displayed          | ME-BRIDGE sec            |
| <b>TPN Bridge Viewer - TDC Status</b>                 |                        |                          |
| System Status – Physical Dirty                        | Set                    | Set                      |
| System Status – Looping                               | Set                    | Set                      |
| System Status – Disconnected                          | Set                    | Set                      |
| System Status – LCNI Media Dead                       | Cleared                | May be Set – this is OK  |
| Notice Status – Multi Alone In Ring                   | Cleared                | May be Set – this is OK  |
| Notice Status – All Other Bits                        | Cleared                | Cleared                  |

## 5. TPN Bridge Hot Insertion

### 5.5. Creating Additional Segments

| Step | Action                                                                                                      |                                          |
|------|-------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 7    | <b>TPN Bridge Viewer – WS Counters</b><br><b>Note: Clear all counters before proceeding with this check</b> |                                          |
|      | Configured Primary                                                                                          | 0                                        |
|      | Running Primary                                                                                             | 0                                        |
|      | Tx Total                                                                                                    | Incrementing 1 every 10 seconds          |
|      | Tx Failures                                                                                                 | 0                                        |
|      | Tx LCN Msg                                                                                                  | 0                                        |
|      | Tx Buddy Msg                                                                                                | 0                                        |
|      | Tx PLAT Msg                                                                                                 | Incrementing 1 every 10 seconds          |
|      | Rx Total                                                                                                    | Increasing                               |
|      | Rx Failures                                                                                                 | 0                                        |
|      | Rx Lost                                                                                                     | 0                                        |
|      | Rx Out of Order                                                                                             | 0                                        |
|      | Rx Duplicated                                                                                               | 0                                        |
|      | Rx LCN Msg                                                                                                  | Increasing                               |
|      | Rx LCN Lost                                                                                                 | 0                                        |
|      | Rx LCN Out of Order                                                                                         | 0                                        |
|      | Rx Buddy Msg                                                                                                | 0                                        |
|      | Rx Buddy Msg Lost                                                                                           | 0                                        |
|      | Rx Buddy Msg Out of Order                                                                                   | 0                                        |
|      | Rx PLAT Msg                                                                                                 | May be increasing                        |
|      | Rx PLAT Msg Lost                                                                                            | 0                                        |
|      | Rx PLAT Msg Out of Order                                                                                    | 0                                        |
|      |                                                                                                             | Incrementing at least 1 every 10 seconds |
|      |                                                                                                             |                                          |
|      |                                                                                                             |                                          |
|      |                                                                                                             |                                          |
|      |                                                                                                             |                                          |
|      |                                                                                                             |                                          |
|      |                                                                                                             |                                          |

Remove the private LCN.

## 5. TPN Bridge Hot Insertion

### 5.5. Creating Additional Segments

---

#### Checking the Bridge Node communications on the LCN

The Bridge node communication tests on the LCN cannot be performed when inserting additional segments. Enabling the new TPN Bridge at this time will cause duplicate LCN messages.

#### Breaking the LCN



#### WARNING

Failure to do the following steps in order and in a timely manner can, and will, cause LCN nodes to go **ISOLATED**.

---

| Step | Action                                                                                                                                                                                                                                                                                                     |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Connect all of the new Bridge MAUs to the LCN cables. Verify there are no errors on the LCN.                                                                                                                                                                                                               |
| 2    | Place the switch on the TPN Bridge Hot Insertion Switchbox in the <b>BREAK/TERMINATE</b> position. The cable may go <b>SUSPECT</b> at this point. This is expected.                                                                                                                                        |
| 3    | Using the <b>TPN Bridge Configuration/Status Snap-In</b> place the new segment in <b>Connected Mode</b> .                                                                                                                                                                                                  |
| 4    | From a US console, refresh the <b>SYSTEM STATUS</b> on each Bridge segment.<br><br>If the Status display responds very slowly (does not respond or redraws only a portion of the <b>SYSTEM STATUS</b> display) the system is going isolated causing timeouts in status gathering. Go to step 6 to resolve. |

---

## 5. TPN Bridge Hot Insertion

### 5.5. Creating Additional Segments

#### Step

#### Action

- Bring up the **Bridge Status**, **TDC Status** and **Node Status** displays on all Bridge nodes. Verify the expected values using the "Connected to LCN (Initial Insertion – 2 segments) Health Check" chart below.

**Connected to LCN (Initial Insertion – 2 Segments) Health Check**

| Value Name                                                        | Primary Expected Value         | Secondary Expected Value        |
|-------------------------------------------------------------------|--------------------------------|---------------------------------|
| <b>TPN Bridge Viewer - Bridge Status</b>                          |                                |                                 |
| Cycle Counter                                                     | Changing                       | Changing                        |
| Load Progress                                                     | 254                            | 254                             |
| OLAN OK                                                           | 3                              | 3                               |
| Buddy OK                                                          | 1                              | 1                               |
| NT → LCN Data (C1)                                                | Changing                       | Changing                        |
| LCN → NT Data (C1)                                                | Changing                       | Changing                        |
| LCN → NT Buffers (C1)                                             | Changing slowly                | Changing slowly                 |
| NT → LCN Buffers (C1)                                             | Changing                       | Changing                        |
| Use NT LAT                                                        | 1                              | 1                               |
| Seconds Since Msg Received (NT)                                   | <11 seconds                    | <11 seconds                     |
| Seconds Since Msg Received (LCNI)                                 | <11 seconds                    | <11 seconds                     |
| <b>TPN Bridge Viewer - Node Status</b>                            |                                |                                 |
| Node [primary node number]                                        | ME-BRIDGE prim                 | BUDDY prim                      |
| Node [secondary node number]                                      | BUDDY sec                      | ME-BRIDGE sec                   |
| All other LCN Node numbers ...                                    | All local segment nodes appear | Appear in list                  |
| <b>TPN Bridge Viewer – TDC Status</b>                             |                                |                                 |
| System Status – Recompute LAT                                     | Cleared                        | May be Set                      |
| System Status – Physical Dirty                                    | Set                            | Set                             |
| System Status – Looping                                           | Set                            | Set                             |
| System Status – Disconnected                                      | Cleared                        | Cleared                         |
| Notice Status – Multi Alone In Ring                               | Cleared                        | Cleared                         |
| Notice Status – All Other Bits                                    | Cleared                        | Cleared                         |
| Time Mastership – all Bits                                        | Cleared                        | Cleared                         |
| <b>TPN Bridge Viewer – WS Counters</b>                            |                                |                                 |
| <i>Note: Clear all counters before proceeding with this check</i> |                                |                                 |
| Configured Primary                                                | 1                              | 0                               |
| Running Primary                                                   | 1                              | 0                               |
| Tx Total                                                          | Changing rapidly               | Incrementing 1 every 10 seconds |

## 5. TPN Bridge Hot Insertion

### 5.5. Creating Additional Segments

---

| Step | Action |  |
|------|--------|--|
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |
|      |        |  |

If all Bridge nodes are OK, go to step 7.

#### 6 IF THE BRIDGES ARE NOT FUNCTIONING:

- Using the TPN Bridge Configuration/Status Snap-In place the new segment in Disconnected Mode.
- Any errors during Disconnect – pull the power on the new Bridge nodes.
- Place the switch on the Insertion Switchbox in the PASSTHROUGH position.
- Troubleshoot the problem. Refer to "Table 4-7" for additional information.

#### 7 Check the LOCAL DRIVER DATA display for the lowest US. If it displays SLAVE or Swap Ticks Left counter is not changing, shut it down. The next lowest US will become the MASTER. Reload the node.

---

| Step | Action                                                                                                                                                                                                                                                       |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8    | While monitoring the <b>TPN Bridge Config/Status Snap-in</b> , right-click the TPN level, select <b>Clear All Errors</b> in the context menu.                                                                                                                |
| 9    | If Bridge errors exist, right-click the Bridge, select <b>All Tasks</b> in the context menu and select <b>Reset Bridge Error Counters</b> to clear the counters. Do this for each Bridge.                                                                    |
| 10   | Using the <b>TPN Bridge Configuration/Status Snap-In</b> enable the <b>Time Sync</b> pre-defined function for all segments.                                                                                                                                  |
| 11   | Use the schematic CLOKSTAT to verify that all the nodes on both segments transition to <b>LISTENER</b> mode within a few minutes. If the CLOKSTAT displays two nodes as <b>MASTER</b> (no <b>SLAVE</b> ) shutdown one of the <b>MASTER</b> nodes and reload. |

### Removing the TPN Bridge Hot Insertion switchbox

---



#### WARNING

The TPN Bridge Hot Insertion Switchbox is a tool used during Bridge insertion and **is not** to be left connected in the LCN.

One of the LCN segments will not be grounded after the Switchbox is removed. That segment **must** be grounded either before removal of the Switchbox with a chassis ground in a cabinet or grounded as the Switchbox is removed using a terminator with a grounding wire.

---

| Step | Action                                                                                                                                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Bring up the schematic \$LNMENU on a TDC Universal Station or GUS.                                                                                    |
| 2    | Choose RESET TIMES.                                                                                                                                   |
| 3    | Choose SELECT TO UPDATE DISPLAY.                                                                                                                      |
| 4    | Choose RESET ALL COUNTERS.                                                                                                                            |
| 5    | Choose MAIN MENU.                                                                                                                                     |
| 6    | Choose NET MEDIA DEADS.<br><br>Observe that the Media Dead counts on all the nodes for several minutes to determine that there are no cable problems. |
| 7    | Choose MAIN MENU.                                                                                                                                     |
| 8    | Bring up the SYSTEM STATUS display on a US or GUS.                                                                                                    |
| 9    | Force the LCN to Cable B.                                                                                                                             |
| 10   | Bring up the schematic \$LNMENU on a TDC Universal Station or GUS.                                                                                    |
| 11   | Choose CABLE SWAP INH.                                                                                                                                |
| 12   | Choose INHIBIT SWAPPING.                                                                                                                              |
| 13   | Determine where the LCN A Cable is to be terminated on each segment.                                                                                  |
| 14   | Determine which segment needs the LCN A Cable grounded.                                                                                               |
| 15   | Disconnect the LCN A Cable from the first segment and terminate with a regular terminator or terminator with a grounding wire (part # 51305492-100).  |

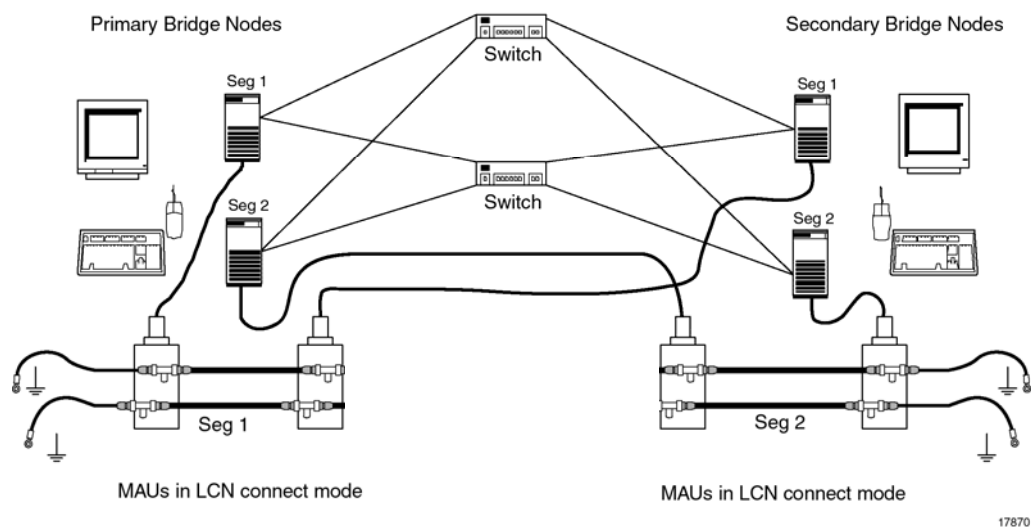
---

| Step      | Action                                                                                                                                                |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>16</b> | Disconnect the LCN A Cable from the second segment and terminate with a regular terminator or terminator with a grounding wire (part # 51305492-100). |
| <b>17</b> | At the INHIBIT SWAPPING display, choose ENABLE SWAPPING.                                                                                              |
| <b>18</b> | Bring up the SYSTEM STATUS display.                                                                                                                   |
| <b>19</b> | Wait until both cable A and cable B are <b>OK</b> .                                                                                                   |
| <b>20</b> | Force the LCN to Cable A.                                                                                                                             |
| <b>21</b> | Bring up the schematic \$LNMENU on a TDC Universal Station or GUS.                                                                                    |
| <b>22</b> | Choose CABLE SWAP INH.                                                                                                                                |
| <b>23</b> | Choose INHIBIT SWAPPING.                                                                                                                              |
| <b>24</b> | Determine where the LCN B Cable is to be terminated on each segment.                                                                                  |
| <b>25</b> | Determine which segment needs the LCN B Cable grounded.                                                                                               |
| <b>26</b> | Disconnect the LCN B Cable from the first segment and terminate with a regular terminator or terminator with a grounding wire (part # 51305492-100).  |
| <b>27</b> | Disconnect the LCN B Cable from the second segment and terminate with a regular terminator or terminator with a grounding wire (part # 51305492-100). |
| <b>28</b> | At the INHIBIT SWAPPING display, choose ENABLE SWAPPING.                                                                                              |
| <b>29</b> | Remove the TPN Bridge Hot Insertion Switchbox and any excess LCN cable.                                                                               |

## 5. TPN Bridge Hot Insertion

### 5.5. Creating Additional Segments

---



**Figure 5-9 Hot Insertion Switchbox Removed and Cables Terminated**

## 5.6 Replacing an LCNE with a TPN Bridge

### Overview

The following procedure will be used when an LCNE is to be replaced with a TPN Bridge.



#### ATTENTION

The TPN Bridge can only be used to replace an LCNE when the following criteria is met:

- The distance between the segments is less than 100 meters.
- The LCN system clock signal is digitally propagated across the LCNE.

### Offline testing of Bridge operation

The next process is to bring up the TPN Bridge nodes so that they are communicating with one another over the Ethernet and then connect them to a private LCN to verify LCN interface functionality.

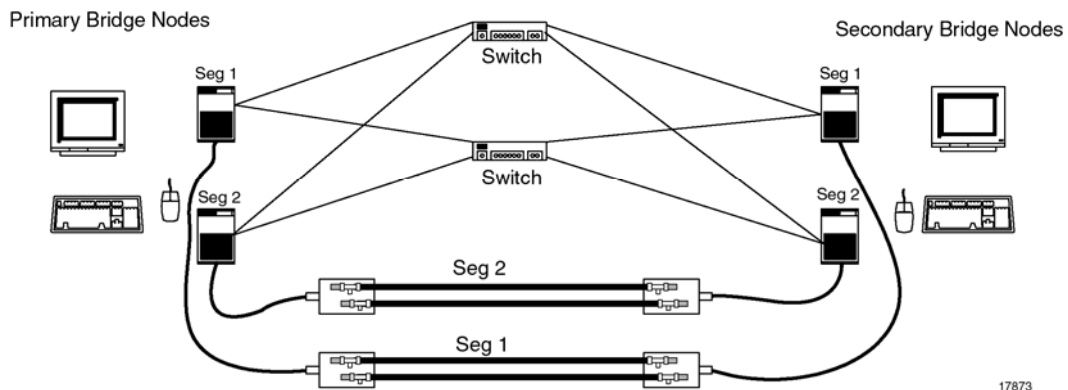


Figure 5-10 Private LCN Setup

## 5. TPN Bridge Hot Insertion

### 5.6. Replacing an LCNE with a TPN Bridge

---

| Step | Action                                                                                                                                                                                                                   |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Using the <b>TPN Bridge Configuration/Status Snap-In</b> make sure that Time Sync has been turned off for all segments.<br><br>Refer to Section 4.11, "Setting TPN Bridge Time Sync Options" for additional information. |
| 2    | Connect each <u>new</u> Bridge Primary to its Bridge Secondary with a private LCN.                                                                                                                                       |
| 3    | Reset each Primary TPN Bridge node <u>twice</u> to establish the LCN ring. The Primary node may fail if not reset twice.                                                                                                 |
| 4    | Using the <b>TPN Bridge Configuration/Status Snap-In</b> make sure that all the segments are in the <b>Connected Mode</b> .                                                                                              |
| 5    | Verify using the <b>LCNP Status</b> display that the TPNB of all Bridge nodes are operational and in the proper state.                                                                                                   |

---

## 5. TPN Bridge Hot Insertion

### 5.6. Replacing an LCNE with a TPN Bridge

| Step | Action |
|------|--------|
|------|--------|

- |   |                                                                                                                                                                      |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 | Bring up the <b>Bridge Status</b> , <b>TDC Status</b> and <b>Node Status</b> displays, verify the expected values using the "Private LCN Health Check " chart below. |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Private LCN Health Check

| Value Name                                                                                                  | Primary Expected Value                   | Secondary Expected Value                 |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------|
| <b>TPN Bridge Viewer - Bridge Status</b>                                                                    |                                          |                                          |
| Cycle Counter                                                                                               | Changing                                 | Changing                                 |
| Load Progress                                                                                               | 254                                      | 254                                      |
| OLAN OK                                                                                                     | 3                                        | 3                                        |
| Buddy OK                                                                                                    | 1                                        | 1                                        |
| Seconds Since Msg Recived (NT)                                                                              | Cycles 0 – 10                            | Cycles 0 – 1                             |
| Seconds Since Msg Recived (LCNI)                                                                            | Cycles 0 – 10                            | Cycles 0 – 1                             |
| <b>TPN Bridge Viewer - Node Status</b>                                                                      |                                          |                                          |
| Node [primary node number]                                                                                  | ME-BRIDGE prim                           | BUDDY prim                               |
| Node [secondary node number]                                                                                | BUDDY sec                                | ME-BRIDGE sec                            |
| <b>TPN Bridge Viewer – TDC Status</b>                                                                       |                                          |                                          |
| System Status – Physical Dirty                                                                              | Set                                      | Set                                      |
| System Status – Looping                                                                                     | Set                                      | Set                                      |
| System Status – Disconnected                                                                                | Cleared                                  | Cleared                                  |
| Notice Status – Multi alone in Ring                                                                         | Cleared                                  | May be Set                               |
| Notice Status – All other bits                                                                              | Cleared                                  | Cleared                                  |
| Time Mastership – All bits                                                                                  | Cleared                                  | Cleared                                  |
| <b>TPN Bridge Viewer – WS Counters</b><br><i>Note: Clear all counters before proceeding with this check</i> |                                          |                                          |
| Configured Primary                                                                                          | 1                                        | 0                                        |
| Running Primary                                                                                             | 1                                        | 0                                        |
| Tx Total                                                                                                    | Incrementing 2 per second                | Incrementing at least 1 every 10 seconds |
| Tx Failures                                                                                                 | 0                                        | 0                                        |
| Tx LCN Msg                                                                                                  | 0                                        | 0                                        |
| Tx Buddy Msg                                                                                                | Incrementing 2 per second                | Incrementing at least 1 every 10 seconds |
| Tx PLAT Msg                                                                                                 | Incrementing at least 1 every 10 seconds | May increment                            |
| Rx Total                                                                                                    | Incrementing at least 1 every 10 seconds | Incrementing 2 per second                |
| Rx Failures                                                                                                 | 0                                        | 0                                        |

## 5. TPN Bridge Hot Insertion

### 5.6. Replacing an LCNE with a TPN Bridge

| Step | Action                    |                                          |
|------|---------------------------|------------------------------------------|
|      | Rx Lost                   | 0                                        |
|      | Rx Out of Order           | 0                                        |
|      | Rx Duplicated             | 0                                        |
|      | Rx LCN Msg                | May increment                            |
|      | Rx LCN Lost               | 0                                        |
|      | Rx LCN Out of Order       | 0                                        |
|      | Rx Buddy Msg              | Incrementing at least 1 every 10 seconds |
|      | Rx Buddy Msg Lost         | 0                                        |
|      | Rx Buddy Msg Out of Order | 0                                        |
|      | Rx PLAT Msg               | May increment                            |
|      | Rx PLAT Msg Lost          | 0                                        |
|      | Rx PLAT Msg Out of Order  | 0                                        |

Refer to "Table 4-7" for additional information.

- 7 Call-up MMC. Right-click **TPN Properties** for LCN Connection page.
- 8 Using the **TPN Bridge Configuration/Status Snap-In** place the new Bridge segment(s) into the **Disconnected Mode**. Verify the expected values using the "Disconnected State Health Check" chart below.

| Disconnected State Health Check          |                        |                          |
|------------------------------------------|------------------------|--------------------------|
| Value Name                               | Primary Expected Value | Secondary Expected Value |
| <b>TPN Bridge Viewer - Bridge Status</b> |                        |                          |
| Cycle Counter                            | Changing               | Changing                 |
| Load Progress                            | 254                    | 254                      |
| OLAN OK                                  | 3                      | 3                        |
| Buddy OK                                 | 0                      | 0 or 1                   |
| <b>TPN Bridge Viewer - Node Status</b>   |                        |                          |
| Node [primary node number]               | ME-BRIDGE prim         | Not displayed            |
| Node [secondary node number]             | Not Displayed          | ME-BRIDGE sec            |
| <b>TPN Bridge Viewer - TDC Status</b>    |                        |                          |
| System Status – Physical Dirty           | Set                    | Set                      |
| System Status – Looping                  | Set                    | Set                      |
| System Status – Disconnected             | Set                    | Set                      |
| System Status – LCN! Media Dead          | Cleared                | May be Set – this is OK  |
| Notice Status – Multi Alone In Ring      | Cleared                | May be Set – this is OK  |
| Notice Status – All Other Bits           | Cleared                | Cleared                  |

## 5. TPN Bridge Hot Insertion

### 5.6. Replacing an LCNE with a TPN Bridge

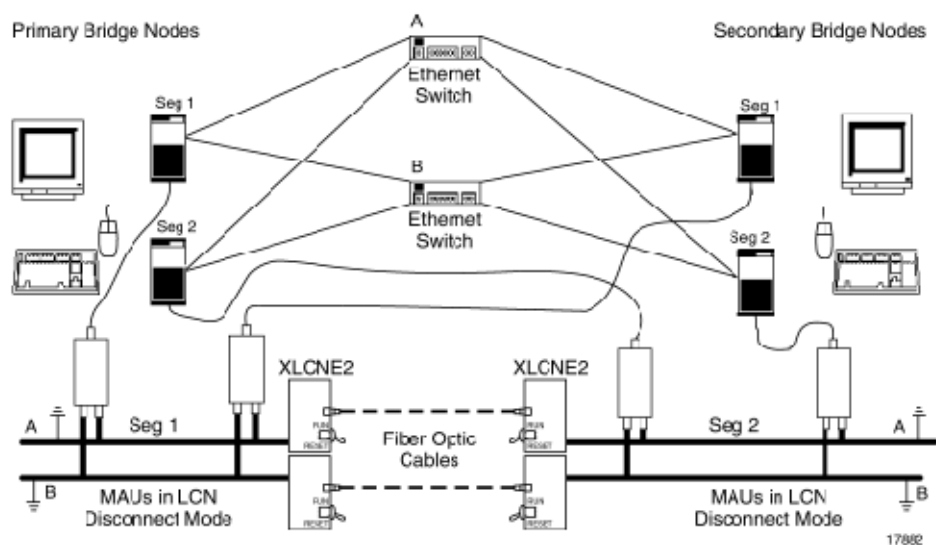
Step

Action

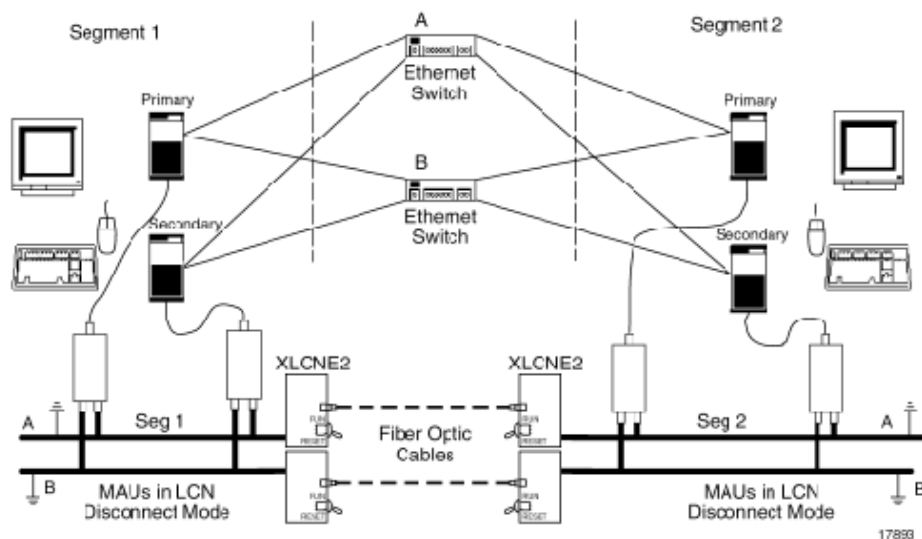
| TPN Bridge Viewer – WS Counters                            |                                 |                                          |
|------------------------------------------------------------|---------------------------------|------------------------------------------|
| Note: Clear all counters before proceeding with this check |                                 |                                          |
| Configured Primary                                         | 1                               | 0                                        |
| Running Primary                                            | 1                               | 0                                        |
| Tx Total                                                   | Incrementing 1 every 10 seconds | 0                                        |
| Tx Failures                                                | 0                               | 0                                        |
| Tx LCN Msg                                                 | 0                               | 0                                        |
| Tx Buddy Msg                                               | 0                               | 0                                        |
| Tx PLAT Msg                                                | Incrementing 1 every 10 seconds | 0                                        |
| Rx Total                                                   | May increment                   | Incrementing at least 1 every 10 seconds |
| Rx Failures                                                | 0                               | 0                                        |
| Rx Lost                                                    | 0                               | 0                                        |
| Rx Out of Order                                            | 0                               | 0                                        |
| Rx Duplicated                                              | 0                               | 0                                        |
| Rx LCN Msg                                                 | May increment                   | May increment                            |
| Rx LCN Lost                                                | 0                               | 0                                        |
| Rx LCN Out of Order                                        | 0                               | 0                                        |
| Rx Buddy Msg                                               | 0                               | 0                                        |
| Rx Buddy Msg Lost                                          | 0                               | 0                                        |
| Rx Buddy Msg Out of Order                                  | 0                               | 0                                        |
| Rx PLAT Msg                                                | May increment                   | Incrementing at least 1 every 10 seconds |
| Rx PLAT Msg Lost                                           | 0                               | 0                                        |
| Rx PLAT Msg Out of Order                                   | 0                               | 0                                        |

## 9 Remove the private LCNs.

### 5.6. Replacing an LCNE with a TPN Bridge



**Figure 5-11 LCNE Replacement — Bridge Equipment Collocated**



**Figure 5-12 LCNE Replacement — Bridge Segments Remote**

### Checking the Bridge node communications on the LCN



#### ATTENTION

If this is not the first segmentation of the LCN, this section is to be skipped.

---



#### ATTENTION

During this communication check, placing BOTH segments in **Connected Mode** at the same time WILL result in LCN node to go **ISOLATED** and **FAILURE**.

---

| Step                                                                                | Action                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                                                                                   | Connect all of the Bridge MAUs to the LCN cables.                                                                                                                                                                                          |
| 2                                                                                   | Check the <b>SYSTEM STATUS</b> display for LCN cable status to ensure no cable problems.                                                                                                                                                   |
| 3                                                                                   | Check the <b>LCN OVERVIEW FROM ALL NODES</b> display to ensure no nodes indicate status of <b>ISOLATED</b> .                                                                                                                               |
|  | <b>ATTENTION</b><br><u>Do not</u> perform the following steps if this is not the first segmentation. Performing these steps on a previously segmented system will cause the status of the nodes to go <b>ISOLATED</b> and <b>FAILURE</b> . |
| 4                                                                                   | Using the <b>TPN Bridge Configuration/Status Snap-In</b> place the <u>first (1<sup>st</sup>)</u> segment in <b>Connected Mode</b> .                                                                                                        |
| 5                                                                                   | Verify the <b>SYSTEM STATUS</b> display indicates a status <b>OK-BRG</b> for the Primary. The Secondary Bridge may indicate <b>OK-BRG</b> or <b>TEST</b> .                                                                                 |
| 6                                                                                   | Using the <b>LOCAL DRIVER DATA</b> display, verify that the lowest US is still functioning as Cable Master and its Swap Ticks Left counter is changing. Reload the node if it is not functioning as a cable Master.                        |

---

## 5. TPN Bridge Hot Insertion

### 5.6. Replacing an LCNE with a TPN Bridge

#### Step

#### Action

- 7 Bring up the **Bridge Status**, **TDC Status** and **Node Status** displays on the connected Bridge nodes. Verify the expected values using the "Connected to LCN (Test Mode - Single Segment) Health Check" chart below.

Connected to LCN (Test Mode – Single Segment) Health Check

| Value Name                               | Primary Expected Value | Secondary Expected Value |
|------------------------------------------|------------------------|--------------------------|
| <b>TPN Bridge Viewer - Bridge Status</b> |                        |                          |
| Cycle Counter                            | Changing               | Changing                 |
| Load Progress                            | 254                    | 254                      |
| OLAN OK                                  | 3                      | 3                        |
| Buddy OK                                 | 1                      | 1                        |
| LCN → NT Data (C1)                       | Changing               | Changing                 |
| NT → LCN Buffers (C1)                    | Changing               | Changing                 |
| Use NT LAT                               | 0                      | 0                        |
| Seconds Since Msg Received (NT)          | Cycling from 0 to 10   | Cycling 0 to 1           |
| Seconds Since Msg Received (LCNI)        | <11 seconds            | <11 seconds              |
| <b>TPN Bridge Viewer - Node Status</b>   |                        |                          |
| Node [primary node number]               | ME-BRIDGE prim         | BUDDY prim               |
| Node [secondary node number]             | BUDDY sec              | ME-BRIDGE sec            |
| All other LCN Node numbers ...           | Appear in list         | Appear in list           |
| <b>TPN Bridge Viewer – TDC Status</b>    |                        |                          |
| System Status – Physical Dirty           | Set                    | Set                      |
| System Status – Looping                  | Set                    | Set                      |
| System Status – Disconnected             | Cleared                | Cleared                  |
| Notice Status – Multi Alone In Ring      | Cleared                | May be Set – this is OK  |
| Notice Status – All Other Bits           | Cleared                | Cleared                  |
| Time Mastership – all Bits               | Cleared                | Cleared                  |

## 5. TPN Bridge Hot Insertion

### 5.6. Replacing an LCNE with a TPN Bridge

Step

Action

| TPN Bridge Viewer – WS Counters                            |                                 |                                 |
|------------------------------------------------------------|---------------------------------|---------------------------------|
| Note: Clear all counters before proceeding with this check |                                 |                                 |
| Configured Primary                                         | 1                               | 0                               |
| Running Primary                                            | 1                               | 0                               |
| Tx Total                                                   | Changing rapidly                | Incrementing 1 every 10 seconds |
| Tx Failures                                                | 0                               | 0                               |
| Tx LCN Msg                                                 | Changing rapidly                | 0                               |
| Tx Buddy Msg                                               | Incrementing 2 per second       | Incrementing 1 every 10 seconds |
| Tx PLAT Msg                                                | Changing – within 10 seconds    | 0                               |
| Rx Total                                                   | Incrementing 1 every 10 seconds | Changing rapidly                |
| Rx Failures                                                | 0                               | 0                               |
| Rx Lost                                                    | 0                               | 0                               |
| Rx Out of Order                                            | 0                               | 0                               |
| Rx Duplicated                                              | 0                               | 0                               |
| Rx LCN Msg                                                 | 0                               | Changing rapidly                |
| Rx LCN Lost                                                | 0                               | 0                               |
| Rx LCN Out of Order                                        | 0                               | 0                               |
| Rx Buddy Msg                                               | Incrementing 1 every 10 seconds | Incrementing 2 per second       |
| Rx Buddy Msg Lost                                          | 0                               | 0                               |
| Rx Buddy Msg Out of Order                                  | 0                               | 0                               |
| Rx PLAT Msg                                                | <11 seconds                     | <11 seconds                     |
| Rx PLAT Msg Lost                                           | 0                               | 0                               |
| Rx PLAT Msg Out of Order                                   | 0                               | 0                               |

Refer to "Table 4-7" for additional information.

## 5. TPN Bridge Hot Insertion

### 5.6. Replacing an LCNE with a TPN Bridge

| Step | Action |
|------|--------|
|------|--------|

- |   |                                                                                                                                                                                                               |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8 | Bring up the <b>Bridge Status</b> , <b>TDC Status</b> , and <b>Node Status</b> displays on the disconnected Bridge nodes. Verify the expected values using the "Disconnected State Health Check" chart below. |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Disconnected State Health Check

| Value Name                                                        | Primary Expected Value          | Secondary Expected Value |
|-------------------------------------------------------------------|---------------------------------|--------------------------|
| <b>TPN Bridge Viewer - Bridge Status</b>                          |                                 |                          |
| Cycle Counter                                                     | Changing                        | Changing                 |
| Load Progress                                                     | 254                             | 254                      |
| OLAN OK                                                           | 3                               | 3                        |
| Buddy OK                                                          | 0                               | 1                        |
| <b>TPN Bridge Viewer - Node Status</b>                            |                                 |                          |
| Node [primary node number]                                        | ME-BRIDGE prim                  | Not displayed            |
| Node [secondary node number]                                      | Not Displayed                   | ME-BRIDGE sec            |
| <b>TPN Bridge Viewer - TDC Status</b>                             |                                 |                          |
| System Status – Physical Dirty                                    | Set                             | Set                      |
| System Status – Looping                                           | Set                             | Set                      |
| System Status – Disconnected                                      | Set                             | Set                      |
| System Status – LCNI Media Dead                                   | Cleared                         | May be Set – this is OK  |
| Notice Status – Multi Alone In Ring                               | Cleared                         | May be Set – this is OK  |
| Notice Status – All Other Bits                                    | Cleared                         | Cleared                  |
| <b>TPN Bridge Viewer - WS Counters</b>                            |                                 |                          |
| <i>Note: Clear all counters before proceeding with this check</i> |                                 |                          |
| Configured Primary                                                | 1                               | 0                        |
| Running Primary                                                   | 1                               | 0                        |
| Tx Total                                                          | Incrementing 1 every 10 seconds | 0                        |
| Tx Failures                                                       | 0                               | 0                        |
| Tx LCN Msg                                                        | 0                               | 0                        |
| Tx Buddy Msg                                                      | 0                               | 0                        |
| Tx PLAT Msg                                                       | Incrementing 1 every 10 seconds | 0                        |
| Rx Total                                                          | Incrementing rapidly            | Incrementing rapidly     |
| Rx Failures                                                       | 0                               | 0                        |
| Rx Lost                                                           | 0                               | 0                        |
| Rx Out of Order                                                   | 0                               | 0                        |
| Rx Duplicated                                                     | 0                               | 0                        |

## 5. TPN Bridge Hot Insertion

### 5.6. Replacing an LCNE with a TPN Bridge

| Step | Action                    |                                          |
|------|---------------------------|------------------------------------------|
|      | Rx LCN Msg                | Incrementing rapidly                     |
|      | Rx LCN Lost               | 0                                        |
|      | Rx LCN Out of Order       | 0                                        |
|      | Rx Buddy Msg              | 0                                        |
|      | Rx Buddy Msg Lost         | 0                                        |
|      | Rx Buddy Msg Out of Order | 0                                        |
|      | Rx PLAT Msg               | Incrementing at least 1 every 10 seconds |
|      | Rx PLAT Msg Lost          | 0                                        |
|      | Rx PLAT Msg Out of Order  | 0                                        |

- 9** Using the **TPN Bridge Configuration/Status Snap-In** place the segment just tested in **Disconnected Mode**.

Both segments should be in the **Disconnected Mode** at this time.

- 10** Verify the **SYSTEM STATUS** display indicates **OFF** for both Bridge LCN addresses.
- 11** Using the **TPN Bridge Configuration/Status Snap-In** place the second (2<sup>nd</sup>) segment in **Connected Mode**.
- 12** Verify the **SYSTEM STATUS** displays the Bridge nodes with **OK-BRG** status.
- 13** Bring up the **Bridge Status**, **TDC Status** and **Node Status** displays on all Bridge nodes. Verify the expected values using the "Connected to LCN (Test Mode – Single Segment) Health Check" chart below.

Connected to LCN (Test Mode – Single Segment) Health Check

| Value Name                               | Primary Expected Value | Secondary Expected Value |
|------------------------------------------|------------------------|--------------------------|
| <b>TPN Bridge Viewer - Bridge Status</b> |                        |                          |
| Cycle Counter                            | Changing               | Changing                 |
| Load Progress                            | 254                    | 254                      |
| OLAN OK                                  | 3                      | 3                        |
| Buddy OK                                 | 1                      | 1                        |
| LCN → NT Data (C1)                       | Changing               | Changing                 |
| NT → LCN Buffers (C1)                    | Changing               | Changing                 |
| Use NT LAT                               | 0                      | 0                        |

## 5. TPN Bridge Hot Insertion

### 5.6. Replacing an LCNE with a TPN Bridge

| Step | Action                                                            |                                 |
|------|-------------------------------------------------------------------|---------------------------------|
|      | Seconds Since Msg Received (NT)                                   | Cycling from 0 to 10            |
|      | Seconds Since Msg Received (LCNI)                                 | Cycling 0 to 1                  |
|      | <b>TPN Bridge Viewer - Node Status</b>                            |                                 |
|      | Node [primary node number]                                        | ME-BRIDGE prim                  |
|      | Node [secondary node number]                                      | BUDDY prim                      |
|      | Node [secondary node number]                                      | BUDDY sec                       |
|      | Node [secondary node number]                                      | ME-BRIDGE sec                   |
|      | All other LCN Node numbers ...                                    | Appear in list                  |
|      | <b>TPN Bridge Viewer - TDC Status</b>                             |                                 |
|      | System Status – Physical Dirty                                    | Set                             |
|      | System Status – Looping                                           | Set                             |
|      | System Status – Disconnected                                      | Cleared                         |
|      | Notice Status – Multi Alone In Ring                               | Cleared                         |
|      | Notice Status – All Other Bits                                    | May be Set – this is OK         |
|      | Time Mastership – all Bits                                        | Cleared                         |
|      | <b>TPN Bridge Viewer – WS Counters</b>                            |                                 |
|      | <b>Note: Clear all counters before proceeding with this check</b> |                                 |
|      | Configured Primary                                                | 1                               |
|      | Running Primary                                                   | 0                               |
|      | Tx Total                                                          | Changing rapidly                |
|      | Tx Failures                                                       | Incrementing 1 every 10 seconds |
|      | Tx LCN Msg                                                        | 0                               |
|      | Tx Buddy Msg                                                      | Changing rapidly                |
|      | Tx PLAT Msg                                                       | Incrementing 2 per second       |
|      | Rx Total                                                          | Incrementing 1 every 10 seconds |
|      | Rx Failures                                                       | Incrementing 1 every 10 seconds |
|      | Rx Lost                                                           | 0                               |
|      | Rx Out of Order                                                   | 0                               |
|      | Rx Duplicated                                                     | 0                               |
|      | Rx LCN Msg                                                        | 0                               |
|      | Rx LCN Lost                                                       | Changing rapidly                |
|      | Rx LCN Out of Order                                               | 0                               |

## 5. TPN Bridge Hot Insertion

### 5.6. Replacing an LCNE with a TPN Bridge

| Step | Action                    |                                          |
|------|---------------------------|------------------------------------------|
|      | Rx Buddy Msg              | Incrementing 1 every 10 seconds          |
|      | Rx Buddy Msg Lost         | 0                                        |
|      | Rx Buddy Msg Out of Order | 0                                        |
|      | Rx PLAT Msg               | Incrementing at least 1 every 10 seconds |
|      | Rx PLAT Msg Lost          | 0                                        |
|      | Rx PLAT Msg Out of Order  | 0                                        |

- 14** Bring up the **Bridge Status**, **TDC Status** and **Node Status** displays on the disconnected Bridge nodes. Verify the expected values using the "Disconnected State Health Check" chart below.

Disconnected State Health Check

| Value Name                               | Primary Expected Value      | Secondary Expected Value    |
|------------------------------------------|-----------------------------|-----------------------------|
| <b>TPN Bridge Viewer - Bridge Status</b> |                             |                             |
| Cycle Counter                            | Changing                    | Changing                    |
| Load Progress                            | 254                         | 254                         |
| OLAN OK                                  | 3                           | 3                           |
| Buddy OK                                 | 0                           | 0 or 1                      |
| <b>TPN Bridge Viewer - Node Status</b>   |                             |                             |
| Node [primary node number]               | ME-BRIDGE prim              | Not displayed               |
| Node [secondary node number]             | Not Displayed               | ME-BRIDGE sec               |
| All other nodes will disappear           | Wait for 120 second timeout | Wait for 120 second timeout |
| <b>TPN Bridge Viewer - TDC Status</b>    |                             |                             |
| System Status – Physical Dirty           | Set                         | Set                         |
| System Status – Looping                  | Set                         | Set                         |
| System Status – Disconnected             | Set                         | Set                         |
| System Status – LCNI Media Dead          | Cleared                     | May be Set – this is OK     |
| Notice Status – Multi Alone In Ring      | Cleared                     | May be Set – this is OK     |
| Notice Status – All Other Bits           | Cleared                     | Cleared                     |

## 5. TPN Bridge Hot Insertion

### 5.6. Replacing an LCNE with a TPN Bridge

---

Step

Action

|  |                                                                                                                  |                                          |                                          |
|--|------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------|
|  | <div>TPN Bridge Viewer – WS Counters</div> <div>Note: Clear all counters before proceeding with this check</div> |                                          |                                          |
|  | Configured Primary                                                                                               | 1                                        | 0                                        |
|  | Running Primary                                                                                                  | 1                                        | 0                                        |
|  | Tx Total                                                                                                         | Incrementing 1 every 10 seconds          | 0                                        |
|  | Tx Failures                                                                                                      | 0                                        | 0                                        |
|  | Tx LCN Msg                                                                                                       | 0                                        | 0                                        |
|  | Tx Buddy Msg                                                                                                     | 0                                        | 0                                        |
|  | Tx PLAT Msg                                                                                                      | Incrementing 1 every 10 seconds          | 0                                        |
|  | Rx Total                                                                                                         | Incrementing rapidly                     | Incrementing rapidly                     |
|  | Rx Failures                                                                                                      | 0                                        | 0                                        |
|  | Rx Lost                                                                                                          | 0                                        | 0                                        |
|  | Rx Out of Order                                                                                                  | 0                                        | 0                                        |
|  | Rx Duplicated                                                                                                    | 0                                        | 0                                        |
|  | Rx LCN Msg                                                                                                       | Incrementing rapidly                     | Incrementing rapidly                     |
|  | Rx LCN Lost                                                                                                      | 0                                        | 0                                        |
|  | Rx LCN Out of Order                                                                                              | 0                                        | 0                                        |
|  | Rx Buddy Msg                                                                                                     | 0                                        | 0                                        |
|  | Rx Buddy Msg Lost                                                                                                | 0                                        | 0                                        |
|  | Rx Buddy Msg Out of Order                                                                                        | 0                                        | 0                                        |
|  | Rx PLAT Msg                                                                                                      | Incrementing at least 1 every 10 seconds | Incrementing at least 1 every 10 seconds |
|  | Rx PLAT Msg Lost                                                                                                 | 0                                        | 0                                        |
|  | Rx PLAT Msg Out of Order                                                                                         | 0                                        | 0                                        |

15

Using the **TPN Bridge Configuration/Status Snap-In** place the segment just tested into the **Disconnected Mode**.

16

Verify the **SYSTEM STATUS** display indicates **OFF** for both Bridge LCN addresses.



## 5. TPN Bridge Hot Insertion

### 5.6. Replacing an LCNE with a TPN Bridge

---

#### Breaking the LCN

---



#### WARNING

Failure to do the following steps in order and in a timely manner can, and will, cause LCN nodes to go **ISOLATED**. All remaining steps in this procedure must be completed within 2 minutes!

---

| Step | Action                                                                                                                                                  |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Locate the LCNE boards for the LCN Extension Set that is being replaced.                                                                                |
| 2    | The <b>RUN/RESET</b> switch on the LCNE boards should be in the <b>RUN</b> position. Reference "Figure 5-13 XLCNE2 Board Assembly" for switch location. |
| 3    | Place the <b>RUN/RESET</b> switch for <u>both</u> LCN cables to the <b>RESET</b> position.                                                              |
| 4    | Using the <b>TPN Bridge Config/Status Snap-in</b> place all segments in <b>Connected Mode</b> .                                                         |
| 5    | Check the <b>SYSTEM STATUS</b> display for LCN cable status to ensure no cable problems.                                                                |

---

## 5. TPN Bridge Hot Insertion

### 5.6. Replacing an LCNE with a TPN Bridge

Step

6

Bring up the **Bridge Status**, **TDC Status** and **Node Status** displays on all Bridge nodes. Verify the expected values using the "Connected to LCN (Initial Insertion – 2 Segments) Health Check" chart below.

Connected to LCN (Initial Insertion – 2 Segments) Health Check

| Value Name                                                        | Primary Expected Value       | Secondary Expected Value        |
|-------------------------------------------------------------------|------------------------------|---------------------------------|
| <b>TPN Bridge Viewer - Bridge Status</b>                          |                              |                                 |
| Cycle Counter                                                     | Changing                     | Changing                        |
| Load Progress                                                     | 254                          | 254                             |
| OLAN OK                                                           | 3                            | 3                               |
| Buddy OK                                                          | 1                            | 1                               |
| NT → LCN Data (C1)                                                | Changing                     | Changing                        |
| LCN → NT Data (C1)                                                | Changing                     | Changing                        |
| LCN → NT Buffers (C1)                                             | Changing slowly              | Changing slowly                 |
| NT → LCN Buffers (C1)                                             | Changing                     | Changing                        |
| Use NT LAT                                                        | 1                            | 1                               |
| Seconds Since Msg Received (NT)                                   | Changing – within 10 seconds | Changing – within 10 seconds    |
| Seconds Since Msg Received (LCNI)                                 | Changing – within 10 seconds | Changing – within 10 seconds    |
| <b>TPN Bridge Viewer - Node Status</b>                            |                              |                                 |
| Node [primary node number]                                        | ME-BRIDGE prim               | BUDDY prim                      |
| Node [secondary node number]                                      | BUDDY sec                    | ME-BRIDGE sec                   |
| All other LCN Node numbers ...                                    | Appear in list               | Appear in list                  |
| <b>TPN Bridge Viewer – TDC Status</b>                             |                              |                                 |
| System Status – Recompute LAT                                     | Cleared                      | May be Set                      |
| System Status – Physical Dirty                                    | Set                          | Set                             |
| System Status – Looping                                           | Set                          | Set                             |
| System Status – Disconnected                                      | Cleared                      | Cleared                         |
| Notice Status – Multi Alone In Ring                               | Cleared                      | Cleared                         |
| Notice Status – All Other Bits                                    | Cleared                      | Cleared                         |
| Time Mastership – all Bits                                        | Cleared                      | Cleared                         |
| <b>TPN Bridge Viewer – WS Counters</b>                            |                              |                                 |
| <b>Note: Clear all counters before proceeding with this check</b> |                              |                                 |
| Configured Primary                                                | 1                            | 0                               |
| Running Primary                                                   | 1                            | 0                               |
| Tx Total                                                          | Changing rapidly             | Incrementing 1 every 10 seconds |

## 5. TPN Bridge Hot Insertion

### 5.6. Replacing an LCNE with a TPN Bridge

| Step | Action                    |                                 |                                 |
|------|---------------------------|---------------------------------|---------------------------------|
|      | Tx Failures               | 0                               | 0                               |
|      | Tx LCN Msg                | Changing rapidly                | 0                               |
|      | Tx Buddy Msg              | Incrementing 2 per second       | Incrementing 1 every 10 seconds |
|      | Tx PLAT Msg               | Incrementing within 10 seconds  | 0                               |
|      | Rx Total                  | Changing rapidly                | Changing rapidly                |
|      | Rx Failures               | 0                               | 0                               |
|      | Rx Lost                   | 0                               | 0                               |
|      | Rx Out of Order           | 0                               | 0                               |
|      | Rx Duplicated             | 0                               | 0                               |
|      | Rx LCN Msg                | Changing rapidly                | Changing rapidly                |
|      | Rx LCN Lost               | 0                               | 0                               |
|      | Rx LCN Out of Order       | 0                               | 0                               |
|      | Rx Buddy Msg              | Incrementing 1 every 10 seconds | Incrementing 2 per second       |
|      | Rx Buddy Msg Lost         | 0                               | 0                               |
|      | Rx Buddy Msg Out of Order | 0                               | 0                               |
|      | Rx PLAT Msg               | Changing within 10 seconds      | Changing within 10 seconds      |
|      | Rx PLAT Msg Lost          | 0                               | 0                               |
|      | Rx PLAT Msg Out of Order  | 0                               | 0                               |

- 7 If all Bridge nodes are OK, go to step 8.
  - If the Bridges are not functioning:
  - Using the TPN Bridge Configuration/Status Snap-In place all new segments in Disconnected Mode.
  - Any errors during Disconnect – pull the power on all new Bridges.
  - Place the RUN/RESET switch for both LCN cables to the RUN position.
  - Troubleshoot the problem. Refer to "Bridge Operation Checklist" in Section 4.4 for additional information.
- 8 Using the **TPN Bridge Config/Status Snap-in** enable the **Time Sync** pre-defined function for each of the two segments.
- 9 Use the schematic CLOKSTAT to verify that all the nodes on both segments transition to LISTENER mode within a few minutes.
- 10 Disconnect the LCNE fiber optic cables from the LCNE boards.



**WARNING**

Failure to disconnect the fiber optic cables could cause an LCN failure if the LCNE **RUN/RESET** switch is ever set to the **RUN** position while the TPN Bridge is running.

---

**Setting Time Sync options**

Set all segment time sync options according to the procedure in Section 4.11, "Setting TPN Bridge Time Sync Options." Also refer to Table 3-2 Clock Sources and Configuration Bits for additional information.

## **5.7 Replacing the Network Card**

### **Overview**

The dual NIC network adapter card originally installed in TPN Bridge nodes has been replaced with a different model. When replacing the TPN Bridge network card, you will need to update driver for the network adapter card and configure the card's MAC addresses. The overall procedures for upgrading the network card are:

- Review the most recent SCN
- Turn off time sync on all segments
- Close all MMC consoles
- Perform proper verification and shut-down procedures
- Remove the computer from the cabinet
- Install the new dual NIC network card
- Update the dual NIC network card driver
- Configure MAC addresses
- Re-enable time sync

### **Replacing the dual NIC network card in a TPN Bridge pair**

If you are replacing the network adapter in both the primary and secondary bridge nodes, it is not necessary to replace both cards at the same time. You may leave one of the nodes up and running while you perform the replacement procedure on the other node.

**IMPORTANT – Perform this procedure before you begin**

Before you begin installing the new card, perform all steps in the following procedure.

| Step | Action                                                                                                                                                                                                                                                                                                                                                                                        |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Have available the latest TPN Bridge SCN.                                                                                                                                                                                                                                                                                                                                                     |
| 2    | <p>If you are replacing a failed dual NIC network card, verify that all other bridges are completely operational before you begin.</p> <p>If you are updating the dual NIC network card, verify that ALL bridges including the one being updated are completely operational before you begin.</p> <p>Refer to “Section 6.1, “TPS LCN System Checkout” for details.</p>                        |
|      | <div><b>CAUTION – Turn off Time Sync on all segments</b><p>The system Time Sync must be turned off on all segments. Refer to the “Turning off the time synchronization option” procedure in Section 4.11. Failure to turn off Time Sync can result in TOTAL collapse of the entire TPS/LCN system.</p></div> |
| 4    | Close all MMC consoles.                                                                                                                                                                                                                                                                                                                                                                       |
| 5    | Use Task Manager on each TPN Bridge to kill all mmc and dllhost applications that are running.                                                                                                                                                                                                                                                                                                |

## 5. TPN Bridge Hot Insertion

### 5.7. Replacing the Network Card

---

#### Remove the computer from the cabinet

---



#### CAUTION

When removing the TPN Bridge computer in the cabinet, DO NOT slide the computer from the cabinet's mounting chassis. Doing so can destroy the foam insulators and cause a complete ISOLATION.

---



#### ESD HAZARD

It is important that you wear a properly connected Electrostatic Discharge (ESD) wrist strap while removing, handling, and installing the network card.

---

| Step | Action                                                                                                                                                    |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Shut down the Windows NT operating system.                                                                                                                |
| 2    | Turn off the ac breaker for the cabinet and disconnect all the ac power cords from the breaker box.                                                       |
| 3    | Disconnect the LCN coax cable T-connector from the MAU assembly.                                                                                          |
| 4    | Remove the front and rear covers for the mounting chassis.                                                                                                |
| 5    | Disconnect the cables from the rear of the computer.                                                                                                      |
| 6    | If the mounting chassis for the computer does not have a node or chassis mounted directly above it, go to Step 9.<br><br>Otherwise, continue with Step 7. |
| 7    | Remove all the screws that are securing the mounting chassis to the cabinet.                                                                              |
| 8    | Remove the mounting chassis (with the computer) from the cabinet.                                                                                         |
| 9    | Remove the screws from the top cover of the mounting chassis and lift the cover off.                                                                      |
| 10   | Remove the piece of isolation foam from the top of the computer.                                                                                          |
| 11   | Lift the computer out of the mounting chassis.                                                                                                            |

---

### Install the dual NIC network card

Use the following procedure to replace the dual NIC network card.

| Step                                                                                                      | Action                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 1                                                                                                         | Identify the A and B ports on the new network adapter card. See Figure 5-14.                                               |
| 2                                                                                                         | Wearing proper ESD protection, remove the existing network adapter card and install the new adapter card in the same slot. |
| 3                                                                                                         | Secure the card in the slot.                                                                                               |
| 4                                                                                                         | Replace the cover and return the computer to the cabinet.                                                                  |
| 5                                                                                                         | Re-connect all computer cables and network cables.                                                                         |
| <b><i>DO NOT re-connect the LCN coax cable T-connector yet.</i></b>                                       |                                                                                                                            |
| If necessary, refer to Section 2, "Installing the TPN Bridge," of the Overview and Implementation manual. |                                                                                                                            |
| 6                                                                                                         | Power up the node.                                                                                                         |

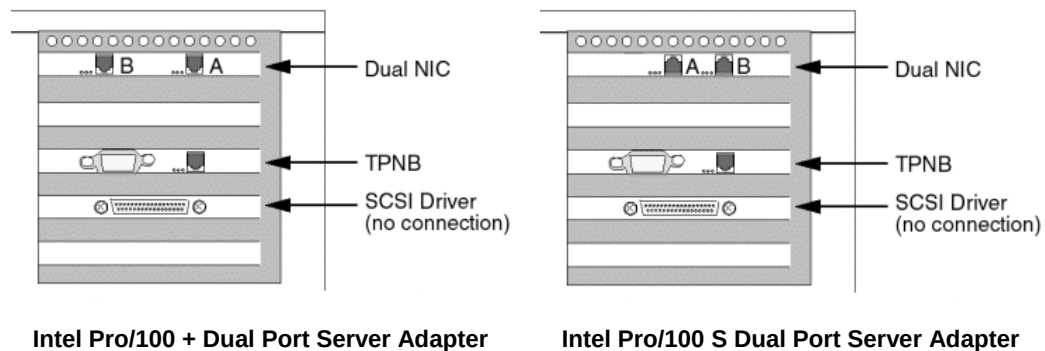


Figure 5-14 Dual NIC Adapter Ports for TPN Bridge

## 5. TPN Bridge Hot Insertion

### 5.7. Replacing the Network Card

---

#### Update Dual NIC network adapter driver

| Step | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Insert the TPN Bridge Software CD into the CD-ROM drive and cancel the automatic install routine.                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2    | Copy the entire Intel folder from the TPN Bridge Software CD to your local hard drive.<br><br><b>ATTENTION:</b> It is very important to copy the entire folder along with its sub-folders.                                                                                                                                                                                                                                                                                                                                            |
| 3    | Select <b>Start &gt; Settings &gt; Control Panel</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4    | Double-click <b>Network</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5    | Select the <b>Adapters</b> tab from the <b>Network</b> dialog.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 6    | Select one of the network adapter ports for the dual NIC adapter card, and click <b>Update</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 7    | From the Windows NT Setup dialog, type in the location you copied the Intel folder to followed by the complete driver location path, which is:<br><br><i>Intel\network\disk1</i><br><br><b>For example:</b> C:\Intel\network\disk1                                                                                                                                                                                                                                                                                                    |
| 8    | Click <b>Continue</b> and then close the <b>Network</b> dialog box.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 9    | Remove the TPN Bridge Software CD.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 10   | Restart the computer when prompted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 11   | To verify the driver was updated, perform the following: <ul style="list-style-type: none"><li>• Select <b>Start &gt; Settings &gt; Control Panel</b>.</li><li>• Double-click <b>Network</b>.</li><li>• Select one of the network adapter ports for the dual NIC adapter card, and click <b>Properties</b>.</li><li>• Select the Network Driver tab from the Intel ProSet dialog.</li><li>• Review the description field under Driver Information to verify the driver version. Refer to the latest SCN for driver details.</li></ul> |

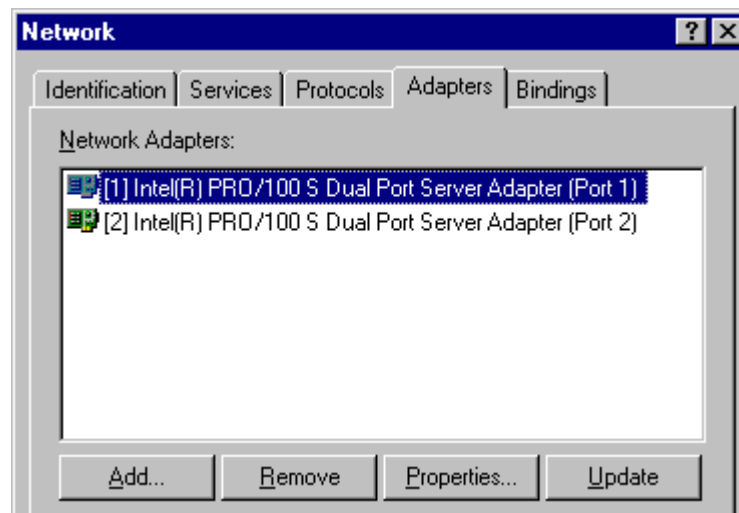
### Configure MAC Addresses for dual NIC network adapter ports



#### ATTENTION

It is very important that you configure both network adapter ports on the dual NIC card to have the same MAC addresses. Doing so will prevent collisions.

| Step | Action                                                                                   |
|------|------------------------------------------------------------------------------------------|
| 1    | From the desktop, right-click <b>Network Neighborhood</b> and select <b>Properties</b> . |
| 2    | Select the <b>Adapters</b> tab.                                                          |
| 3    | Select the adapter for Port 1 and click <b>Properties</b> .                              |



- 4 From the **Intel PROSet** window, select the **General** tab.

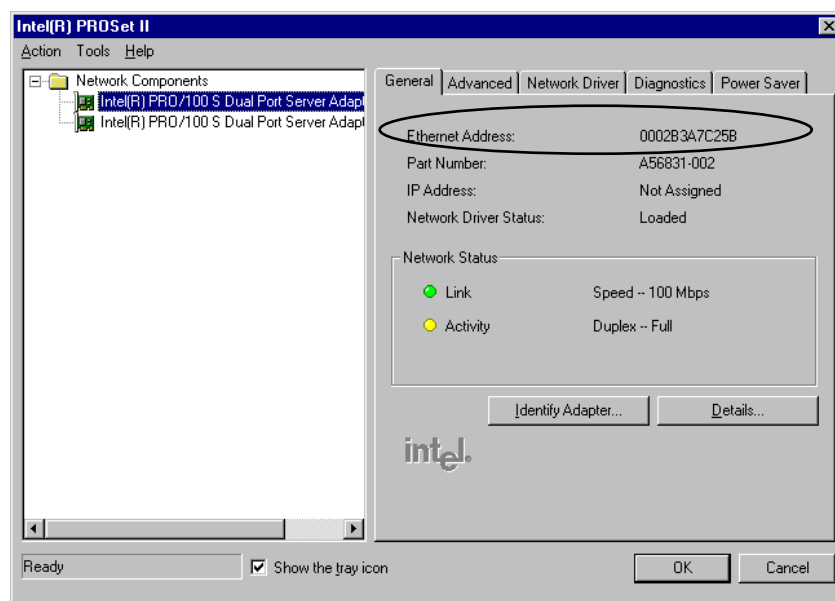
## 5. TPN Bridge Hot Insertion

### 5.7. Replacing the Network Card

---

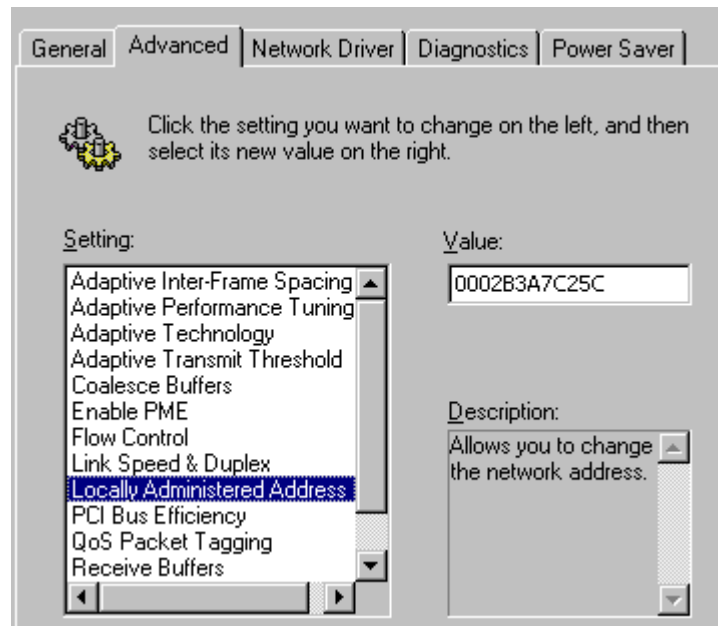
| Step | Action                                                            |
|------|-------------------------------------------------------------------|
| 5    | Identify the Ethernet Address for the Adapter and record it here: |

**Note:** This is the same as the port's physical MAC address.



- Click **Cancel** to return to the Network window.
- Select the adapter for Port 2 and click **Properties**.
- From the **Intel PROSet** window, select the **Advanced** tab.

| Step | Action                                                                                                                                         |
|------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 9    | In the <b>Setting</b> area, click the <b>Locally Administered Address</b> field to bring up Adapter 2's MAC address in the <b>Value</b> field. |

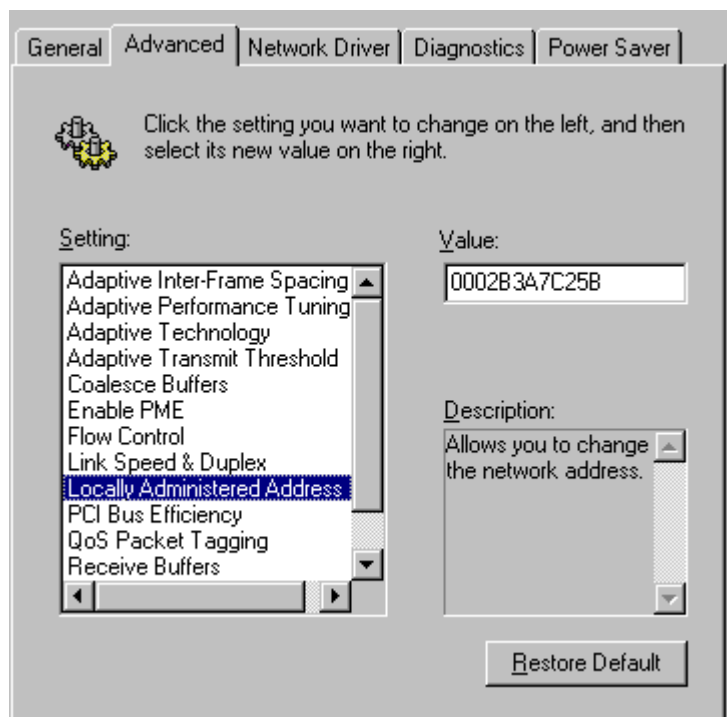


## 5. TPN Bridge Hot Insertion

### 5.7. Replacing the Network Card

---

| Step | Action                                                                                                                                |
|------|---------------------------------------------------------------------------------------------------------------------------------------|
| 10   | Clear the address in the <b>Value</b> field and type in the address for Adapter port 1 that you recorded in Step 5 of this procedure. |



- |    |                                                           |
|----|-----------------------------------------------------------|
| 11 | Click <b>OK</b> .                                         |
| 12 | Close the <b>Network</b> dialog and reboot when prompted. |
-

**Verify LCNP and TPN Bridge status**

| Step     | Action                                                                                                                                                                                                                                                |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b> | Check the LCNP Status:<br><br>If the LCNP State is NOT OK, you will need to resolve the problem before re-connecting the LCN cables.<br><br>If the LCNP state is OK, shut down and power down the node and re-connect the LCN coax cable T-connector. |
| <b>2</b> | Power up the node.                                                                                                                                                                                                                                    |
| <b>3</b> | Open the TPN Bridge Viewer and select TPN Bridge View.                                                                                                                                                                                                |
| <b>4</b> | Verify that the values are what they should be. Refer to Table 6-1.                                                                                                                                                                                   |
| <b>5</b> | Re-enable the time sync option. Refer to Section 4.11, "Setting TPN Bridge Time Sync Options."                                                                                                                                                        |

## 5. TPN Bridge Hot Insertion

### 5.8. Replacing a Bridge and Restoring Current Configuration

---

## 5.8 Replacing a Bridge and Restoring Current Configuration

### Overview

This procedure will restore a single Bridge node that requires a complete reinstallation of all software (injection) or replacement of an existing Bridge computer with a spare node that has been previously injected. You would also use these procedures if you are replacing the network card.



#### WARNING

Do not replace a TPN Bridge with a Hot Spare TPN Bridge PC without first re-injecting and re-configuring the Hot Spare. ***Replacing a failed TPN Bridge PC with the Hot Spare that is not configured properly can collapse the entire network.*** You must perform a complete injection of the TPN Bridge software and properly configure the replacement TPN Bridge PC before using it in the network.

---

### Recovering the hard drive

This procedure utilizes the Customized Boot Disk, created earlier, and the TPN Bridge CD to restore a Bridge node. This procedure will repartition and completely overwrite the hard drive. This procedure is required if replacing a hard drive or if performing a disk recovery. A complete injection of the hard drive is required. If a boot disk is not available, create the disk on another Bridge node. See "Create a Customized Boot Disk" in Section 4.9 for procedure.



#### ATTENTION

The following procedure will repartition and completely overwrite the existing hard drive.

---

| Step | Action                                                                                                                                                                  |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Disconnect the TPN Bridge node being restored from both the LAN and the LCN. <u>Do not</u> break the LCN cables. Do not reconnect any cables until instructed to do so. |
| 2    | Insert the Customized Boot Disk (archived diskette) into the floppy drive. Insert the TPN Bridge CD into the CD drive.                                                  |

| Step                                                                                | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3                                                                                   | <p>Reboot the computer.</p> <p>The computer will reboot from the CD (TPN Bridge is configured for bootable CD). The customer data will be read from the customized diskette. You will be asked to verify if the system should be repartitioned and overwritten.</p> <p>Press any key to continue.</p> <p>The hard drive is now repartitioned and formatted.</p> <p>The system copies the temporary installation files from the CD onto the hard drive. The console screen displays filenames as they are copied.</p> <p>This step will take approximately 5 to 10 minutes.</p> |
| 4                                                                                   | <p>When all the files have been copied, the screen will turn blue and the Windows NT Setup screen will be displayed as additional files are copied to the hard disk.</p> <p>The process of copying the files from the CD to the hard disk will take from 20 to 30 minutes.</p>                                                                                                                                                                                                                                                                                                 |
| 5                                                                                   | Remove the boot diskette and the CD.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6                                                                                   | Reboot the computer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|  | <p><b>ATTENTION</b></p> <p>If you did not remove the boot diskette and the CD, the system will reboot from the boot diskette or the CD, and drive A:\ prompt displays. If this happens, remove the boot diskette and/or the CD, and press the “reset” button and proceed to the next step.</p>                                                                                                                                                                                                                                                                                 |
| 7                                                                                   | The process is automated and the system should not be disturbed until a message is displayed indicating that the load is complete.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|  | <p><b>ATTENTION</b></p> <p>Several reboots will occur and several screens that appear to be prompting for input may be presented. Never touch the keyboard or the mouse until the message in step 7 displays.</p>                                                                                                                                                                                                                                                                                                                                                              |
| 8                                                                                   | <p>The following text displays:</p> <p><b>Messenger Service</b></p> <p><b>Injection Process Complete - Shutdown and Reboot this computer</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 9                                                                                   | Confirm that no error messages are displayed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 10                                                                                  | Click the <b>Ok</b> button to close the load complete message box.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## 5. TPN Bridge Hot Insertion

### 5.8. Replacing a Bridge and Restoring Current Configuration

---

| Step      | Action                                                                                                                                                                      |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>11</b> | Select the following:<br><br><b>Start &gt; Shut Down</b>                                                                                                                    |
| <b>12</b> | Select the <b>Shut down the computer ?</b> radio button.                                                                                                                    |
| <b>13</b> | Click the <b>Yes</b> button to verify the shutdown and reboot the system.<br><br>Select <b>Restart</b> to allow the computer to restart                                     |
| <b>14</b> | This completes the injection and recovery process, and the system has now been configured to the state in which TPS Systems ship from the Honeywell Manufacturing facility. |

---

Restore the Bridge configuration

| Step | Action                                                                                                                                                                                                                                                                 |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Disconnect the bridge from the LAN and LCN                                                                                                                                                                                                                             |
| 2    | If an injection was not performed, reinstall the software using the "TPN Bridge and NT Configuration" procedure in Section 4.9. Enter the <b>Date/Time</b> and <b>Time Zone</b> information.                                                                           |
| 3    | Go to each TPN Bridge node and bring up <b>Task Manager &gt; Processes</b> . Ensure that the <b>Image Names</b> column does not have MMC.exe and DLLHOST.exe running. If these two images are running, select each one and click <b>End Process</b> before continuing. |
| 4    | Configure the computer name and network addresses.                                                                                                                                                                                                                     |

Connect the Bridge to the LAN while the computer reboots.



ATTENTION:

If you are performing this procedure because you replaced your Dual NIC card, verify that you are connecting the LAN cables to the correct adapter ports. See Figure 5-14.

- 5 Using a configured Bridge node (not the Bridge node being restored), start the MMC console program. (See Section 4.9, "TPN Bridge and NT Configuration" for procedure.)

| Bridge   | Node/Type      | Segment | LEDs | State | Cycles | Progress              | OLAN Status | Bridge Errors |
|----------|----------------|---------|------|-------|--------|-----------------------|-------------|---------------|
| BRIDGE01 | 95 - Primary   | Seg 1   | 95   | OK    | 17426  | Running / Time Sync'd | OK          | 817           |
| BRIDGE02 | 0 - Secondary  | Seg 1   | ???  | ???   | ???    | ???                   | ???         | ???           |
| BRIDGE03 | 95 - Primary   | Seg 2   | 95   | OK    | 15491  | Running / Time Sync'd | OK          | 1             |
| BRIDGE04 | 96 - Secondary | Seg 2   | 96   | OK    | 18316  | Running / Time Init   | OK          | 41            |

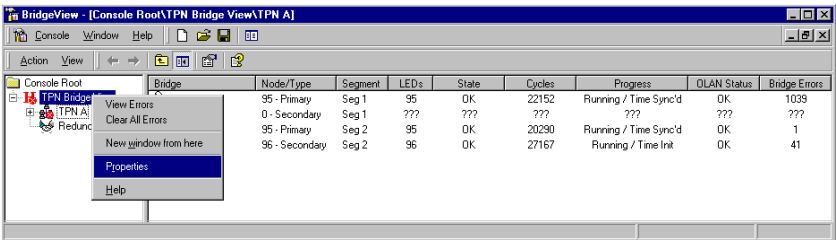
**Note:** For this scenario, **BRIDGE02** has been reinitialized and will be restored.

- 6 Right-click on **TPN Bridge View** level and select **Properties**.

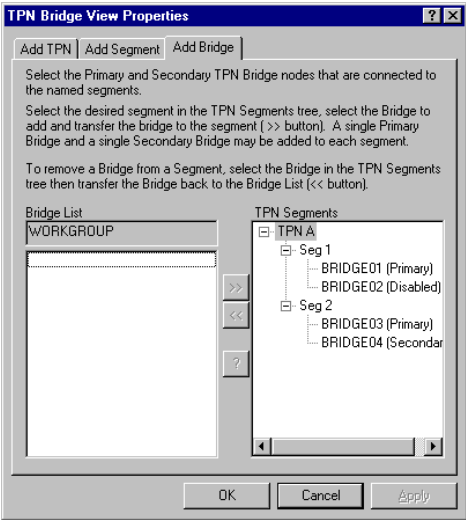
5. TPN Bridge Hot Insertion

5.8. Replacing a Bridge and Restoring Current Configuration

Step Action



- 7 Select the **Add Bridge** tab. The TPN Bridge View Snap-in will search for additional Bridges and find no new bridges.



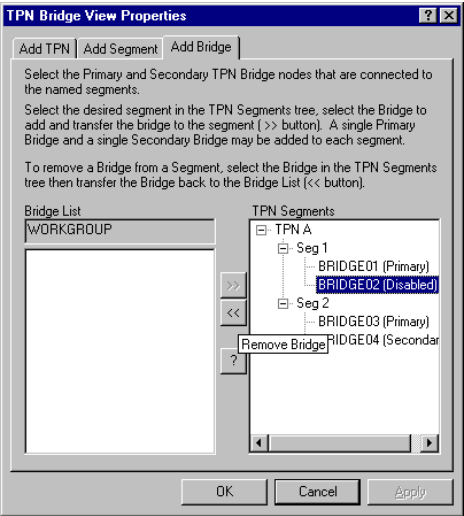
- 8 Select the Bridge being restored (**BRIDGE02** in example) and click **Remove Bridge** button. Click **Apply**.

## 5. TPN Bridge Hot Insertion

### 5.8. Replacing a Bridge and Restoring Current Configuration

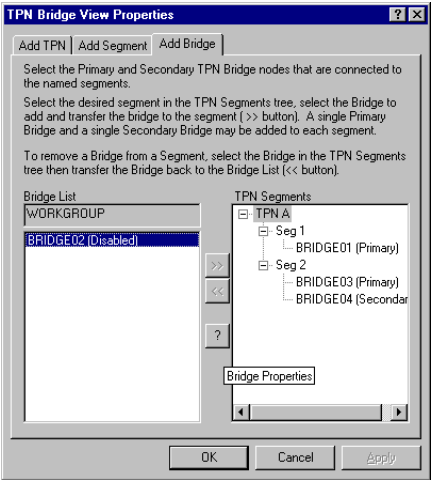
#### Step

#### Action



- 9 The TPN Bridge View snap-in will remove the Bridge and search the WORKGROUP for new bridges. The removed Bridge will be displayed in the **Bridge List**.

Select **BRIDGE02** in the **Bridge List** and click ? (**Bridge Properties**).



## 5. TPN Bridge Hot Insertion

### 5.8. Replacing a Bridge and Restoring Current Configuration

| Step | Action |
|------|--------|
|------|--------|

- |    |                                                                                                                                                                    |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10 | Set the LCN address to the correct Primary or Secondary Bridge LCN address and click <b>OK</b> . Select <b>OK</b> in the displayed Applying Properties dialog box. |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Bridge Properties                                                                           | Bridge Properties                                                                           |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Name: BRIDGE02                                                                              | Name: BRIDGE02                                                                              |
| Domain: WORKGROUP                                                                           | Domain: WORKGROUP                                                                           |
| Bridge Number: 0                                                                            | Bridge Number: 0                                                                            |
| Bridge Alias: _bridge0                                                                      | Bridge Alias: _bridge0                                                                      |
| TCP/IP Address: 192.168.1.217                                                               | TCP/IP Address: 192.168.1.217                                                               |
| Node Number: 0                                                                              | Node Number: 96                                                                             |
| Bridge Type:<br><input type="radio"/> Primary<br><input checked="" type="radio"/> Secondary | Bridge Type:<br><input type="radio"/> Primary<br><input checked="" type="radio"/> Secondary |
| OK Cancel                                                                                   | OK Cancel                                                                                   |

- |    |                                                                                      |
|----|--------------------------------------------------------------------------------------|
| 11 | Select the segment and the Bridge being restored and click <b>Add Bridge</b> button. |
|----|--------------------------------------------------------------------------------------|

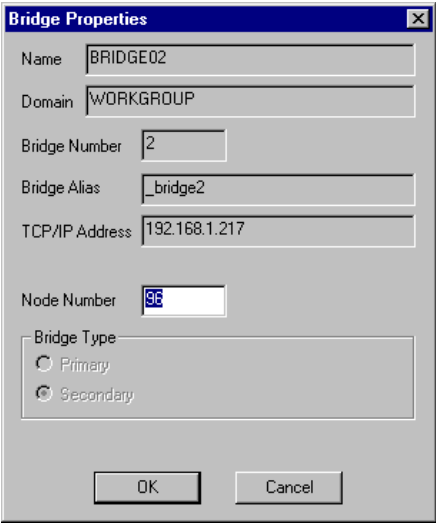
The TPN Bridge View Properties dialog box has three tabs: 'Add TPN', 'Add Segment', and 'Add Bridge'. The 'Add Bridge' tab is active. It contains instructions: 'Select the Primary and Secondary TPN Bridge nodes that are connected to the named segments.', 'Select the desired segment in the TPN Segments tree, select the Bridge to add and transfer the bridge to the segment (>> button). A single Primary Bridge and a single Secondary Bridge may be added to each segment.', and 'To remove a Bridge from a Segment, select the Bridge in the TPN Segments tree then transfer the Bridge back to the Bridge List (<< button)'. On the left is the 'Bridge List' with 'WORKGROUP' and 'BRIDGE02 (Secondary)'. On the right is the 'TPN Segments' tree with 'TPN A' expanded, showing 'Seg 1' (containing 'BRIDGE01 (Primary)') and 'Seg 2' (containing 'BRIDGE03 (Primary)' and 'BRIDGE04 (Secondary)'). Between the lists are '>>' and '<<' buttons, an 'Add Bridge' button, and a '?' button. At the bottom are 'OK', 'Cancel', and 'Apply' buttons.

- |    |                                                                                                                                       |
|----|---------------------------------------------------------------------------------------------------------------------------------------|
| 12 | The <b>Bridge Properties</b> dialog displays with the restored <b>Bridge Number</b> and <b>Bridge Alias</b> values. Click <b>OK</b> . |
|----|---------------------------------------------------------------------------------------------------------------------------------------|

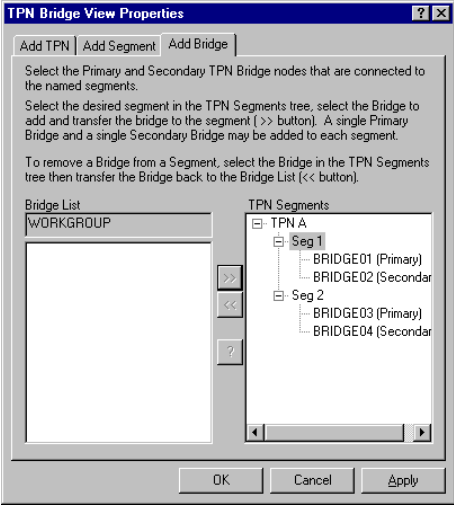
5. TPN Bridge Hot Insertion

5.8. Replacing a Bridge and Restoring Current Configuration

| Step | Action |
|------|--------|
|------|--------|



13 The Bridge displays in the **TPN Segments** tree. Click **OK**.



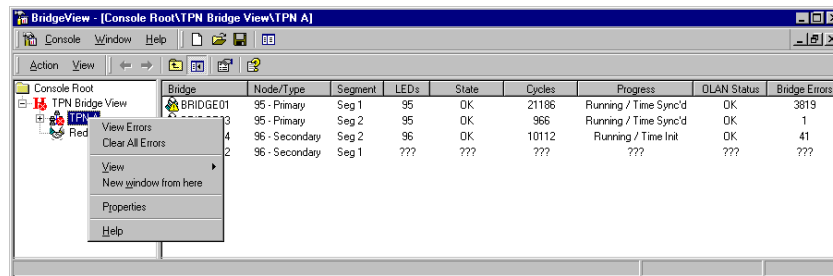
14 The bridge will appear in the Console window. Right-click the **TPN A** level and select **Properties**.

## 5. TPN Bridge Hot Insertion

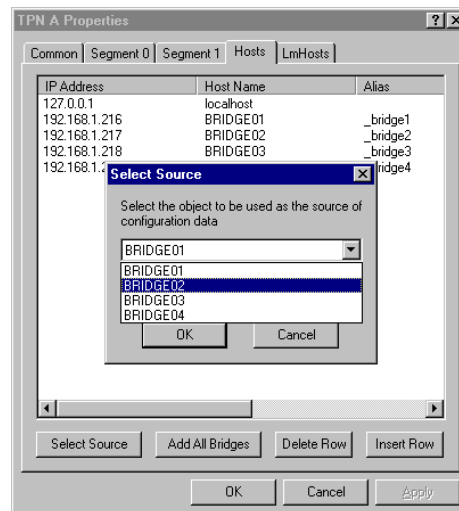
### 5.8. Replacing a Bridge and Restoring Current Configuration

#### Step

#### Action



- 15 Select the **Hosts** tab and click **Select Source** button. A **Select Source** pop-up appears containing a drop-down list of Bridges. Select the Bridge being restored (**BRIDGE02** in example). Click **OK**.



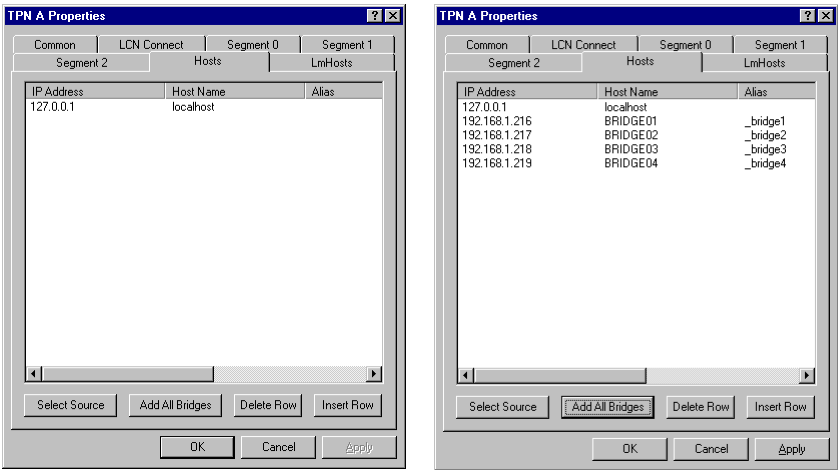
- 16 The Host file of the Bridge being restored displays. Click **Add All Bridges** button and click **Apply**.

5. TPN Bridge Hot Insertion

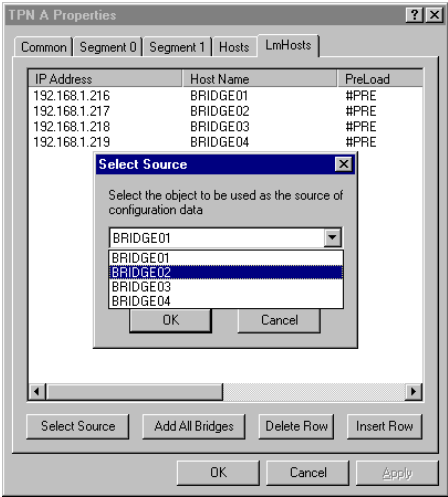
5.8. Replacing a Bridge and Restoring Current Configuration

Step

Action



- 17 Select the **LmHosts** tab and click **Select Source** button. A **Select Source** pop-up appears containing a drop-down list of Bridges. Select the Bridge being restored (**BRIDGE02** in example) and click **OK**.

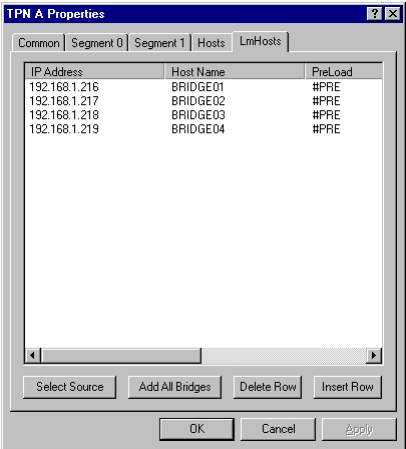
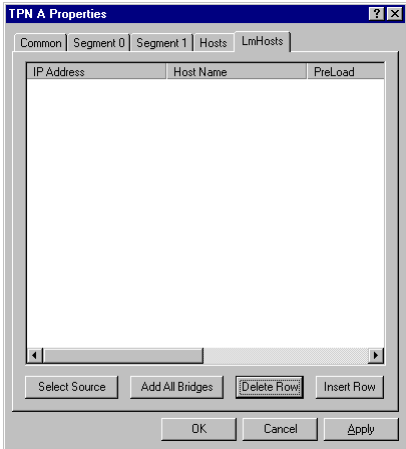


- 18 Select **Add All Bridges** and click **Apply**.

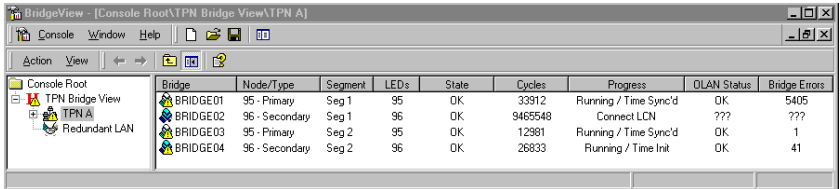
5. TPN Bridge Hot Insertion

5.8. Replacing a Bridge and Restoring Current Configuration

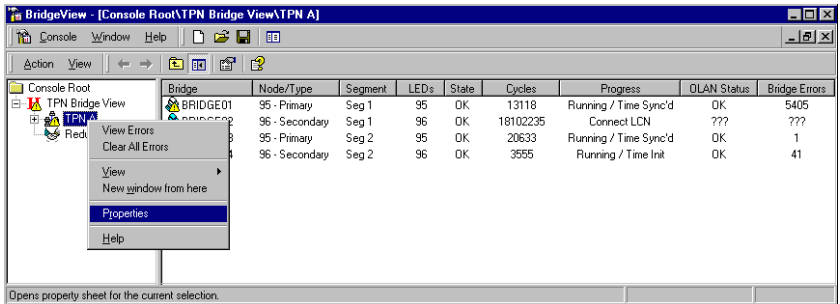
Step Action



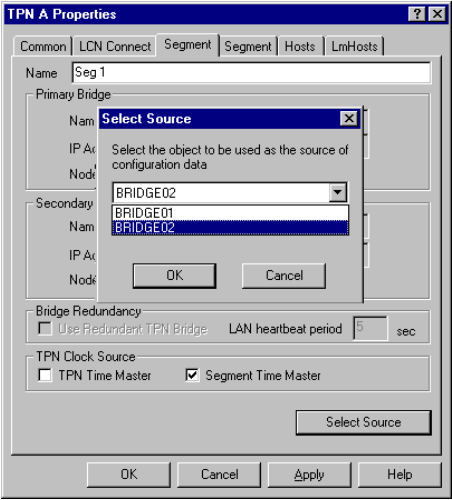
- 19 Close the MMC console without saving the MSC console file.
- 20 Reboot the restored TPN Bridge node and logon.
- 21 Using a configured TPN Bridge node (not the one being restored), start the MMC Console program. The status of the restored bridge will be in the **Connect LCN** state.



- 22 Right-click **TPN A** level and select **Properties**.



| Step | Action                                                                                                                                                                                                                                                        |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23   | Select the Segment tab containing the Bridge being restored and click <b>Select Source</b> button. A <b>Select Source</b> pop-up appears containing a drop-down list of Bridges. Select the Bridge (BRIDGE02 in example) being restored and click <b>OK</b> . |



## 5. TPN Bridge Hot Insertion

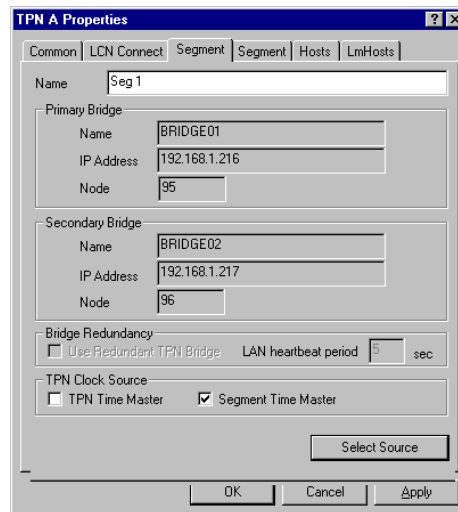
### 5.8. Replacing a Bridge and Restoring Current Configuration

---

| Step | Action |
|------|--------|
|------|--------|

---

- |    |                                                                                                                                                                       |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24 | The current settings from the Bridge being restored will be displayed. Set the LCN clock source value to the appropriate settings for this segment. Click <b>OK</b> . |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|



- |    |                                                                                                                         |
|----|-------------------------------------------------------------------------------------------------------------------------|
| 25 | The Bridge restoration is now complete. Shutdown only the Bridge being restored.                                        |
| 26 | Power off the Bridge node and reconnect the LCN cables.                                                                 |
| 27 | Power on the Bridge. The Bridge should now be operating normally.                                                       |
| 28 | Bring up the <code>SYSTEM STATUS</code> display on a US or GUS and ensure the TPN Bridges display <code>OK-BRG</code> . |
| 29 | Verify the Bridge operation using Table 4-7 and Table 4-8.                                                              |
-

## 5.9 Removing and Replacing the Ethernet Switch

You will perform this procedure when the Ethernet Switch requires removal and replacement due to a failure.

### Replacing the TPN Bridge System ethernet switch

| Step                                                                                | Action                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                                                                                   | In the <b>TPN Bridge Configuration/Status Snap-in</b> , each TPN Bridge will display <b>Suspect</b> under <b>LAN Status</b> column.                                                                                                                                                                                                                                              |
| 2                                                                                   | Expand the <b>Redundant LAN</b> item and look at status-column for details. The <b>Redundant LAN</b> will display <b>LAN A</b> or <b>LAN B</b> as <b>Failed</b> .                                                                                                                                                                                                                |
| 3                                                                                   | Ensure all TPN Bridge nodes are using the correct Ethernet cable (i.e., the cable running to the switch that IS NOT being replaced).                                                                                                                                                                                                                                             |
| 4                                                                                   | To force the TPN Bridges to the correct cable, right-click on the <b>Redundant LAN</b> item and select <b>Force Cable X</b> , where X is the good Ethernet Switch cable.                                                                                                                                                                                                         |
| 5                                                                                   | Remove the LAN cables (A or B) from the failing Ethernet Switch.                                                                                                                                                                                                                                                                                                                 |
| 6                                                                                   | Power off (unplug power cord from outlet) the Ethernet Switch that is to be replaced.                                                                                                                                                                                                                                                                                            |
|  | <b>WARNING</b><br>The Ethernet Switch does not have a power on/off switch. Removal of the power cord is the only way to turn off power to this device. The power cord must always be connected in a location that can be accessed quickly and safely in case of an emergency.                                                                                                    |
|  | <b>WARNING</b><br>Verify snooping has been turned off on the switch. See the "Disable "snooping" on Nortel (BayStack) switch" procedure in Section 4.7.                                                                                                                                                                                                                          |
| 7                                                                                   | Install the new Ethernet Switch.                                                                                                                                                                                                                                                                                                                                                 |
| 8                                                                                   | Reinstall all LAN cables (A and B) and restore power to the Ethernet Switch (plug power cord into outlet). Verify it passes self-test by observing the Power and Status LEDs turn on. Check the Port LEDs. The port status LEDs will flash various patterns and then remain on. If failures occur, consult vendor manual or customer service from whom you purchased the switch. |
| 9                                                                                   | Bring up the <b>SYSTEM STATUS</b> display on a US or GUS and ensure the TPN Bridges display <b>OK-BRG</b> for all TPN Bridges on all segments.                                                                                                                                                                                                                                   |

## 5. TPN Bridge Hot Insertion

### 5.9. Removing and Replacing the Ethernet Switch

---

| Step | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10   | In the <b>TPN Bridge Configuration/Status Snap-in</b> , expand the <b>Redundant LAN</b> item. Verify the <b>Redundant LAN</b> status has <b>Active</b> and <b>Backup</b> on all TPN Bridges.                                                                                                                                                                                                                                                                   |
| 11   | Right-click the <b>Redundant LAN</b> item and select <b>Clear All Errors</b> . Clear errors on <u>all</u> TPN Bridges.                                                                                                                                                                                                                                                                                                                                         |
| 12   | Test the replaced Ethernet Switch by forcing all TPN Bridge nodes to <u>each</u> cable connected to the replaced switch. <ul style="list-style-type: none"><li>• Right-click on the <b>Redundant LAN</b> item, select <b>Force Cable A</b>. Wait one minute, and verify all cables switch to Cable A.</li><li>• Right-click on the <b>Redundant LAN</b> item, select <b>Force Cable B</b>. Wait one minute, and verify all cables switch to Cable B.</li></ul> |
| 13   | This verifies that the new Ethernet Switch is running correctly.                                                                                                                                                                                                                                                                                                                                                                                               |

---

## 6. Health Check of TPS LCN System

### 6.1 TPS LCN System Checkout

Importance of



#### CAUTION

Before performing any TPN Bridge upgrade process, it is critical to confirm that the TPS LCN system is functioning correctly. Failure to do so could result in LCN node failure(s) and isolation of segments. **Check the LCN System Status display on all segments to ensure the LCN nodes are working properly.**

#### Checking TPS LCN system

| Step | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <p>Verify CLOKSTAT display indicates the following:</p> <p>Every node displays LISTENER with the exception to the configured TPS LCN CLOCK MASTER and SLAVE. If the nodes do not display the correct information, the problem must be resolved before the TPN Bridges can be upgraded.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2    | <p>Verify the LCN Diagnostics displays do not display any known problems using the \$LNMENU:</p> <ul style="list-style-type: none"><li>• NET MEDIA DEADS – All configured nodes should display a value of 0 for CBL-A and CBL-B. If there are any other value, resolve the problem before the TPN Bridge can be upgraded.</li><li>• CABLE A&amp;B RATES – All configured nodes should display a value of 0 for CBL-A and CBL-B. If there are any other value, resolve the problem before the TPN Bridge can be upgraded.</li><li>• LOCAL DRVR DATA – Call up this display and enter the LOWEST US or GUS node number and verify the display indicated Cable Master value is "Master". If this is not the case, shutdown and reload the US or GUS and verify it now displays "Master".</li><li>• CROSSED CABLES – The lowest node should display a value of one higher than the rest of the nodes. This is depending on whether the system is an all K2/K4 system. If the values are not correct, resolve the problem before the TPN Bridge can be upgraded.</li></ul> |

## 6. Health Check of TPS LCN System

### 6.1. TPS LCN System Checkout

---

| Step | Action                                                                                                                                                                                                                                                              |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3    | Verify the System Status display indicated "CABLE A: OK" and "CABLE B: OK" and displayed in green with one of the cables back-lit in solid green. If the cables are not displayed correctly, resolve the problem before upgrading the TPN Bridge software           |
| 4    | Verify the System Status display has "SYS VERS: displaying a software version, not "INIT LCN RECONNECT". If it displays "INIT LCN RECONNECT", resolve the problem before the TPN Bridge can be upgraded.                                                            |
| 5    | If all of the information above meets the requirements, call up the \$LNMENU display, "RESET TIMES". And select "RESET ALL COUNTERS". By doing this, you now have a known status as of NOW, to determine if problems are encountered during the TPN Bridge upgrade. |

---

## 6.2 Normal Operation TPN Bridge Health Check

Expected values for TPN Bridge status

Table 6-1 TPN Bridge Status: Expected Values

| Value Name                                                                                                                       | Primary Expected Value | Secondary Expected Value |
|----------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------------------|
| TPN Bridge Viewer – Bridge Status                                                                                                |                        |                          |
| Cycle Counter                                                                                                                    | Changing               | Changing                 |
| Load Progress                                                                                                                    | 254*                   | 254*                     |
| We Primary                                                                                                                       | 1                      | 0                        |
| OLAN OK                                                                                                                          | 3                      | 3                        |
| Buddy OK                                                                                                                         | 1                      | 1                        |
| NT -> LCN Data (C1)                                                                                                              | Changing               | Changing                 |
| LCN -> NT Data (C1)                                                                                                              | Changing               | Changing                 |
| LCN -> NT Buffers (C1)                                                                                                           | Changing Slowly        | Changing Slowly          |
| NT -> LCN Buffers (C1)                                                                                                           | Changing               | Changing                 |
| Use NT LAT                                                                                                                       | 0                      | 0                        |
| Seconds Since Msg Received (NT)                                                                                                  | less than 11 seconds   | less than 11 seconds     |
| Seconds Since Msg Received (LCNI)                                                                                                | less than 11 seconds   | less than 11 seconds     |
| *254 is the expected value if Time Synchronization is Disabled.<br>255 is the expected value if Time Synchronization is Enabled. |                        |                          |

**6. Health Check of TPS LCN System**  
6.2. Normal Operation TPN Bridge Health Check

---

**Expected values for TPN Bridge**

**Table 6-2 Expected Values**

| TPN Bridge TDC Status                                      |                                |                                 |
|------------------------------------------------------------|--------------------------------|---------------------------------|
| Value Name                                                 | Primary Expected Value         | Secondary Expected Value        |
| TPN Bridge Viewer – TDC Status                             |                                |                                 |
| System Status – Recompute LAT                              | May be set                     | May be set                      |
| System Status – Physical Dirty                             | Set                            | Set                             |
| System Status – Looping                                    | Set                            | Set                             |
| System Status – Disconnected                               | Cleared                        | Cleared                         |
| Notice Status – Multi Alone In Ring                        | Cleared                        | Cleared                         |
| Notice Status – All Other Bits                             | Cleared                        | Cleared                         |
| Input Loop Scan                                            | Set                            | Set                             |
| TPN Bridge Viewer – WS Counters                            |                                |                                 |
| Note: Clear all counters before proceeding with this check |                                |                                 |
| Configured Primary                                         | 1                              | 0                               |
| Running Primary                                            | 1                              | 0                               |
| Tx Total                                                   | Changing rapidly               | Incrementing 1 every 10 seconds |
| Tx Failures                                                | 0                              | 0                               |
| Tx LCN Msg                                                 | Changing rapidly               | 0                               |
| Tx Buddy Msg                                               | Incrementing 2 per second      | Incrementing 1 every 10 seconds |
| Tx PLAT Msg                                                | Incrementing within 10 seconds | 0                               |
| Rx Total                                                   | Changing rapidly               | Changing rapidly                |
| Rx Failures                                                | 0                              | 0                               |
| Rx Lost                                                    | 0                              | 0                               |
| Rx Out of Order                                            | 0                              | 0                               |
| Rx Duplicated                                              | 0                              | 0                               |
| Rx LCN Msg                                                 | Changing rapidly               | Changing rapidly                |

**Table 6-2      Expected Values**

| TPN Bridge TDC Status     |                                 |                            |
|---------------------------|---------------------------------|----------------------------|
| Value Name                | Primary Expected Value          | Secondary Expected Value   |
| Rx LCN Lost               | 0                               | 0                          |
| Rx LCN Out of Order       | 0                               | 0                          |
| Rx Buddy Msg              | Incrementing 1 every 10 seconds | Incrementing 2 per second  |
| Rx Buddy Msg Lost         | 0                               | 0                          |
| Rx Buddy Msg Out of Order | 0                               | 0                          |
| Rx PLAT Msg               | Changing within 10 seconds      | Changing within 10 seconds |
| Rx PLAT Msg Lost          | 0                               | 0                          |
| Rx PLAT Msg Out of Order  | 0                               | 0                          |

## 6.3 TPN Bridge Hints and Troubleshooting

### Hints

The following table provides a list of recommended actions to take when using the TPN Bridge status displays.

| No. | Recommended action                                                                                                                                    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Check Bridge viewer statistics using the chart.                                                                                                       |
| 2   | Check for excessive CPU or memory usage in Task Manager, and close (end process) the mmc.exe or DLLHOST.exe if they are running.                      |
| 3   | Close the MMC console after each use.<br><b>Note:</b> It is located under My Administrative Tools.                                                    |
| 4   | Periodically clear the System and Application logs in Event Viewer.<br><b>Note:</b> Event logs cannot be written to when they contain too many items. |

## Troubleshooting

The following table provides a list of actions to take under certain circumstances.

| If this occurs . . .                                                                    | Do this . . .                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary bridge failure when there is a solid LCN cable fault.                           | Reset the primary TPNB (LCNP).<br><br><b>Cause:</b> The secondary bridge node continues to try and communicate on both LCN cables.<br><br>The cable problem(s) must be resolved as soon as possible for the Primary bridge to stay operational. |
| TPN Bridge node isolation                                                               | Always fix any LCN Cable problems as soon as possible.<br><br><b>Cause:</b> If the system runs for a long period of time with a bad cable, the system can become ISOLATED if double cable faults are present.                                   |
| TPN Bridge locks up when Disk Administrator is opened.                                  | Immediately push the reset button.<br><br>Avoid calling up the Disk Administrator until Honeywell has a fix in place for this problem.                                                                                                          |
| Every 10 to 30 minutes the LCN status for the bridges goes to Test for several seconds. | Right-click on the LAN status in the MMC and view the errors to identify the faulty switch.                                                                                                                                                     |

## **6. Health Check of TPS LCN System**

### **6.3. TPN Bridge Hints and Troubleshooting**

---

## 7. Recovery Guidelines

### 7.1 Overview

#### Purpose

These scenario recovery procedures are intended as *guidelines only*. They are not meant to supplant a customer determined recovery plan, but to support one. They have relevant and critical information presented in a general format, intended to be used as templates for creating a specific recovery plan.

#### Responsibilities

It is the responsibility of the customer to create step-by-step RECOVERY procedures and to ensure that the recovery plan is assigned to the appropriate personnel.

Discuss and determine responsibilities for Hardware, Equipment, Software, etc. Areas of responsibility include Operations, Process Control Engineer, Maintenance, etc. and Honeywell (if applicable).

#### TPS LCN recovery requirements

The following are requirements for a TPS LCN Recovery Plan:

- One person who is assigned as the LEAD (must be the Customer).
- All Network Configurations are to be completed; no engineering work during this transition.
- Complete HM (all HM's) Backups
- FAST LOAD Cartridges available
- For Standard US or UxS, ZIP or Bernoulli drives are available.  
**NOTE**: Must be LCN formatted. ZIP or Bernoulli drives on a US cannot be PC formatted.
- For GUS, an Emulated Disk on the local hard drive or as a ZIP drive is available.  
**NOTE**: Must be PC formatted. ZIP drive on a GUS cannot be LCN formatted. Bernoulli drives are not available on GUS.

## 7. Recovery Guidelines

### 7.1. Overview

---

#### Required files for FAST LOAD cartridges

FAST LOAD cartridges MUST contain the following files:

- Personality Files for US/GUS/HG/NIM/UCN Devices/AM/CG
  - &UNP (Universal Personality)
  - &HGO (Hiway Gateway)
  - &NMO (Network Interface Module)
  - &UCN (UCN Device (PMM/APM/HPM/SMM/SM/FSC/LMM))
  - &CIO (Computer Gateway)
  - &AMO (Application Module)
- All AREA DATA BASE files
- AREAx.x.DA (where xx represents the appropriate area)
- Button file
- Schematic Volumes and/or Directories to include all schematics or GUS Graphics
- All (&CUS and &CLX) backplane software applications files
- Process Network checkpoint(s)
- Hiway Gateway Network for ALL BOXES
  - UCN and ALL UCN Devices (APM, PMM, HPM, LMM, SM, SMM, FSC)
- FAST LOAD cartridges available at each CONSOLE/AREA
- Latest NCF files (up-to-date)
  - (&ASY .SE and .SP files, and any other customer support files)
  - The system must be clear of LCN communication problems. If they exist, they must be resolved. (Check system displays, system journals, etc.)
- Checkpoint ALL AM/APP/PHD to removable for the FAST LOAD cartridges or emulated disks.

- If AM's/APP/PHD use Totalizers or Accumulated application values, the AM's MUST be checkpointed to the FAST LOAD cartridges before the AM's are shutdown. The LAST values in the checkpoint will then be restored to these entity types when the AM's are reloaded using the HOT LOAD.
- Checkpoint performed for all CG/PCIM/CLM etc. to the FAST LOAD cartridges
- IF CG/PCIM/CLM etc. exist, the upper-level processor (VAX etc.) may require operations available.
- Have operations available (on standby) for worse case scenario.
- Someone available at each LCN segment to monitor the system during the installation.
- TOPOLOGY MAP available (Up-to-date)
- Telephone #'s available for each location.
- Reset all LCN statistics.
- ESD protection is mandatory
- Hiway Gateway Network for ALL BOXES
- You may consider BASIC CONTROL
  - You may consider setting Process Modules to a safe state.
  - You may consider TIME BASE related applications
- ALL UCN Devices (APM, PMM, HPM, LMM, SM, SMM, FSC)
  - You may consider BASIC CONTROL
  - You may consider setting Process Modules to a safe state.
  - You may consider OUPUTS of either HOLD or UNPOWER
  - You may consider TIME BASE related applications

## 7.2 Scenario Recovery Procedures

### Overview

These procedures are **NOT** to be used as the Customer's recovery plan. These procedures are to be used for reference only. All Customers must document their own site-specific requirements.

#### Scenario # 1: No problems encountered

No problems encountered. No recovery plan required. Monitor the system to ensure it remains stable. Notify parties involved of the status.

#### Scenario # 2: Some TPS nodes go ISOLATED



#### ATTENTION

If the INIT LCN RECONNECT is not on the SYSTEM STATUS display, select each node one at a time to determine the best view to the process and initiate the LCN reconnect from that segment and use that segment to load the ISOLATED nodes. If this is not followed, there could be a loss of view to the process or loss of control. If at any time, you issue the LOAD ISOLATE, and you decide you want it to be terminated, select the CANCEL LOAD. No further nodes will be loaded until another load is issued.

---

**Table 7-1 Recovery Guidelines for ISOLATED TPS Nodes (some)**

| No. | Guideline                                                                                                                                             |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Notify Operations immediately on the system status.                                                                                                   |
| 2   | Identify the most critical process TPS/LCN segment: all nodes must be reloaded from that segment.                                                     |
| 3   | If the System HM is accessible, and there are non-redundant nodes involved (HG or NIM) you can use the LOAD ISOLATE depending on the plant situation. |

**Table 7-1      Recovery Guidelines for ISOLATED TPS Nodes (some)**

| No.      | Guideline                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>4</b> | <p>If the System HM is not available on the most critical segment, perform either of the following precautions:</p> <p>From the segment that can see the System HM, select it and perform a SHUTDOWN NODE and wait for it to go to QUALIF State. Go to the critical process segment and using either "&amp;Z1" (CD, ZIP, EMULATED DISK), or any prepared loadable HM personality disk), load the HM to the HMON personality.</p> <p>If there is no US(s) that has VIEW to the System HM, go to the HM node electronic and press the RESET button (1) time to allow it to be loaded from the critical process segment. Verify on the critical process segment that the HM displays QUALIF. If not, resolve problem before continuing. This must be initiated from the most critical segment of the plant to retain VIEW to the process.</p>                                                                                                         |
| <b>5</b> | <p>If redundant nodes HG(s) or EPLCG(s) exist on the system and their status is ISOLATED, perform the following:</p> <ol style="list-style-type: none"> <li>a) If you have view from a segment other than the critical segment, select the BACKUP HG and perform a SHUTDOWN. The HG running as PRIMARY will display WARNING because it has no BACKUP.</li> <li>b) Wait for the HG to go to QUALIF status.</li> <li>c) Go to the critical process segment and load the HG using the "LOAD SFT UPGRADE" target.</li> <li>d) Once VIEW has been retained on both segments, go to the segment that has view to the PRIMARY HG and shut it down.</li> <li>e) Wait for QUALIF or a loadable state.</li> <li>f) Go to the critical process segment LOAD the redundant partner</li> <li>g) Once it goes to BACKUP, select the HG that displays UPGRADE and "ENABLE UPGRADE" target.</li> <li>h) Wait for the redundant pair to go to OK/BACKUP.</li> </ol> |

## 7. Recovery Guidelines

### 7.2. Scenario Recovery Procedures

---

**Table 7-1      Recovery Guidelines for ISOLATED TPS Nodes (some)**

| No.       | Guideline                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>6</b>  | <p>If redundant nodes are NIM(s) and their status is ISOLATED, perform the following:</p> <ul style="list-style-type: none"><li>a) If you have view from a segment other than the critical segment, select the BACKUP NIM and perform a SHUTDOWN.</li><li>b) Wait for the NIM to go to QUALIF status. The NIM running as PRIMARY will display WARNING because it has no BACKUP.</li><li>c) Go to the critical process segment and load the NIM using the "LOAD SFT UPGRADE" target.</li><li>d) Once VIEW has been retained on both segments, go to the segment that has view to the PRIMARY NIM and shut it down.</li><li>e) Wait for QUALIF or a loadable state.</li><li>f) Go to the critical process segment and perform the "ENABLE UPGRADE" function.</li><li>g) Wait for the status to go to WARNING.</li><li>h) LOAD the redundant partner..</li><li>i) Wait for the redundant pair to go to OK/BACKUP.</li></ul> |
| <b>7</b>  | <p>If the nodes that went to the ISOLATED State are non-critical, (GUS) RESET the LCNP. Go back to the US or GUS and perform a LOAD PWR_ON.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>8</b>  | <p>If the nodes that went to the ISOLATED State are non-critical RESET the LCNP. Go back to the US or GUS and perform a LOAD PWR_ON. You may use the LOAD ISOL from the SYSTEM STATUS display.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>9</b>  | <p>If the node(s) are AM/APP/PHD, perform MANUAL checkpoints to the removable media and Shutdown the AM/APP/PHD and load it using the critical process segment and the new checkpoints if Totalizers and accumulative values are to be restored.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>10</b> | <p>If the node(s) are CG/PLNM/CLM etc., the node and the upper level processor must be prepared for a reload.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>11</b> | <p>Once all nodes have been reloaded (that were ISOLATED), the INIT LCN RECONNECT on the system status display will go away after 2 minutes.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>12</b> | <p>Using up to THREE US's or GUS stations to perform loads of the TPS system, select the most critical nodes and load them to allow VIEW and CONTROL to the</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

**Table 7-1 Recovery Guidelines for ISOLATED TPS Nodes (some)**

| No. | Guideline                                                                                              |
|-----|--------------------------------------------------------------------------------------------------------|
|     | process.                                                                                               |
| 13  | Once all nodes have been loaded, check all control strategies to ensure the plant is running properly. |
| 14  | Notify ALL parties involved of the status.                                                             |

**Scenario # 3: Segment(s) ISOLATED from each other**



**ATTENTION**

If the INIT LCN RECONNECT is not on the SYSTEM STATUS display, select each node one at a time to determine the best view to the process and initiate the LCN reconnect from that segment and use that segment to load the ISOLATED nodes. If this is not followed, there could be a loss of view to the process or loss of control. If at any time, you issue the LOAD ISOLATE, and you decide you want it to be terminated, select the CANCEL LOAD. No further nodes will be loaded until another load is issued.

**Table 7-2 Recovery Guidelines for ISOLATED Segments**

| No. | Guideline                                                                                        |
|-----|--------------------------------------------------------------------------------------------------|
| 1   | Notify Operations immediately on the system status.                                              |
| 2   | Decide which segment will be used to reload the remote segment(s).                               |
| 3   | Identify the most critical process TPS/LCN segment: all nodes must be reloaded from this segment |
| 4   | All ISOLATED nodes visible from one segment, must be used to SHUTDOWN these nodes.               |
| 5   | If the System HM is accessible, you can use the LOAD ISOLATE depending on the plant situation    |

## 7. Recovery Guidelines

### 7.2. Scenario Recovery Procedures

---

**Table 7-2 Recovery Guidelines for ISOLATED Segments**

| No. | Guideline                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6   | <p>If the System HM is not available on the most critical segment, perform either of the following:</p> <ol style="list-style-type: none"><li>From the segment that can see the System HM, select it and perform a SHUTDOWN NODE and wait for it to go to QUALIF State. Go to the critical process segment and using either "&amp;Z1" (CD, ZIP, EMULATED DISK), or any prepared loadable HM personality disk), load the HM to the HMON personality.</li><li>If there is no US(s) that has VIEW to the System HM, go to the HM node electronic and press the RESET button (1) time to allow it to be loaded from the critical process segment. Verify on the critical process segment that the HM displays QUALIF. IF not, resolve problem before continuing. This must be initiated from the most critical segment of the plant to retain VIEW to the process.</li></ol>                                  |
| 7   | <p>If redundant nodes HG(s) or EPLCG(s) exist on the system and their status is ISOLATED, perform the following:</p> <ol style="list-style-type: none"><li>If you have view from a segment other than the critical segment, select the BACKUP HG and perform a SHUTDOWN. The primary HG will go to WARNING due to it not having a BACKUP.</li><li>Wait for the HG to go to QUALIF status.</li><li>Go to the critical process segment and load the HG using the "LOAD SFW UPGRADE" target.</li><li>Once VIEW has been retained on both segments, go to the segment that has view to the PRIMARY HG and shut it down.</li><li>Wait for QUALIF or a loadable state.</li><li>Go to the critical process segment LOAD the redundant partner</li><li>Once it goes to BACKUP, select the HG that displays UPGRADE and "ENABLE UPGRADE" target.</li><li>Wait for the redundant pair to go to OK/BACKUP.</li></ol> |

**Table 7-2      Recovery Guidelines for ISOLATED Segments**

| No.       | Guideline                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>8</b>  | <p>If redundant nodes are NIM(s) and their status is ISOLATED, perform the following:</p> <ol style="list-style-type: none"> <li>a) If you have view from a segment other than the critical segment, select the BACKUP NIM and perform a SHUTDOWN.</li> <li>b) Wait for the NIM to go to QUALIF status. The Primary NIM will go to WARNING due to it not having a BACKUP.</li> <li>c) Go to the critical process segment and load the NIM using the "LOAD SFT UPGRADE" target.</li> <li>d) Once VIEW has been retained on both segments, go to the segment that has view to the PRIMARY NIM and shut it down.</li> <li>e) Wait for QUALIF or a loadable state.</li> <li>f) Go to the critical process segment and perform the "UPGRADE ENABLE" function.</li> <li>g) Wait for the status to go to WARNING.</li> <li>h) LOAD the redundant partner.</li> <li>i) Wait for the redundant pair to go to OK/BACKUP.</li> </ol> |
| <b>9</b>  | <p>If the nodes that went to the ISOLATED State are non-critical, (GUS) RESET the LCNP. Go back to the US or GUS and perform a LOAD PWR_ON.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>10</b> | <p>If the node(s) are AM/APP/PHD, perform MANUAL checkpoints to the removable media and Shutdown the AM/APP/PHD and load it using the critical process segment and the new checkpoints if totalizers and accumulative values are to be restored.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>11</b> | <p>If the node(s) are CG/PLNM/CLM etc., the node and the upper-level processor must be prepared for reload.</p> <ul style="list-style-type: none"> <li>• If the nodes that went ISOLATED are non-critical (US's) go to each node electronics and press the RESET button. Go back to the GUS or US and perform a LOAD PWR_ON.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>12</b> | <p>If there are nodes that are ISOLATED and are non-critical, RESET the LCNP. Go back to the US or GUS and perform a LOAD PWR_ON.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>13</b> | <p>Once all nodes have been reloaded (that were ISOLATED), the INIT LCN RECONNECT on the system status display will go away after 2 minutes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## 7. Recovery Guidelines

### 7.2. Scenario Recovery Procedures

---

**Table 7-2 Recovery Guidelines for ISOLATED Segments**

| No. | Guideline                                                                                                                                                         |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14  | Using up to THREE US's or GUS stations to perform loads of the TPS system, select the most critical nodes and load them to allow VIEW and CONTROL to the process. |
| 15  | Once all nodes have been loaded, check all control strategies to ensure the plant is running properly.                                                            |
| 16  | Notify ALL parties involved of the status.                                                                                                                        |

---

#### Scenario # 4: All nodes are ISOLATED or in the FAIL state

**Table 7-3 Recovery Guidelines for All Nodes in ISOLATED or FAIL State**

| No. | Action                                                                                                                                                                                                                                             |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Notify Operations immediately on the system status.                                                                                                                                                                                                |
| 2   | Decide if the new installation should be removed from the system at this time.                                                                                                                                                                     |
| 3   | FAST LOAD cartridges will have to be used in order to load nodes.                                                                                                                                                                                  |
| 4   | Go to each node on the LCN. If the LCNI card (transmit LED amber in color) is flashing, and the address LED indicates it is failed, ALL nodes must be powered off. Once all nodes are powered off, go to each node and power them up one at a time |
| 5   | Load a US or a GUS using the FAST LOAD cartridge(s) or emulated disk(s)                                                                                                                                                                            |
| 6   | Enter the correct date/time.                                                                                                                                                                                                                       |
| 7   | Check to ensure the System HM is accessible.                                                                                                                                                                                                       |
| 8   | Using up to THREE US's or GUS stations to perform loads of the TPS system, select the most critical nodes and load them to allow VIEW and CONTROL to the process.                                                                                  |
| 9   | Once all nodes have been loaded, check all control strategies to ensure the plant is running properly                                                                                                                                              |
| 10  | Notify ALL parties involved of the status.                                                                                                                                                                                                         |

---



**Honeywell**

---

Honeywell International  
Industry Solutions  
2500 West Union Hills  
Phoenix, AZ 85027